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ECO





OPERATION AND MAINTENANCE MANUAL

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Please have the system's serial number available (printed on the rating label on the front of the cabinet).

Visit us on the web: www.nuvonic.com

To order a further copy of this manual, use the contact details above quoting the code on the front cover.

Warranty

Nuvonic warrants their UV systems against defective materials and manufacturing defects for a period of 18 months from date of receipt.

Refer to our website for full details of the terms and conditions of warranty for systems, UV lamps, and other consumables.

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Passwords

The default passwords required to gain access to the different system menus are listed below.

Login	Password
user	—
admin	admin
factory	(supplier use only)

System parameters are factory-set to suit the requirements of the process. Changing these parameters can lead to insufficient treatment levels or malfunctioning of the entire system, and may prevent the minimum performance requirements being met.

The configuration of the system should only be changed by personnel who are fully aware of the implications of the changes. For technical assistance and details of training courses, please consult technical support using the contact details given on the inside front cover of this manual.

Remove this page from the manual when appropriate to prevent unauthorised access to the software menus.

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1. Introduction

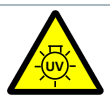
This manual is intended to provide detailed guidance on the installation, commissioning, operation, and maintenance of your UV system.

Please make this manual available for reference by any persons working with the system.

A section has been provided (Appendix A) to note updates, modifications and operating data. Ensuring this section is used and kept up to date is the responsibility of the equipment owner and will assist in ensuring the system is maintained in a good operating condition.

A printed Installation Guide is also provided with this system.

The following symbols are used in this manual (les symboles suivants sont utilisés dans ce manuel):

	This symbol indicates a requirement or warning. Indique une nécessité ou un avertissement.
	This symbol indicates a recommendation. Indique un conseil.
	This symbol indicates an electrical hazard. Indique un danger électrique.
	This symbol indicates a hazard due to ultraviolet (UV) light. Indique un danger en raison de la lumière ultraviolette (UV).
	This symbol indicates a hazard due to pressurised liquid within the chamber. Indique un danger dû à un liquide sous pression dans la chambre.
	This symbol indicates a hazard due to high temperature. Indique un danger en raison de températures élevées.

1.1. Glossary of Abbreviations/Terms

316L	A grade of 316 stainless steel with reduced carbon content
CFD	Computational Fluid Dynamics
COSHH	Control of Substances Hazardous to Health regulations
Dose	UV energy incident on unit surface area, measured in mJ/cm^2; also known as fluence
Drg	Drawing
ID	Internal Diameter
Intensity	UV power incident on unit surface area, measured in W/m^2
IPA	Iso-propyl alcohol (2-propanol)
LCD	Liquid Crystal Display
LED	Light-Emitting Diode

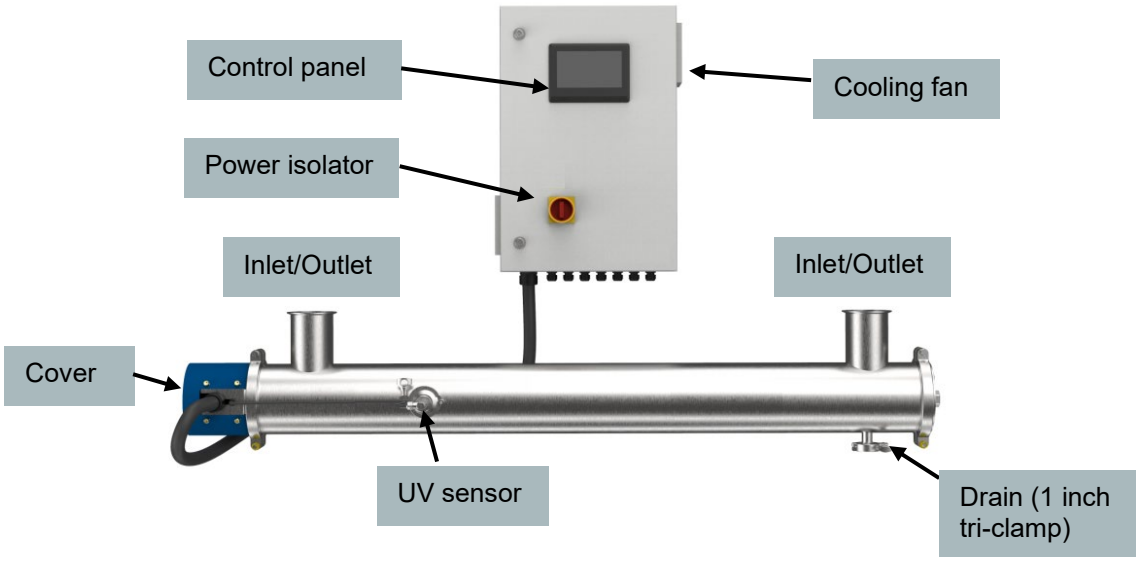
OD	Outer Diameter
P/N	Part number
RED	Reduction Equivalent Dose – measured UV dose in mJ/cm^2 delivered by the system
Tube	A quartz sleeve (open both ends) which surrounds the UV lamp
UV	Ultraviolet
UVT	Ultraviolet Transmittance; the percentage of UV light transmitted through a material, usually over a distance of 10 mm (0.4 inches); also abbreviated as TX
VFC	Volt-Free Contact
VP	Variable Power

1.2. System Description

The UV system consists of a UV chamber, control panel and connecting cables.

The primary components of the system are shown in Figure 1.

Figure 1: System Overview



1.2.1. UV chamber

The chambers are manufactured out of 316L stainless steel. They are available with a variety of inlet and outlet connection interfaces to suit customers' requirements (the sizes are fixed for each model).

The UV lamp is housed within a quartz tube to prevent direct contact with the fluid being treated. The quartz tube and the lamp can only be installed from one end of the chamber, adjacent to the outlet connection.

The UV lamps used in the system are of the amalgam type, which emit ultraviolet light at a wavelength of approximately 254 nm. A UV sensor, fitted to the chamber, detects the

intensity of UV light being emitted from the UV lamp measured after passage through the process stream. The UV intensity is displayed on the control panel. The output is also available as a 4–20 mA active output (representing UV dose).

A temperature sensor fitted to the surface of the chamber will signal the control panel to shut down the system in the event of the chamber overheating under conditions of low or zero process flow.

Depending on the application and the model, the chamber may also be fitted with a motor-driven wiper (section 1.2.4) if deposition is likely to occur on the quartz tube.

Where systems are used in critical applications, additional parts and documentation are available, including higher internal chamber finishes.

1.2.2. Control panel

The main functions of the control panel are to:

- Supply power to the UV lamp.
- Provide a customer control interface.
- Provide dose and fault indication.

The control panel is fitted with:

- A door isolator switch – for isolating the UV system during maintenance and service.
- Front panel “power on” indicator.
- Electronic ballast – for supplying high frequency voltage to the UV lamp.
- Customer interface terminals – for the customer to use for sensors, alarm warnings, remote starting and stopping of the system.
- Touchscreen display indicating operation and status of system, with ‘soft’ buttons for local start/stop of system and menu access.

1.2.3. System identification

- Control Panel – the rating plate on the front of the cabinet lists:
  Model name
  Part number
  Serial number

Other useful information on the required supply voltage, phase, frequency and circuit protection are also listed. The software version can also be seen on the display at start-up.

- Chamber – the rating plate lists:
  Model name
  Part number
  Serial number

Other useful information on the maximum water temperature and working pressure is also listed, as well as the lamp and quartz tube part numbers.

1.2.4. Options

An automatic cleaning mechanism for the quartz tube is available on certain models for processes where quartz fouling is likely to occur.

This device, operated by an electric motor, causes two elastomeric rings (fixed to a stainless steel leadscrew) to be drawn over the outer surfaces of the quartz tube to remove deposits (see section 5.6).

1.2.5. UL and FCC certification

These UV systems are UL-certified for sale into the U.S.A. The manufacturer holds the UL file reference which can be viewed online at www.ul.com (reference number ZDUZ.E533312). The file reference gives the models to which a UL listed mark can be applied.

As part of this certification, the manufacturer certifies that these products conform to the applicable provisions of the Code of Federal Regulations (CFR) including Chapter I, Subchapter J, *Radiological Health*.

The systems also comply with part 15B of the FCC Rules.

2. Safety

2.1. Risks Associated with the System

The protection that the equipment provides may be impaired if used in a manner not specified in this manual.

Incorrect operation or use for other than the intended purpose may:

- endanger the health of the operator;
- damage the unit and other equipment;
- prevent efficient operation of the unit.

All persons concerned with the installation, commissioning, operation, maintenance and repair of the unit:

- **must** be suitably qualified;
- **must** observe the operating instructions;
- **should** attend a specific training course on the system's operation and maintenance.

Take the time to read and understand the Safety AND the System Operation sections in this manual before attempting to use the system in order to ensure correct and safe operation.

	**IF IN DOUBT - ASK.**

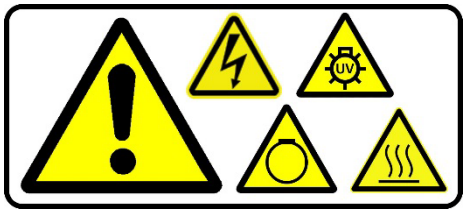
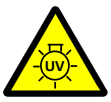
	**System parameters are factory-set to suit the requirements of the process. Changing these parameters can lead to insufficient treatment levels or malfunctioning of the entire system, and may prevent the minimum performance requirements being met.** **The configuration of the system should only be changed by personnel who are fully aware of the implications of the changes. For technical assistance and details of training courses, please consult technical support using the contact details given on the inside front cover of this manual.**

2.2. System Warning Labels

2.2.1. Cabinet

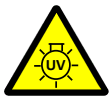
	Labels affixed on front of cabinet. General warning, caution to be taken. Access allowed only by qualified personnel. Risk of electric shock. Disconnect power and refer to the Operation and Maintenance Manual before commencing any servicing procedures. Étiquettes apposées sur le devant de l'armoire et sur le côté du ballast. Avertissement général, précautions à prendre. Accès autorisé au personnel qualifié uniquement. Risque de choc électrique. Coupez l'alimentation et reportez-vous au manuel d'utilisation et d'entretien avant toute intervention.

2.2.2. Chamber

<table>
<tr><td rowspan="6">  </td><td colspan="2">Label provides warnings for:
Les étiquettes préviennent contre les dangers suivants :</td></tr>
<tr><td>  </td><td>Chamber to be accessed only by authorised personnel.
Accès au réacteur restreint aux personnes autorisées uniquement.</td></tr>
<tr><td>  </td><td>Electrical hazard
High voltages; risk of electric shock. DO NOT remove any covers or fittings whilst the UV lamp is ON.
Danger électrique
Hautes tensions ; risque de choc electrique. Ne pas retirer de caches ou de raccords avec la lampe UV allumée.</td></tr>
<tr><td>  </td><td>Optical radiation hazard (UV)
UV light is harmful to eyes and exposed skin. DO NOT remove any covers or fittings whilst the UV lamp is ON.
Danger rayonnement optique (UV)
Les rayons UV sont nocifs pour les yeux et la peau exposée. Ne pas retirer de caches ou de raccords avec la lampe UV allumée.</td></tr>
<tr><td>  </td><td>Pressurised system warning
Chamber may be under pressure. Release liquid pressure and drain before servicing.
Avertissement : système sous pression
Le réacteur peut être sous pression. Relâcher la pression et vidanger avant entretien.</td></tr>
<tr><td>  </td><td>High temperature (Hot surface)
The outside surface of the chamber may be hot. Care should be taken when handling.
Température élevée (surface chaude)
La surface extérieure du réacteur peut être chaude. Prendre ses précautions lors de la manipulation.</td></tr>
<tr><td>  </td><td colspan="2">Label affixed to protective cover on chamber. Provides warning of UV light hazard (see above) and risk of electric shock. Access allowed only by qualified personnel.
Etiquette apposée sur le réacteur de manière adjacente au point d’accès à la lampe. Fournit un avertissement du danger lié au rayonnement UV et au risque de choc électrique. Accès au réacteur restreint aux personnes autorisées uniquement.</td></tr>
</table>

2.3. Ultra Violet Light

The UV lamp emits wavelengths principally in the 200–315 nm region. The emissions in this region are invisible and highly damaging to eyes and exposed skin.



DO NOT ATTEMPT TO OPERATE THE UV LAMP OUTSIDE THE CHAMBER.

DO NOT ATTEMPT TO LOOK AT AN OPERATING UV LAMP.

N’ESSAYEZ PAS DE FAIRE FONCTIONNER LA LAMPE UV EN DEHORS DE LA CHAMBRE.

N’ESSAYEZ PAS DE REGARDER UNE LAMPE UV EN FONCTIONNEMENT.

2.4. Pressurised Liquid

During operation, the pressure of the liquid in the chamber can reach up to 10 bar (1000 kPa / 145 psi). If the quartz tube fitting is removed before the system is depressurised, the tube can be ejected violently from the chamber and may cause serious injury to personnel. This could also occur with other parts connected to the chamber.



ALWAYS ISOLATE AND DRAIN THE CHAMBER BEFORE CHANGING THE QUARTZ TUBE OR ANY OTHER WETTED PARTS, AND ENSURE NO-ONE IS STANDING IN LINE WITH THE TUBE WHILE THE QUARTZ TUBE FITTING IS REMOVED.

Refer to section 5.4 for further details.

TOUJOURS ISOLEZ ET VIDANGEZ LA CHAMBRE AVANT DE CHANGER LE MANCHON DE QUARTZ OU TOUT AUTRE PIÈCES MOUILLÉES, ET VÉRIFIEZ QUE PERSONNE NE SE TIENT EN LIGNE AVEC LE MANCHON EN QUARTZ PENDANT QUE LE RACCORD EST SUPPRIMÉE.

Reportez-vous au section 5.4 pour plus de détails.



2.5. Earthing



THE EQUIPMENT MUST BE EARTHED CORRECTLY USING A CONDUCTOR OF ADEQUATE CROSS-SECTION, AND THE SUPPLIED LEADS.

Refer to section 3 for installation details.

L'ÉQUIPEMENT DOIT ÊTRE CORRECTEMENT MIS À LA TERRE À L'AIDE D'UN CONDUCTEUR DE SECTION TRANSVERSALE ADÉQUATE AINSI QUE LES CÂBLES FOURNIS.

Pour plus de détails concernant l'installation, reportez-vous à la section 3.

2.6. High Voltages



ENSURE INSTALLATION HAS BEEN CARRIED OUT IN ACCORDANCE WITH THIS MANUAL BEFORE OPERATING EQUIPMENT.

There may be voltages in excess of 300 V in the cabinet and UV lamp connections.

Wait a minimum of one (1) minute after the isolator has been turned off before gaining access to the cabinet, ballast or UV chamber connections.

Before attempting to service a treatment chamber, ensure that the power supply is completely disconnected by means of the door isolator. The isolator should be

	padlocked in the OFF position, and local lock-off procedures followed. Never reach inside the cabinet through any opening in the base when the power supply is connected, as you may touch live cables. **AVANT DE FAIRE FONCTIONNER L'ÉQUIPEMENT, ASSUREZ-VOUS QUE L'INSTALLATION A BIEN ÉTÉ RÉALISÉE CONFORMÉMENT AUX INSTRUCTIONS DE CE MANUEL.** L'armoire et les connexions de la lampe UV peuvent présenter des tensions supérieures à 300 V. Après extinction de l'isolateur, attendre un minimum d'une (1) minute avant d'avoir accèss à l'armoire ou aux connexions de la chambre de traitement. Avant d'intervenir sur une chambre de traitement, s'assurer que l'alimentation électrique est complètement déconnectée par le sectionneur de la porte. Le sectionneur doit être cadenassé en position OFF et les procédures locales de verrouillage doivent être respectées. N'atteignez jamais l'intérieur de l'armoire par une ouverture dans la base lorsque l'alimentation est connectée, car vous pouvez toucher des câbles sous tension.

	**ALL INSTALLATION AND MAINTENANCE WORK SHOULD BE CARRIED OUT BY QUALIFIED PERSONNEL.** **TOUTES LES TÂCHES D'INSTALLATION ET DE MAINTENANCE DOIVENT ÊTRE RÉALISÉES PAR DES EMPLOYÉS QUALIFIÉS.**

2.7. High Temperatures

The UV lamp wall reaches high temperatures during normal operation and the lamp retains heat for a considerable time after it has been switched off.

	**TAKE EXTREME CARE WHEN HANDLING HOT UV LAMPS.** **FAITES EXTRÊMEMENT ATTENTION LORSQUE VOUS MANIPULEZ DES LAMPES UV CHAUDS.**

2.8. Mercury

UV lamps contain a small quantity of mercury (<25 mg / 0.001 oz).

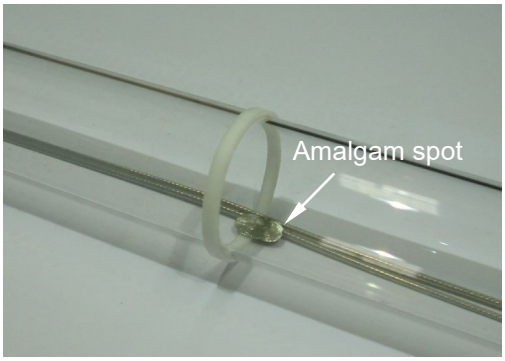
Cotton gloves should be worn at all times when handling lamps to prevent the glass being marked with fingerprints, and care should be taken to avoid breakage. New lamps should be stored in their original packaging until required for use.

Old UV lamps must be disposed of safely. On request, your supplier representative can supply details of suitable disposal companies.

2.8.1. Mercury spillage procedure

In the event of a UV lamp breakage, carry out the following procedure.

Check List
Items needed to clean up a small mercury spill:
Zip-Lock type plastic bags Rubber or latex gloves Paper towels Cardboard or squeegee Eyedropper, syringe or pipette Duct tape, or shaving cream and small paint brush Mercury Spillage Cleaning kit (optional)

1.	Put on rubber or latex gloves.
2.	If there are any small broken pieces of glass or sharp objects, pick them up with care. Place all broken objects on a paper towel. Fold the paper towel and place in a zip lock bag, secure and label as broken glass.
3.	Locate visible mercury beads or amalgam spots. Use a squeegee or cardboard to gather mercury beads. Use slow sweeping motions to keep mercury from becoming uncontrollable.  **Note:** Mercury can move surprising distances on hard-flat surfaces, so be sure to inspect the entire room when "searching."
4.	Use the eyedropper to collect or draw up the mercury beads. Slowly and carefully squeeze mercury onto a damp paper towel. Place the paper towel in a zip lock bag, secure and label as spilt mercury.
5.	After you have removed the larger beads, put shaving cream on top of a small paint brush and gently "dot" the affected area to pick up smaller hard-to-see beads. Alternatively, use duct tape to collect smaller hard-to-see beads. Place the paint brush or duct tape in a zip lock bag and secure.
6.	OPTIONAL STEP: It is optional to use commercially available mercury cleaning kits that contain powdered sulphur to absorb the beads that are too small to see. The sulphur performs two functions: (1) it makes the mercury easier to see since there may be a colour change from yellow to brown and (2) it binds to the mercury so that it can be easily removed and suppresses the vapour of any residual mercury. Mercury spill kits can normally be purchased from laboratory, chemical supply and hazardous materials supply manufacturers.
7.	Mercury waste should be properly disposed of in accordance with local laws for hazardous waste.

2.9. Breakage of the Tube

The tube that encapsulates the UV lamp is made from quartz, a highly purified form of silica. Cotton gloves should be worn at all times when handling tubes and care should be taken to avoid breakage. The tube may crack or shatter if subjected to sudden impact forces from sharp edges. The edges of the pieces of broken quartz will be extremely sharp and should be handled very carefully. Ensure that the chamber is thoroughly cleaned out after a breakage to remove quartz fragments.

Quartz is able to withstand considerable compressive forces when loaded evenly, so the tube is very unlikely to break due to the force exerted by the pressurised liquid in the chamber. It can also withstand high thermal gradients, and will not be affected by the temperature difference between the operating UV lamp and the liquid being treated.

2.10. Risk Assessment

OPERATION OR PROCEDURE: Commissioning, Servicing, Maintaining and Operating Low Pressure/ Amalgam UV Treatment Systems

	Hazards	Persons at Risk	Controls and Precautions
	High Voltages (Cabinet and UV lamp)	Service Engineer Operator, Installer	1. Door switch safety interlock. 2. 'High voltage' warning signs. 3. Circuit breakers in OFF position. 4. Statement in this guide referring to high voltages (section 2.6). 5. Protective cover over lamp connections during operation.
	Ultraviolet light (UV lamp)	Service Engineer Operator, Installer	1. Hazard warning label on UV chamber. 2. Instruction to switch off, isolate system and lock off isolator prior to replacing UV lamp. 3. Statement in this guide referring to exposure to eyes and skin (section 2.3).
	Pressurised liquid (UV chamber)	Service Engineer Operator, Installer	1. Hazard warning label on UV chamber. 2. Instruction to switch lamp off, isolate and drain chamber prior to replacing tube. 3. Instruction to stand to one side of chamber when removing tube.
	High temperatures (UV chamber)	Service Engineer Operator, Installer	1. Hazard warning label on UV chamber. 2. Statement in this guide referring to high chamber temperatures (section 2.2.2), with recommendation for risk assessment.
	High temperatures (UV lamp)	Service Engineer Operator, Installer	1. Instruction to allow lamp to cool before removal. 2. Statement in this guide referring to high temperatures (section 2.7).
	Mercury (UV lamp)	Service Engineer Operator, Installer	1. Follow COSHH guidelines. 2. Mercury spillage procedure (section 2.8.1).

	Heavy weights (Cabinet and UV chamber)	Installer	1. Weights given in Table 1. 2. Recommendation to use suitable lifting equipment when installing.
	Noise	Operator	1. The noise level of the machinery is below 70 dB(A) at any time during all foreseeable uses of this machine.

3. Installation and Commissioning



ALL INSTALLATION WORK MUST BE CARRIED OUT BY QUALIFIED PERSONNEL USING APPROPRIATE SAFE WORKING PRACTICES.*

READ THE SAFETY SECTION BEFORE COMMENCING INSTALLATION.

TOUT TRAVAIL D'INSTALLATION DOIT ÊTRE EFFECTUÉ PAR DU PERSONNEL QUALIFIÉ EN UTILISANT DES PRATIQUES DE TRAVAIL SURES ET APPROPRIÉES.*

VEUILLEZ LIRE LA SECTION TRAITANT DE LA SÉCURITÉ AVANT DE COMMENCER L'INSTALLATION.

* Refer to local safety standards and guidelines as appropriate, such as NFPA 70E in the USA and the current edition of the IEE Wiring Regulations in the UK.

3.1. Parts Supplied

The UV system components are:

- 1 × UV control panel
- 1 × UV chamber with optional auto-wiper mechanism
- Quartz tube(s) with O-ring (pre-installed in the chamber)
- UV lamp(s)
- Wiring harness (pre-wired to the cabinet)
- 1 × Installation Guide (printed copy)
- 1 × Operation and Maintenance Manual (electronic copy)
- 1 × Certification pack (optional)
- 1 × Panel locking key

Also required, but not supplied with the UV system are:

- An adjustable wrench with jaw opening at least 36 mm (1.4 inches) for removing the quartz tube
- Set of hexagonal head drivers
- Cross-headed and flat-headed screwdrivers
- A section of plastic pipe, 1.5–2 metres (4.9–6.6 feet) long × 25 mm (1 inch) diameter, to assist with quartz tube removal

Tools required for the auto-wiper option:

- Circlip pliers
- M8 socket driver

3.2. Handling and Storage Instructions

1.	Prior to installation, store the equipment in a clean dry area at -10ºC to 50ºC (14ºF to 122ºF) with a relative humidity of less than 95% (non-condensing).
2.	The UV system equipment is delivered in protective packaging but it should be protected from shocks and heavy weights. When unpacking the equipment, check that all components are present and undamaged. Contact your supplier representative if there is any damage.

3.	The cabinet and chamber can be large and heavy (refer to Table 1 for details). Before unpacking the equipment, perform a risk assessment to identify a safe method of lifting and moving the items. Use suitable and properly maintained lifting equipment operated by trained personnel.
4.	**Do NOT touch the quartz tube or UV lamp with bare hands** – use cotton or non-marking disposable gloves to prevent the glass being marked with fingerprints.

3.3. Installation Requirements

The UV chamber MUST be installed horizontally. It should be installed in a readily accessible location to enable maintenance of the system, with access at both ends of the chamber.



Access space equal to or greater than the chamber length MUST be allowed behind the protective cover at the end of the chamber for UV lamp and quartz tube replacement.

À préciser toutefois que l'on DOIT prévoir un dégagement égal ou supérieur à la longueur de la chambre derrière le couvercle de protection à l'extrémité de la chambre pour le remplacement de la lampe UV et de le manchon de quartz.

Other requirements when installing the system are as follows:

1.	**Install the equipment in an indoor environment**, out of direct sunlight, with a minimum clearance of 150 mm (5.9 inches) around the control panel. Leave a gap of at least 25 mm (1 inch) behind the control panel to allow for cooling. The chamber and control panel should be located to prevent public access to them. Suitable lighting to allow operation and maintenance should be installed around the equipment.
2.	**Do NOT install a UV chamber upstream from the process pump**, as negative pressure can cause damage to the internal components, in particular the quartz tube.
3.	The **ambient temperature** in the installation area should be between **5ºC and 40ºC (41ºF to 104ºF)** with a **relative humidity** of less than **95% (non-condensing)**.
4.	Insulate the unit from extreme temperatures.
5.	The cabinet and the chamber can be large and heavy; refer to Table 1 for details. Before installing the system: • **Identify a suitable fixture or structure** against which the items can be mounted which will safely support their weight. • **Identify suitable fittings** for attaching the items to the supporting fixtures. • **Perform an assessment of the safest method of lifting and supporting the items** while they are attached to the supporting fixtures. Use suitable and properly maintained lifting equipment operated by trained personnel.
6.	**Ensure that the chamber can be safely isolated and drained for servicing**. We recommend that lockable valves are fitted at the inlet and outlet of the chamber, and that a bypass is installed to allow the process liquid to flow during servicing.

7. **If the installation uses plastic pipework**, we recommend using stainless steel pipework for the first metre at the inlet and outlet to minimise the exposure of the plastic pipework to UV light and to prevent light from inside the chamber being visible. Any bends installed as light traps should be at least 2 pipe diameters from the connection point.

8. **Ensure that the displays on the control panel are at eye level** or otherwise easily viewable.

9. **Mount the control cabinet clear of the floor** as a precaution against water ingress and ensure that the cabinet fan vents are not blocked.

10. **Ensure that the chamber is easily accessible** in case access to the sensors is required during maintenance of the system.

11. **Perform a risk assessment and take suitable precautions to prevent accidental contact with the surface of the chamber** in case it becomes hot during operation. This is particularly important when the alarm temperature is set above 50ºC (122ºF) or if the system ever operates without liquid flow.

12. The **standard cable length** between system components is 3 metres (9.8 feet). The cables are fed through a length of conduit to minimise mechanical stress and exposure to water and other fluids. If longer cables are required, please contact your supplier representative.

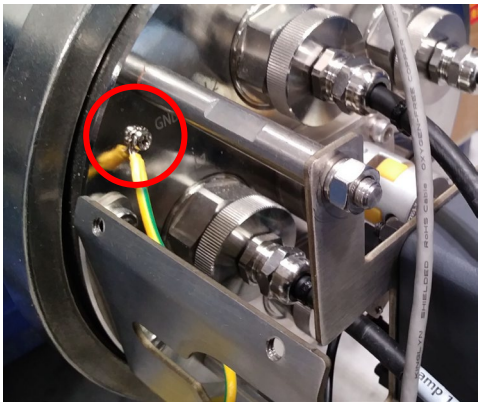
    Install cables in accordance with local regulations.

13. Remove the protective cover on the chamber (see section 3.4) and connect the leads in the wire harness from the cabinet to the chamber as follows:
    - Screw each numbered black lamp cable to the correct quartz tube fitting, following the markings on the chamber end plate, and hand tighten.
    - Connect the cables labelled motor, limit switch 1, limit switch 2 and temperature to the flying leads attached to the sensors in the chamber.
    - Connect the UV sensor cable to the sensor and hand tighten, then insert the sensor into the sensor housing (see section 5.5.3).

    Note: The leads used for the auto-wiper are labelled and wrapped separately. If the chamber is unwiped, you can leave these leads wrapped and ignore them.

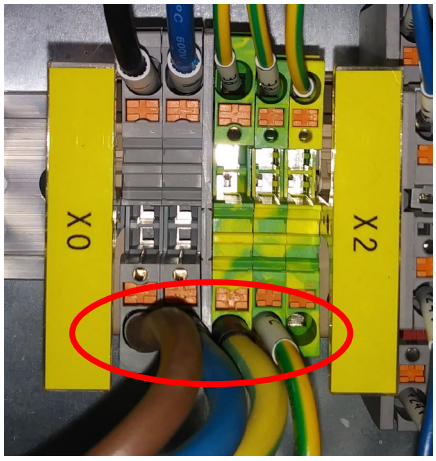
14. Ensure that the **earth lead** (green/yellow) from the cabinet is fitted to the UV chamber earthing point under the protective cover (Figure 2).

    Figure 2: Earth Terminal on Chamber



15. The mains supply cable to the control cabinet must be wired on site. This is connected to the cabinet through one of the available inlet ports on the bottom of the panel and is wired to the X0 terminals in the cabinet as shown in Figure 3 and in Appendix C. Refer to Table 2 for supply information.

Figure 3: Incoming Supply Cable Connections

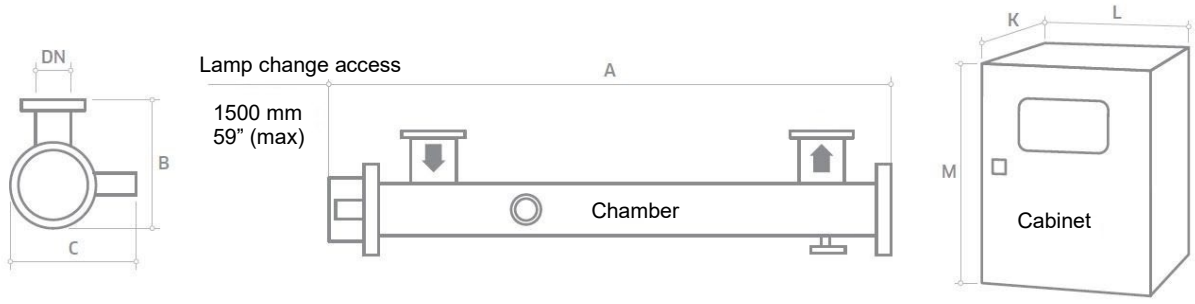


16. **Check the system for leaks BEFORE fitting the UV lamp.**

17. When installing the lamps, ensure that the **UV lamp amalgam spot** is on the underside of each lamp and a wire retaining ring is over the spot (see section 5.3.2).

18. Ensure that the **panel locking key** is kept in a convenient, safe location for future access to the cabinet components.

Table 1: Chamber and Cabinet Dimensions and Weights



Model	Chamber Dimensions (mm / inches)					Chamber Weight (kg / lb)	
	A without wiper	A with wiper	B	C	DN	without wiper	with wiper
4	745 / 29.3	n/a	210 / 8.3	245 / 9.5	25 / 1	10.5 / 23.1	n/a
8	1245 / 49.0	n/a	230 / 9.1	255 / 10.0	50 / 2	18.5 / 40.7	n/a
15	1245 / 49.0	n/a	230 / 9.1	255 / 10.0	50 / 2	18.5 / 40.7	n/a
32	1785 / 70.3	n/a	230 / 9.1	275 / 10.7	80 / 3	24.3 / 53.6	n/a
40	1785 / 70.3	1895 / 74.6	335 / 13.2	305 / 12.0	80 / 3	42.7 / 94.1	49.0 / 107.9
85	1785 / 70.3	1895 / 74.6	335 / 13.2	320 / 12.6	100 / 4	42.7 / 94.1	49.0 / 107.9
125	1785 / 70.3	1895 / 74.6	335 / 13.2	305 / 12.0	125 / 5*	43.0 / 94.8	48.1 / 106.0
215	1785 / 70.3	1895 / 74.6	455 / 17.9	355 / 14.0	150 / 6	66.4 / 146.4	73.7 / 162.5
225	1785 / 70.3	1895 / 74.6	455 / 17.9	355 / 14.0	150 / 6	66.0 / 145.5	74.1 / 163.4
330	1790 / 70.5	1900 / 74.8	525 / 20.7	440 / 17.3	200 / 8	114.0 / 251.3	123.0 / 271.2
350	1790 / 70.5	1900 / 74.8	525 / 20.7	440 / 17.3	200 / 8	115.0 / 253.5	124.0 / 273.4
380	1790 / 70.5	1900 / 74.8	525 / 20.7	440 / 17.3	200 / 8	116.0 / 255.7	125.0 / 275.6

* DN is 150 / 6 when connection is Tri-clamp

Model	IP54 (NEMA 12) Cabinet (mm / inches)			IP56 (NEMA 4X) Cabinet (mm / inches)			Cabinet Weight (kg / lb)
	K	L	M	K	L	M	
4	180 / 7.1	470 / 18.5	535 / 21.1	240 / 9.4	547 / 21.5	655 / 25.7	20.5 / 45.2
8	180 / 7.1	470 / 18.5	535 / 21.1	240 / 9.4	547 / 21.5	665 / 26.2	20.5 / 45.2
15	180 / 7.1	470 / 18.5	535 / 21.1	240 / 9.4	547 / 21.5	665 / 26.2	20.5 / 45.2
32	180 / 7.1	470 / 18.5	535 / 21.1	240 / 9.4	547 / 21.5	665 / 26.2	20.5 / 45.2
40	180 / 7.1	470 / 18.5	535 / 21.1	240 / 9.4	547 / 21.5	665 / 26.2	20.5 / 45.2
85	210 / 8.3	490 / 19.3	600 / 23.6	270 / 10.6	557 / 21.9	750 / 29.5	27.6 / 60.8
125	210 / 8.3	490 / 19.3	600 / 23.6	270 / 10.6	557 / 21.9	750 / 29.5	29.1 / 64.2
215	210 / 8.3	500 / 19.7	700 / 27.6	270 / 10.6	567 / 22.3	850 / 33.4	35.5 / 78.3
225	210 / 8.3	500 / 19.7	700 / 27.6	270 / 10.6	567 / 22.3	850 / 33.4	36.9 / 81.4
330	210 / 8.3	500 / 19.7	700 / 27.6	270 / 10.6	567 / 22.3	850 / 33.4	36.9 / 81.4
350	210 / 8.3	510 / 20.1	800 / 31.5	270 / 10.6	567 / 22.3	950 / 37.4	44.5 / 98.1
380	210 / 8.3	510 / 20.1	800 / 31.5	270 / 10.6	567 / 22.3	950 / 37.4	45.9 / 101.2

Table 2: System Power Supply Requirements

Model	System Power (W)	Voltage (±10%) / Frequency	Full Load Current (A)	Supply Circuit Breaker
4	90	110 VAC / 60 Hz	1.0	D6A
8	150		1.6	D6A
15	250		2.6	D6A
32	360		3.7	D6A
40	380		3.9	D6A
85	750		7.6	D10A
125	1080		11.0	D16A
215	1420		14.4	D20A
225	1750	220 VAC / 60 Hz	8.9	D16A
330	1750		8.9	D16A
350	2100		10.7	D16A
380	2440		12.4	D20A
4	90	220 VAC / 50 Hz	0.5	D6A
8	150		0.8	D6A
15	250		1.3	D6A
32	360		1.9	D6A
40	380		2.0	D6A
85	750		3.8	D10A
125	1080		5.5	D10A
215	1420		7.2	D10A
225	1750		8.9	D16A
330	1750		8.9	D16A
350	2100		10.7	D16A
380	2440		12.4	D20A

Notes: All models require a single-phase power supply (2W+G). Install the system in accordance with local electrical safety guidance and requirements, such as the National Electric Code (NEC) in the USA and the current edition of the IEE Wiring Regulations in the UK, and refer to these for methods to incorporate derating factors such as cable type, temperature and bunching, etc. Circuit breakers are to be of the 'D' or 'K' curve type (suitable for high current inrush).

The **mains power supply must be stable to within half a cycle** and have a voltage tolerance within the range shown in Table 2. Systems installed in locations with power supplies that do not meet these criteria must be equipped with an Uninterruptible Power Supply (UPS) operating continuously. Failure to observe this requirement could result in the lamp stopping frequently, causing the system to trip.



The system must only be connected to the correct mains power supply as indicated on the rating label on the front of the control cabinet. If the correct voltage is not available, DO NOT connect the supply to the cabinet. Contact your supplier representative.

Le système doit être raccordé à l'alimentation secteur adéquate comme indiqué sur la plaque signalétique placée sur le devant de l'armoire. Si la tension adéquate n'est pas disponible, NE raccordez PAS l'alimentation à l'armoire. Contactez le représentant de votre fournisseur.

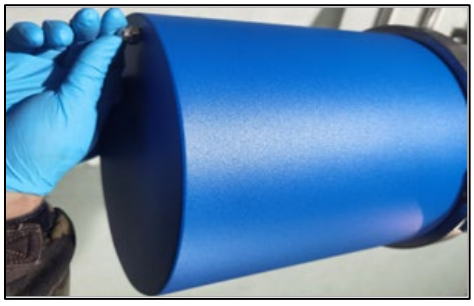
3.4. Installing the UV Lamp

	WEAR COTTON OR NON-MARKING DISPOSABLE GLOVES WHEN HANDLING THE UV LAMP AND SPACER TO PREVENT THE GLASS BEING MARKED WITH FINGERPRINTS. **PORTEZ DES GANTS DE COTON OU DES GANTS JETABLES NE MARQUANT PAS POUR MANIPULER LA LAMPE UV ET LA VIROLE EN QUARTZ POUR EMPÊCHER QUE LE VERRE SOIT MARQUÉ PAR DES EMPREINTES DIGITALES.**

1. Remove the four screws on the protective cover using a cross-headed screwdriver.

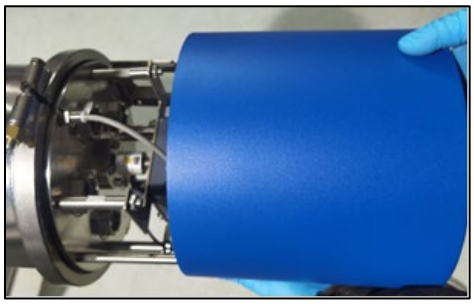


2. Unscrew and remove the nut on the end face by hand.



3. Remove the protective cover.

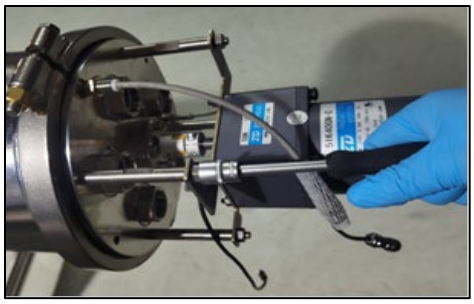
   If an auto-wiper is fitted, go to step 4; otherwise go to step 8.



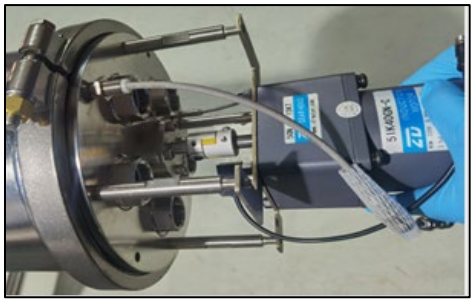
4. Disconnect the cables for the wiper motor, limit switch 1 and limit switch 2.



5. Use an M8 socket wrench to remove the four nuts securing the wiper motor to the bracket.



6. Lift the motor off the bracket.
   The coupling should remain in place.



7. Disconnect the temperature sensor connector from the cable in the wiring harness.

8. Loosen the cable gland on each lamp cable, then unscrew the lamp cables if they are attached to the quartz fittings.



9. Insert a UV lamp into each tube.
   Note: Ensure the amalgam spot is on the underside of the lamp (towards the bottom of the UV chamber) and the white ring is positioned over the spot.

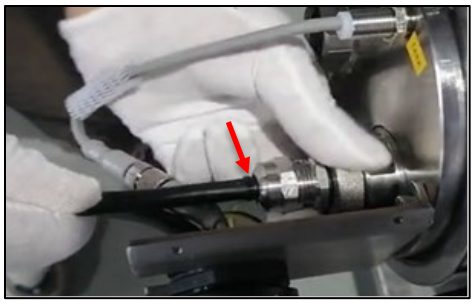


10. Connect the ceramic connectors on the lamps and the lamp cables, being careful to align the metal pins on the lamp with the holes on the connector.



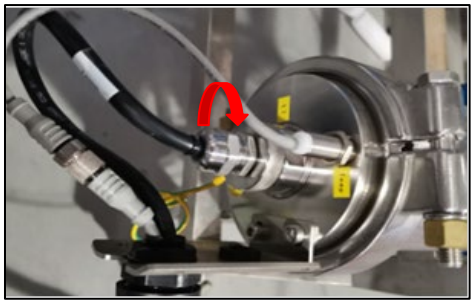
11. Gently push the lamp into the tube and screw the cable on to the quartz fitting using the knurled collar, until it is hand tight.

    Make sure that the marker on the cable (marked in image) is next to the cable gland to position the lamp correctly inside the chamber.



12. Carefully tighten the cable gland, making sure that the cable is not twisted as you do so.

    The cable must not rotate while you tighten the gland, as this will cause the lamp to rotate and the amalgam spot to move from its optimum position.



13. Refit the wiper motor (if required) and the protective cover by reversing steps 1–6.

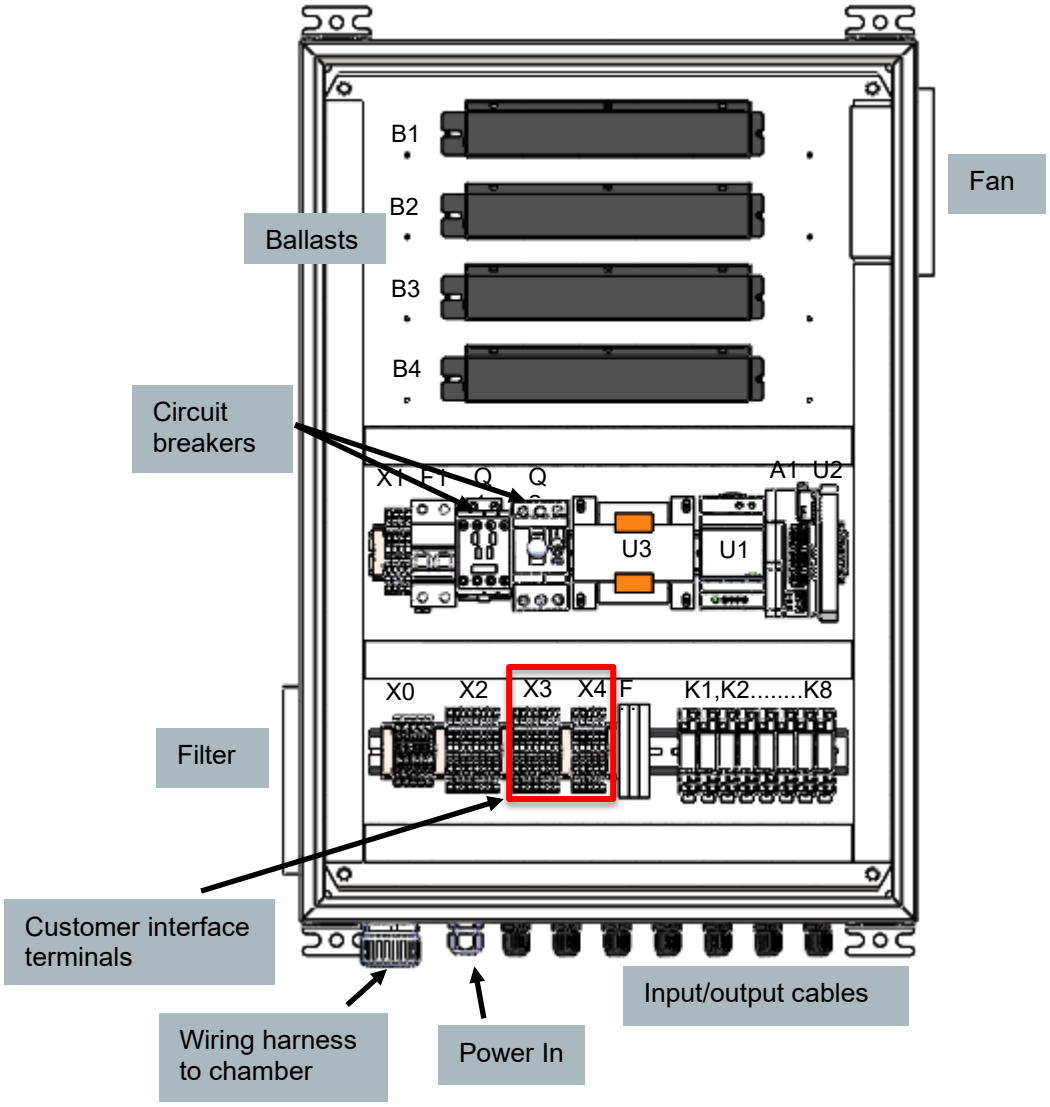
DO NOT COIL LAMP CABLES.

If the cables provided are longer than required, we recommend that they are shortened by a competent electrician.

3.5. Interface Wiring

Remote control and monitoring of the UV system is achieved using the user's interface terminals located in the control cabinet (Figure 4). Table 3 gives an overview of all inputs and outputs that might be required for external control of the UV system using these terminals. Please also refer to the circuit diagram in Appendix C.

Figure 4: Connections to Control Cabinet



Notes:

1. Cabinet shown as an example only (layout varies for different models).
2. Circuit breaker Q2 is only used with wiped chambers.

Table 3: Available Inputs and Outputs

Label on Site Wiring Diagram	Terminals	I/O Type	Function
Low Dose Alarm	X3, 11/12	Volt-free (dry) output for passing information to user's control system.	Initially LOW (open) and goes HIGH (closed) if the measured UV signal falls below the warning threshold.
Lamp Fail Alarm	X3, 21/22	Volt-free (dry) output for passing information to user's control system.	Initially LOW (open) and goes HIGH (closed) if a lamp has not started successfully.
Lamp Ready	X3, 31/32	Volt-free (dry) output for passing information to user's control system.	Initially LOW (open) and goes HIGH (closed) when the lamp is warmed up to operating temperature.
Any Warning	X3, 41/42	Volt-free (dry) output for passing information to user's control system.	Initially LOW (open) and goes HIGH (closed) after the system has a non-critical fault (e.g., UV Dose Low).
Any Fault (Trip)	X3, 51/52	Volt-free (dry) output for passing information to user's control system.	Initially LOW (open) and goes HIGH (closed) after the system has a critical fault (e.g., Chamber too hot).
Lamp Preheat	X3, 61/62	Volt-free (dry) output for passing information to user's control system.	Initially LOW (open) and goes HIGH (closed) when the lamp is warming up, then goes LOW (open) when lamp is ready.
Remote control (on/off)	X3, 71/72	Digital input (volt-free contact closure) from user's control system.	Closing a volt-free (dry) contact across these terminals will start the lamp when in Remote mode. Opening the contact will stop the lamp.
Flow Rate	X4, 31/32	4–20 mA input to control panel.	Can be direct from a flow meter or from the user's control system.
UV Transmittance	X4, 21/22	4–20 mA input to control panel.	T_{10} value from an online transmittance meter or from the user's control system.
UV Dose	X4, 11/12	4–20 mA output for providing information to user's control system.	Active analog output of dose calculated from dose model.

Note: There is no Remote Reset input available. Alarms can only be reset using the button on the control panel (see section 4.4).

3.5.1. Configuring analog inputs

If the customer is using an external flow sensor and/or transmittance meter, they must connect the outputs from these instruments to the appropriate analog inputs in the control panel (see Table 3).

These analog inputs must also be configured in the software, as follows:

1. Log in to the software as a user and go to the User Setting screen (section 4.5).
2. Tap the **Flow Meter OFF** button to enable the analog input.



3. Enter the maximum flow that can be measured by the flow meter and select the flow units.



Repeat these steps for the transmittance meter settings if required.

3.6. Commissioning Procedure

Commissioning of the UV system should be carried out in accordance with the recommendations of this manual with particular attention to safety (see section 2).

Follow the procedure in the commissioning checklist below, referring to the relevant sections of this manual as required. Section 6 gives information on the alarm conditions and advice for fault-finding.

In order to check system parameters and settings see the Acceptance Test Record supplied with the system.

Procedure	Tick if OK
The testing of the outputs in the steps below requires access to the control panel while the system is operational, and should only be performed by engineers trained and approved by the system supplier.	
Ensure the system is free from damage caused during transit or installation, and contains all required components (section 3.1).	
Check that the installed chamber and control panel are located satisfactorily with respect to operation and safety (section 3.3).	
Check that the chamber is installed horizontally with inlet and outlet pipework securely connected.	
Check that connections to ALL terminals inside the cabinet are secure.	
Check that the control panel is mounted at a suitable height and location for operation and above ground level to prevent water ingress in the event of a leak.	
Check that all interconnecting wiring is correct.	
Check that the supply voltage and frequency are as shown on the panel rating label.	
Check that the power supply wiring and circuit breaker are as recommended in Table 2.	
Turn on the process stream and check for leaks from the chamber and quartz tube.	
If applicable, check that bleed valves are fitted correctly.	
Install the UV lamp (section 3.4).	
Switch on the power to the control panel and the chamber using the door isolator.	
Start the lamps (section 4.2) and check that no faults are indicated.	
Observe if there are any 'hot spots' on the chamber due to air pockets, and bleed the chamber if necessary.	
If a flow meter is available: • Ensure the flow meter is connected to the correct analog inputs in the control panel (Table 3). • Follow the steps in section 3.5.1 to configure the analog input. • Check the Home screen to confirm that the meter is delivering the correct data.	
If a transmittance meter is available, follow the same steps as for the flow meter.	
Connect the UV dose output signal (Table 3) and configure the analog output in the User Settings screen as required by the site operator.	
Perform a wipe cycle if an auto-wiper is fitted.	
If required by the customer, set the system to **Remote** mode in the Home screen.	
Close a volt-free contact across terminals 71 and 72 in terminal block X3 (Remote Control On/Off) and check that the lamp switches on.	
Confirm that the operator is trained and is competent to take over operation of the system.	
Confirm that the operator is competent to replace a UV lamp and quartz tube.	
Ensure that the operator is aware of all health and safety considerations (section 2).	
Remarks:	

SIGNATURE (Supplier): **Commissioning and Training Completed**	**NAME (Print)**	**DATE**
SIGNATURE (Client): **Training Completed**	**NAME (Print)**	**DATE**

4. System Operation

This section provides detailed information on operation of the system.

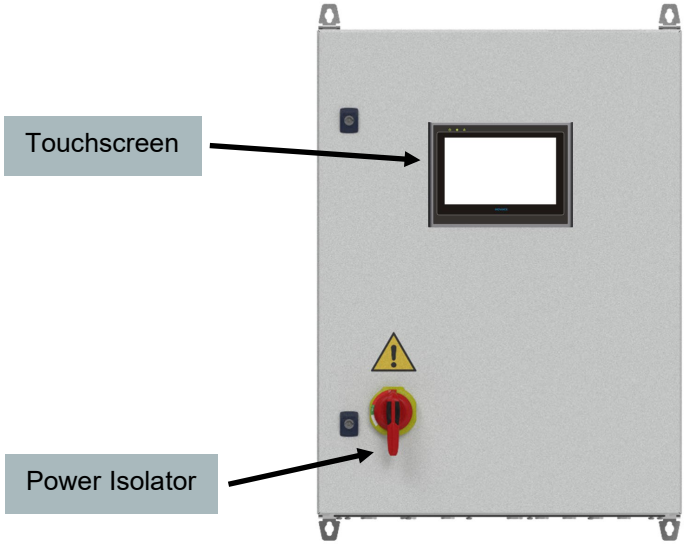
DO NOT USE THE EQUIPMENT IN A MANNER OTHER THAN THAT SPECIFIED IN THIS MANUAL, OTHERWISE THE PROTECTION PROVIDED BY THE EQUIPMENT MAY BE IMPAIRED.

N'UTILISEZ PAS L'ÉQUIPEMENT D'UNE MANIÈRE AUTRE QUE CELLE SPÉCIFIÉE DANS CE MANUEL, AUTREMENT LA PROTECTION FOURNIE PAR L'ÉQUIPEMENT PEUT ÊTRE ALTÉRÉE.

4.1. Overview of the Control Panel

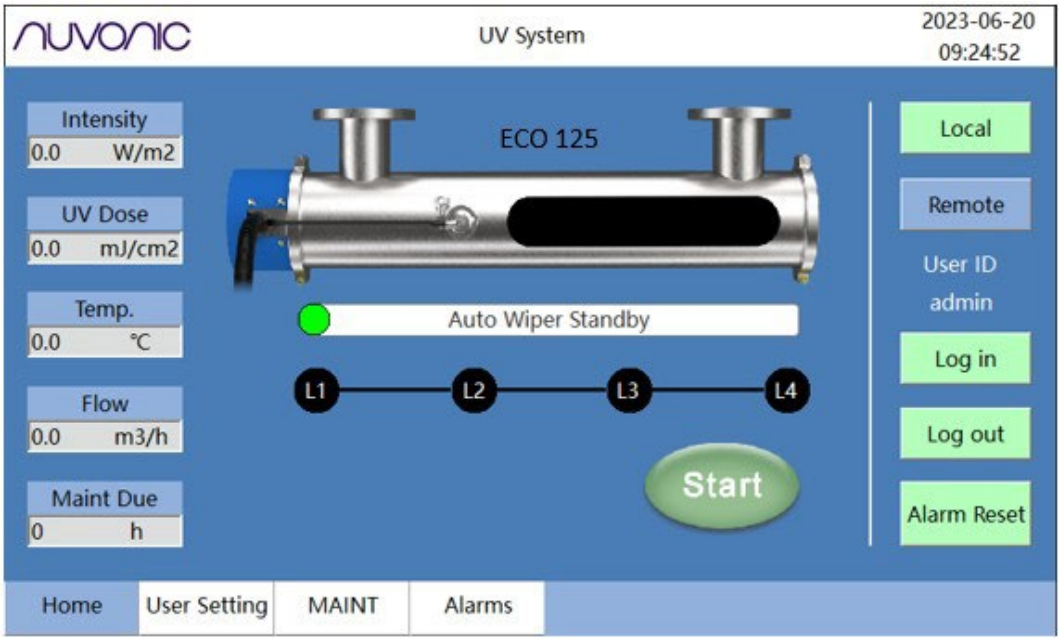
The system is controlled using the touchscreen on the front of the control cabinet.

Figure 5: Control Panel



When you switch on the power using the main isolator, the software will load (this should take less than a minute).

Figure 6: Home Screen

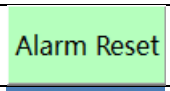
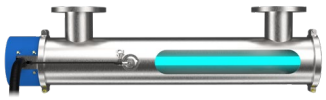


From this screen you can navigate to all areas of the software using the tabs at the bottom of the screen:

- User Setting: Use this screen to set the operating parameters of the system (section 4.5).
- MAINT: Use this screen to view the maintenance status of the lamps and quartz tubes, and reset the running time counters when parts are serviced (section 4.6).
- Alarms: View a list of recent alarms on the system (section 4.4)

The features of the Home screen are as follows:

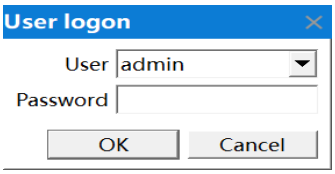
Intensity 0.0 W/m2	UV intensity as measured by the sensor.
UV Dose 0.0 mJ/cm2	UV dose (RED) calculated by the software.
Temp. 0.0 ℃	Temperature of the chamber surface.
Flow 0.0 m3/h	Flow rate of liquid; either a fixed value entered manually or a measured value from a flow meter.
Maint Time 0 h	The time remaining until the next scheduled service.
Log in	Log in to the software to adjust the settings.
Log out	Log out to protect the software settings from being changed.
Local	Select this mode to switch the lamps on and off from the touchscreen.
Remote	Select this mode to use a remote signal to switch the lamps on and off.

	Tap to reset alarms that are no longer active.
	Tap to switch on the lamps (in Local mode).
	Tap to switch off the lamps (in Local mode).
	Indicates lamps are off.
	Indicates lamps are on.
	Indicates status of each lamp (blue is on, black is off).
	Position and progress indicator for wipe cycles (if optional wiper fitted).

4.1.1. Logging In and Out

Any user is able to start and stop the system using the buttons on the Home screen (when in Local mode), and reset alarms that are no longer active. To change the user settings, you must log in as an admin user:

1. Tap **Log in** to display the Login dialog.



2. Select **admin** from the drop-down list, and enter the password. Passwords are listed at the front of this manual.
3. Tap **OK**.

The Factory login is only to be used by the supplier of the system.

4.2. Switching On

Before attempting to start the system, ensure that the chamber is full of water.

Avant de tenter de démarrer le système, assurez-vous que la chambre est pleine d'eau.

To start the system in Local mode:

➢ Tap 

To start the system in Remote mode:

1. Close a volt-free contact across terminals X3,71 and 72.
2. Tap 

4.3. Auto-Wiper Operation (option)

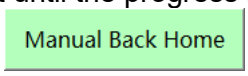
On systems fitted with an automatic wiper, a wiper yoke containing a set of rings is propelled along the quartz tube by a rotating leadscrew, driven by an electric motor. The wiper yoke is located at the end of the chamber nearest to the motor when the auto-wiper is not operating. During a wipe cycle, the yoke first travels away from the motor for a specific number of revolutions of the leadscrew (counted by switch 1) or until a programmed time has elapsed, whichever occurs first; this is the *away* position. The wiper mechanism then returns along the chamber towards the motor until detected by switch 2; this is the *home* position.

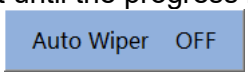
The control panel automatically wipes the quartz tubes at regular intervals, as set in the User Settings screen (every 3 hours by default). All tubes are wiped simultaneously. While wiping, the system will show a progress bar on the Home screen.

The wiper will also perform a single wipe cycle if a Low UV dose alarm occurs. See section 6 for possible causes if the Low UV Dose signal remains after the system is wiped.

If, for any reason, a wiping action has to be carried out at a given moment, you can run a manual wipe cycle after first disabling the auto-wiper, as follows:

1. Tap 

2. Wait until the progress bar shows that the wiper has reached the away position.
3. Tap 

4. Wait until the progress bar shows that the wiper has reached the home position.
5. Tap 

to enable the auto-wiper again.

When the wiper is in operation, it cannot be stopped until it reaches one end of the chamber. If you tap any of the buttons, a message is displayed to show that the wiper is in operation. Tap **Confirm** to remove the message.

	**You can adjust the wiper interval in the User Settings screen (section 4.5), but be aware that reducing the interval below 180 minutes may cause the wiper to require servicing more frequently than in the recommended service schedule (section 5.1).** **Vous pouvez régler l'intervalle du nettoyeur dans l'écran User Settings (Paramètres utilisateur) (section 4.5), mais sachez que la réduction de l'intervalle en-dessous de 180 minutes peut entraîner que le nettoyeur nécessitera un entretien plus fréquent que celui du calendrier de service recommandé (section 5.1).**

4.4. Warnings and Trips

Malfunctions result in warning or trip messages. If a warning occurs, it is possible to continue the operation of the system. If a trip occurs, the lamps are switched off and cannot be restarted until the alarm condition is cleared.

The fault conditions that cause a trip are:

- Chamber over-temperature
- Temperature sensor fault
- UV sensor fault

All other fault conditions generate a warning message.

➢ Tap the **Alarms** tab to view a list of recent warnings and trips on the system, with any information available on their cause.

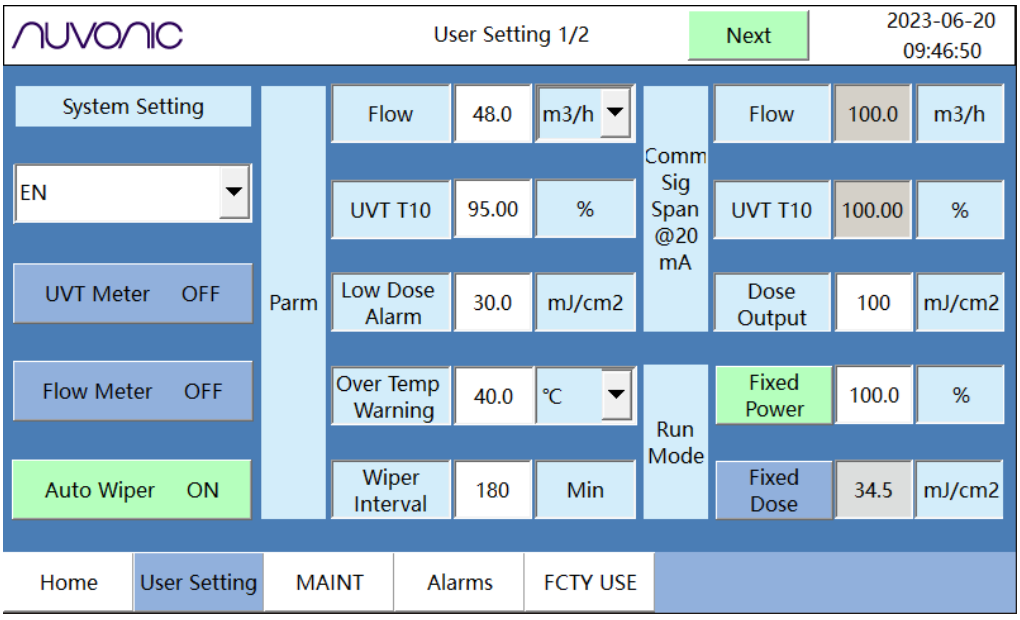
➢ When an alarm is resolved, tap Alarm Reset to reset the alarms and allow the system to operate again.
This is the only way to reset alarms in both Local or Remote mode.

Refer to section 6 for further information on alarms.

4.5. Operation Settings

➢ From the Home screen, tap the **User Setting** tab to display the system's operation settings (on two screens).

Figure 7: User Settings Screen 1



The features of the screen are as follows:

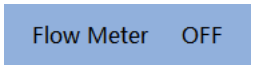
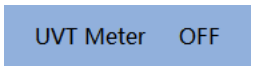
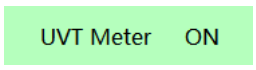
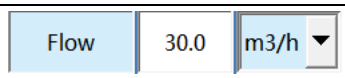
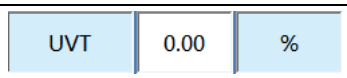
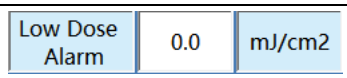
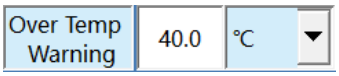
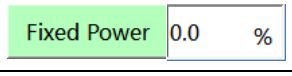
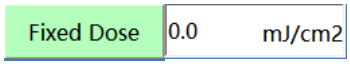
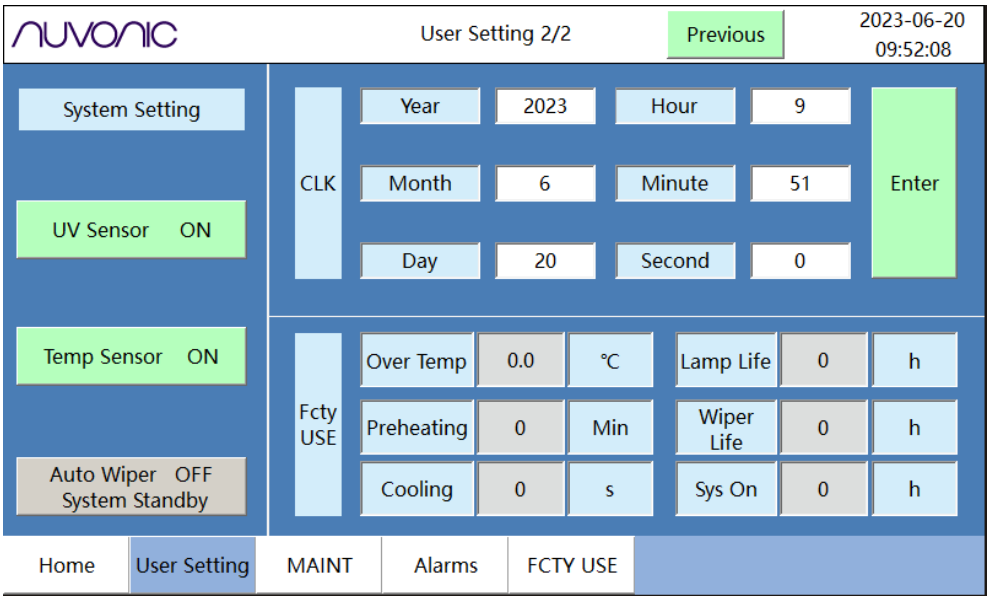
	Select the display language from the drop-down list.
	Flow input signal is disabled (no flow meter connected); fixed flow value used in UV dose calculations.
	Flow meter connected and flow input signal used in UV dose calculations.
	UV transmittance input signal is disabled (no UVT meter connected); fixed UVT value used in UV dose calculations.
	UV transmittance meter connected and UVT input signal used in UV dose calculations.
	Auto-wiper disabled (wiper can only be operated manually).
	Auto-wiper enabled (wiper cannot be operated manually).
	Set fixed flow rate and units (m^3/h or U.S. gallons per minute (GPM)).
	Set fixed UVT (T_{10}) as a percentage.
	Set minimum UV dose (alarm is triggered when dose is below this value).
	Set warning temperature limit (system displays high temperature warning when this value is exceeded), and select temperature units from drop-down list (used throughout software).
	Set time between automatic wipe cycles (minimum 5 minutes). If less than 180 minutes, screen displays warning that frequent wiping increases maintenance requirements.
	Set maximum UV intensity value of analog input from sensor (20 mA).
	Set maximum temperature value of analog input from sensor (20 mA).
	Set maximum flow rate value of analog input from sensor (20 mA).
	Set maximum UVT value of analog input from sensor (20 mA).
	Set maximum UV dose value of analog output (20 mA). We recommend using double the normal dose delivered by the chamber.
	Select Fixed Power run mode and enter the ballast output (50–100%). (System ignores UV dose.)
	Select Fixed Dose run mode and enter target UV dose (make sure this is higher than Low Dose Alarm value above). (System adjusts lamp output to deliver required dose.)

Figure 8: User Settings Screen 2



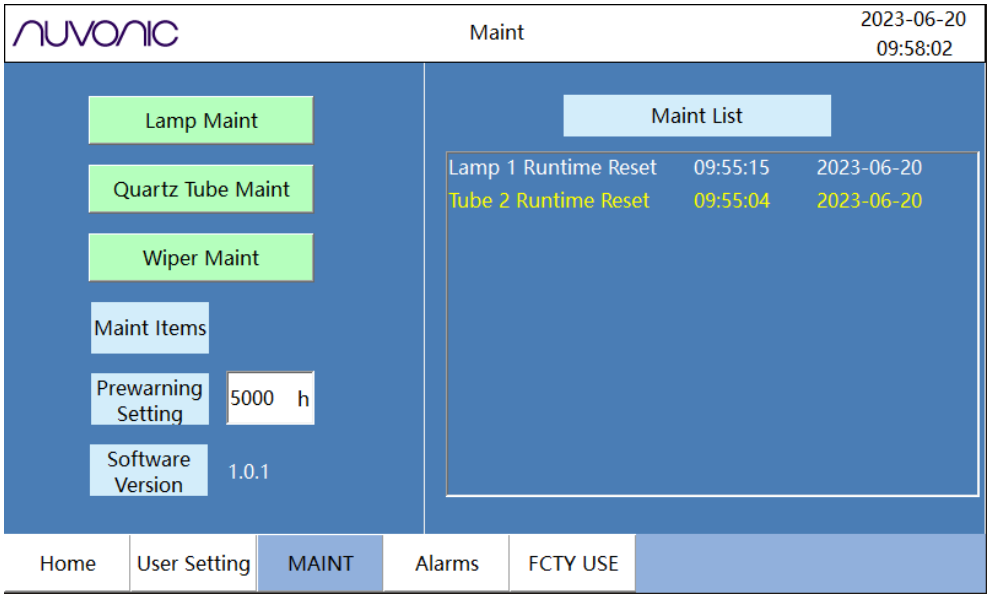
The features of the screen are as follows:

UV Sensor OFF	UV sensor errors ignored by software (use when sensor is faulty).
UV Sensor ON	UV sensor operating normally.
Temp Sensor OFF	Temperature sensor errors ignored by software (use when sensor is faulty).
Temp Sensor ON	Temperature sensor operating normally.
Auto Wiper OFF System Standby	Auto-wiper does not operate when system is in standby mode (lamps are not switched on).
Auto Wiper ON System Standby	Auto-wiper performs wipe cycles when system is in standby mode (lamps are not switched on).
CLK	Set the current date and local time, then tap **Enter**.
Fcty USE	For supplier use only (requires Factory user login).

4.6. Maintenance Settings

➢ From the Home screen, tap the **MAINT** tab to display the system's maintenance overview screen.

Figure 9: Maintenance Overview Screen



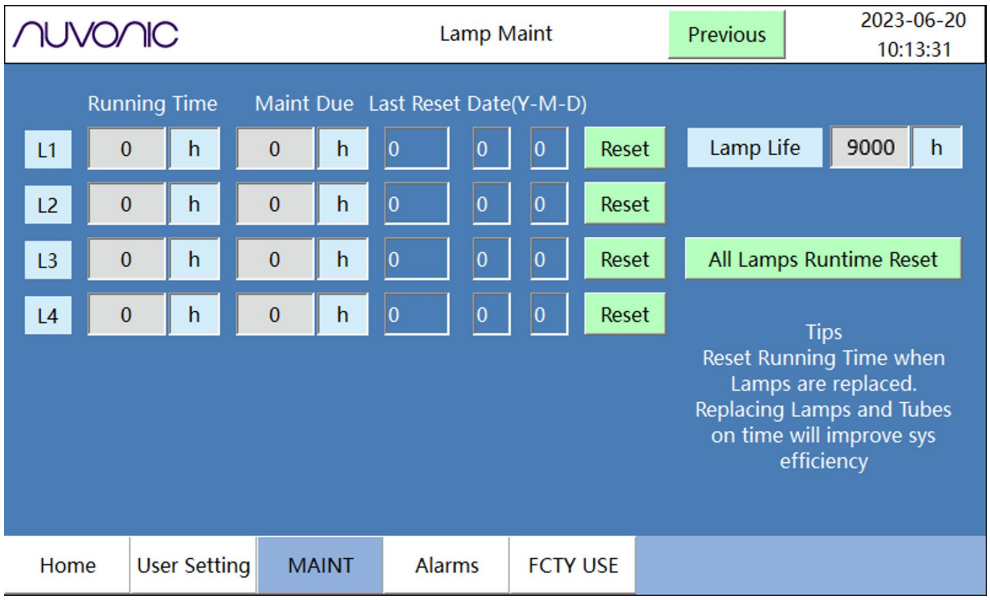
Lamp Maint	Tap to display Lamp Maintenance screen (section 4.6.1).
Quartz Tube Maint	Tap to display Tube Maintenance screen (section 4.6.2).
Wiper Maint	Tap to display Wiper Maintenance screen (section 4.6.3).
Maint Items	Displays the next item scheduled to be serviced.
Prewarning Setting 5000 h	Set the service hours for the consumable parts (lamp, quartz tube) at which a warning is displayed that a service is due soon.
Software Version 1.0	The current version of software installed in the panel.
Maint List	A log of previous replacements of consumable parts.

4.6.1. Lamp Maintenance

Use the **Reset** buttons on this screen to reset the running hours for each lamp once it has been replaced (or reset the hours for all the lamps in the chamber by tapping **All Lamps Runtime Reset**).

You can also view (but not change) the expected lamp life, the current running hours for each lamp, how long until they are due for replacement, and when they were last changed.

Figure 10: Lamp Maintenance Screen

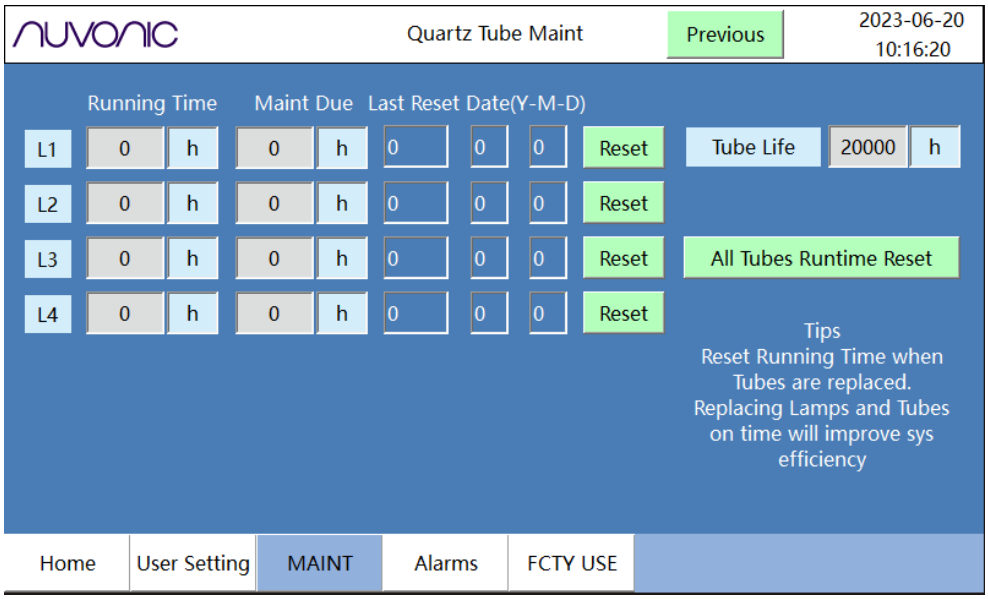


4.6.2. Quartz Tube Maintenance

Use the **Reset** buttons on this screen to reset the running hours for each quartz tube once it has been replaced (or reset the hours for all the tubes in the chamber by tapping **All Tubes Runtime Reset**).

You can also view (but not change) the expected tube life, the current running hours for each tube, how long until they are due for replacement, and when they were last changed.

Figure 11: Quartz Tube Maintenance Screen



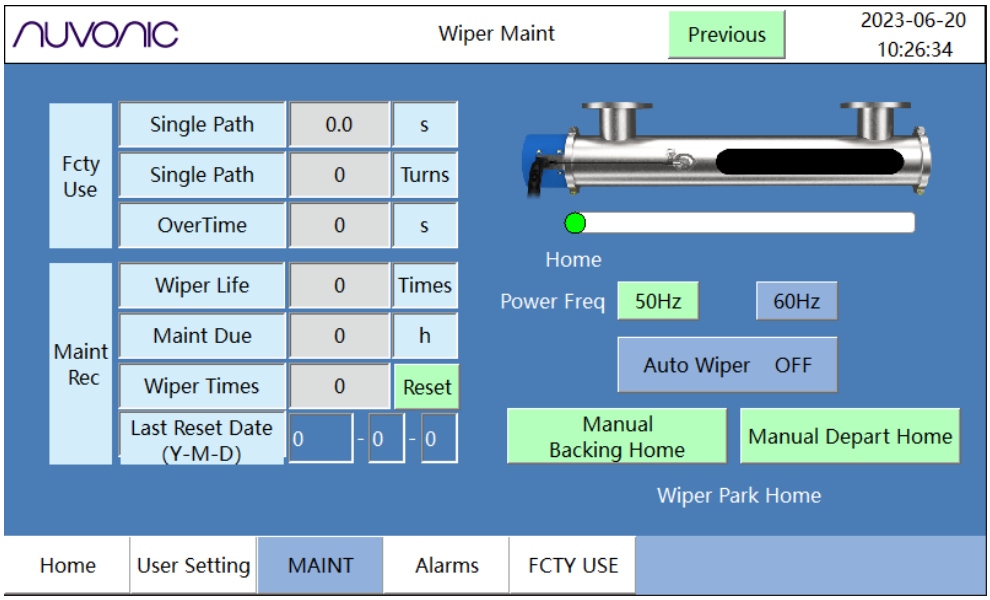
4.6.3. Wiper Maintenance (option)

In this screen, you can view (but not change) the expected life for the wiper rings, the current number of wipe cycles performed, how long until the parts are due for replacement, and

when they were last changed. In **Fcty Use** you can see the timings used to control the wiper (these cannot be changed by the user).

You can also control the wiper manually (see below).

Figure 12: Wiper Maintenance Screen



Auto Wiper OFF	Indicates wiper can only be operated manually; tap to enable auto-wiper.
Auto Wiper ON	Indicates auto-wiper is enabled and cannot be operated manually; tap to disable auto-wiper.
Manual Back Home	Tap to move wiper yoke to home position (closest to motor).
Manual Backing Home	Displays while yoke moving to home position.
Manual Depart Home	Tap to move wiper yoke to away position (furthest from motor).
Manual Departing Home	Displays while yoke moving to away position.
Power Freq 50Hz 60Hz	Adjust frequency of power supply to wiper motor.

5. Maintenance

5.1. Service Schedule

Period	Maintenance Procedure
Daily	Check front panel display for alarms
Weekly	Clean dust /debris from system
2 months	Inspect and clean cabinet air filters (replace if required)
9000 hrs or Low UV *	Lamp replacement
12 months	Clean quartz tube
	Clean UV sensor housing window
	Verify UV sensor **
	Replace wiper parts including seals***
24 months	Replace quartz tube (including seals) and UV sensor housing
	Inspect electrical connections in cabinet (pull test on terminals, heat signature)

* Whichever is sooner.
** Check sensor performance using a reference sensor at least every 12 months.
*** Based on a wiper interval of 3 hours; a shorter interval may cause the wiper to need more frequent servicing.

Record the maintenance procedure tasks carried out and their dates in the table in Appendix A.



PRIOR to carrying out ALL maintenance tasks on the UV chamber or control panel, ensure that the isolator on the front of the control panel is switched to OFF and LOCKED OFF.

See Figure 13 and section 2.

AVANT de réaliser TOUTE tâche de maintenance sur la chambre UV ou sur le panneau de commande, assurez-vous que l'isolateur situé à l'avant du panneau de commande est ÉTEINT et qu'il est DÉVERROUILLÉ.

Reportez-vous à la Figure 13 et section 2.

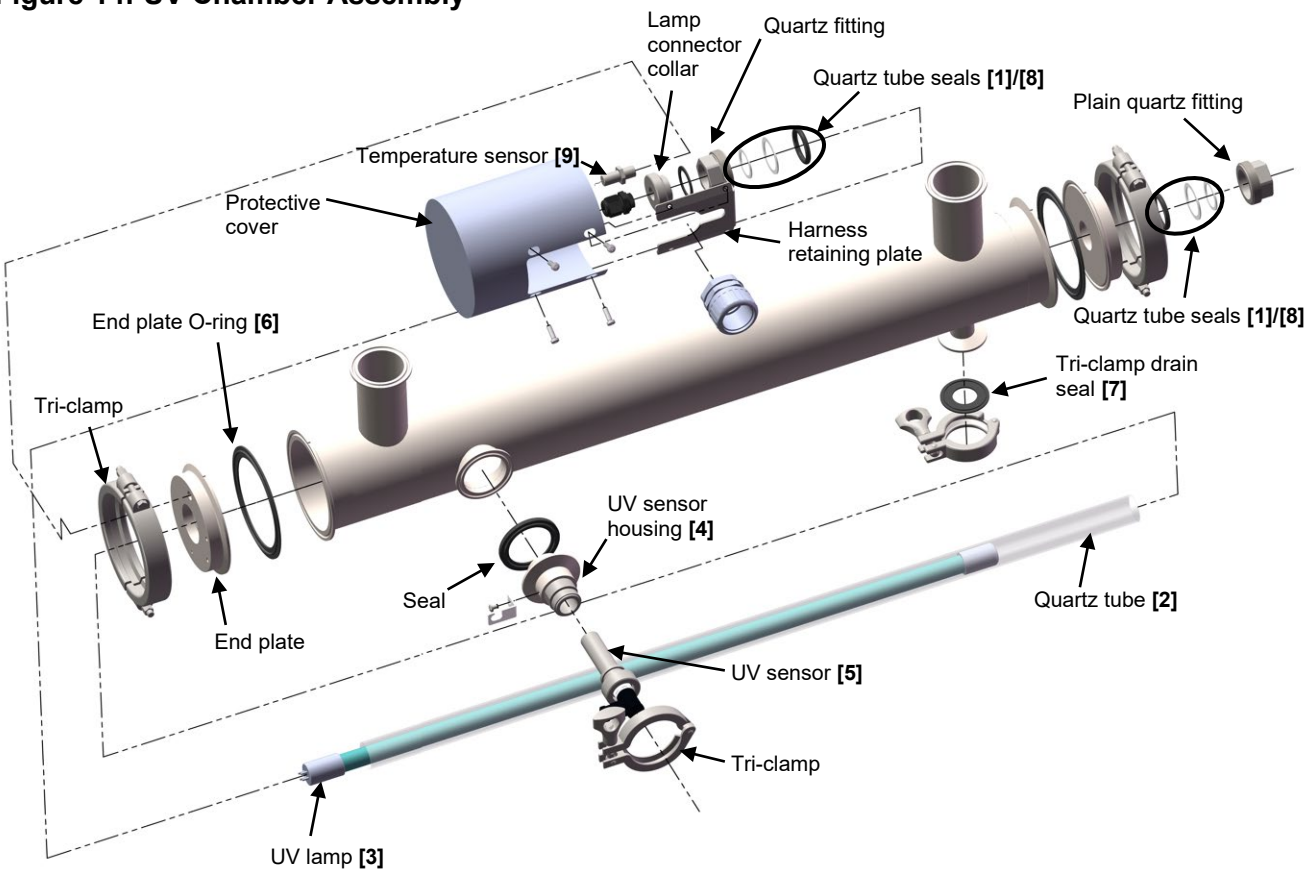
Figure 13: Panel with Power 'Locked Off'



5.2. UV Chamber

	Wear suitable cotton or non-marking disposable gloves, e.g., neoprene rubber, to protect the optical components from being contaminated with fingerprints. Portez des gants de coton ou des gants jetables ne marquant pas pour manipuler la lampe UV et la gaine de quartz pour empêcher que le verre soit marqué par des empreintes digitales.
	Ensure that there is a clean, dry and safe working space to set down components and carry out cleaning operations with iso-propyl alcohol and water. Assurez-vous de disposer d'un espace de travail propre, sec et sûr pour y déposer les composants et réalisez les opérations de nettoyage.

Figure 14: UV Chamber Assembly



Note: Assembly shown as an example. Numbers in square brackets refer to the table of spare parts in section 5.11.

Sufficient spare quartz tubes, UV lamps and seal service kits should be kept available in order to service the system with the minimum downtime.

5.3. UV Lamp

5.3.1. Removing the UV lamp



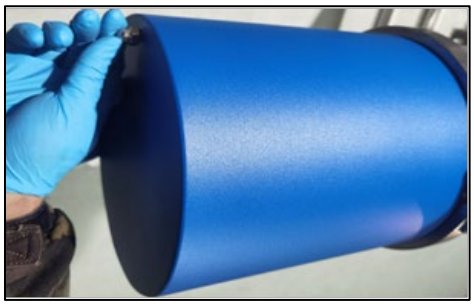
CAUTION! WHEN REMOVING THE UV LAMP IT MAY BE HOT.
NEVER TOUCH A UV LAMP WITH BARE FINGERS. HOLD WITH ACID-FREE TISSUE OR COTTON/NON-MARKING DISPOSABLE GLOVES.
ATTENTION! LA LAMPE PEUT ÊTRE CHAUD AU MOMENT OÙ VOUS L'ENLEVEZ.
NE TOUCHEZ JAMAIS UNE LAMPE UV À MAINS NUES. PRENEZ-LE AVEC UN TISSU NON ACIDE OU AVEC DES GANTS DE COTON OU DES GANTS JETABLES NE MARQUANT PAS.

1. Ensure all lamps are switched off.

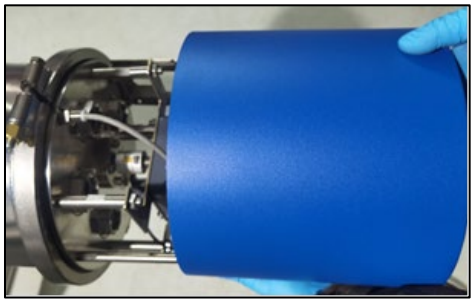
2. Remove the four screws on the protective cover using a cross-headed screwdriver.



3. Unscrew and remove the nut on the end face by hand.



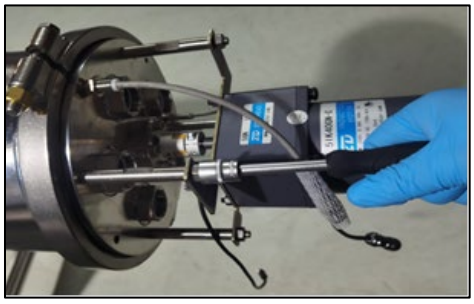
4. Remove the protective cover.
   If an auto-wiper is fitted, go to step 5; otherwise go to step 9.



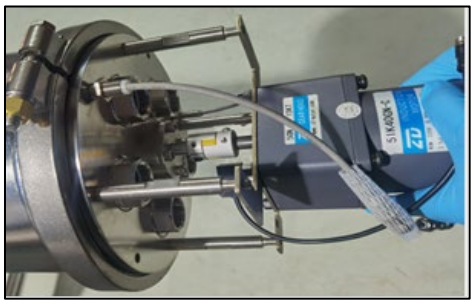
5. Disconnect the cables for the wiper motor, limit switch 1 and limit switch 2.



6. Use an M8 socket wrench to remove the four nuts securing the wiper motor to the bracket.



7. Lift the motor off the bracket.
   The coupling should remain in place.



8. Disconnect the temperature sensor connector from the cable in the wiring harness.

9. Loosen the cable gland on each lamp cable, then unscrew the knurled collar from the quartz fittings.
   Note: Hold the lamp cable while unscrewing the collar to avoid straining the electrical connections.



10. Disconnect the ceramic connectors on the lamps and the lamp cables.



11. Withdraw the lamp from the chamber.



5.3.2. Fitting the UV lamp

These instructions apply to fitting a new lamp or reinstalling a used lamp. We recommend that the quartz tube is cleaned before fitting a new UV lamp (see section 5.4.2).

NEVER TOUCH A UV LAMP WITH BARE FINGERS. HOLD WITH ACID-FREE TISSUE OR COTTON/NON-MARKING DISPOSABLE GLOVES.

NE TOUCHEZ JAMAIS UNE LAMPE UV À MAINS NUES. PRENEZ-LE AVEC UN TISSU NON ACIDE OU AVEC DES GANTS DE COTON OU DES GANTS JETABLES NE MARQUANT PAS.

To remove accidental finger marks, clean the UV lamp with iso-propyl alcohol. When the tube is cold, mercury may condense in patches on the inner surfaces to look like a finger mark; this is normal and does not indicate a fault.

1. Ensure that the quartz tube is installed and the fitting has been fully tightened.



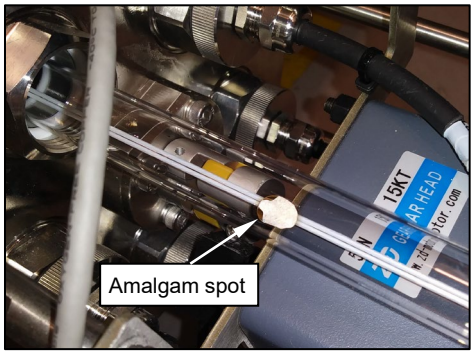
2. Ensure that the drain plug is installed or the drain valve is closed.



3. Ensure that the system has been checked for leaks if the quartz tube has been removed and refitted.

4. Present the end of the UV lamp without the white electrical connection to the quartz tube and gently slide the lamp into place.

   Note: Ensure the amalgam spot is on the underside of the lamp (towards the bottom of the UV chamber) and the white ring is over that spot.

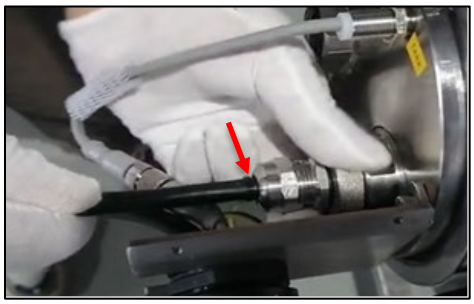


5. Before pushing the lamp fully into the quartz tube, connect the ceramic connector on the lamp cable to the connector on the UV lamp.



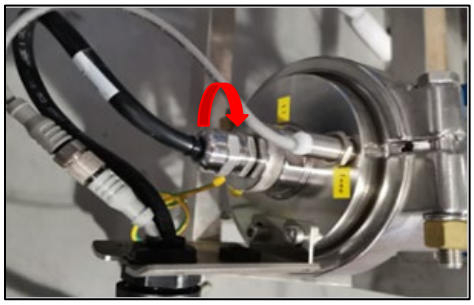
6. Gently push the lamp into the tube and screw the cable on to the quartz fitting using the knurled connector collar, until it is hand tight.

   Make sure that the marker on the cable (shown in image) is next to the cable gland to position the lamp correctly inside the chamber.



7. Carefully tighten the cable gland, making sure that the cable is not twisted as you do so.

   The cable must not rotate while you tighten the gland, as this will cause the lamp to rotate and the amalgam spot will move from its optimum position.



8. When all the required lamps are replaced, reverse steps 2–8 in section 5.3.1 to refit the wiper motor (if required) and cover.

9. Go to the Lamp Maint screen and tap the **Reset** button for the replaced lamp(s), or tap **All Lamps Runtime Reset** to set all lamp operating hours to zero.

5.3.3. Cleaning the UV lamp

The UV lamp should be handled with cotton or non-marking disposable gloves. To remove accidental finger marks, clean the UV lamp with iso-propyl alcohol. When the tube is cold, mercury may condense in patches on the inner surfaces to look like a finger mark; this is normal and does not indicate a fault.

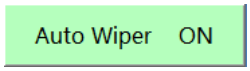
5.4. Quartz Tube

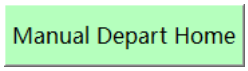
5.4.1. Removing the quartz tube

If an auto-wiper is fitted to the chamber, the wiper yoke needs to be sent to the away position (furthest from the motor) to make it easier to replace the quartz tubes.

1. Check that the wiper is in the home position in the Home screen by the location of the green circle.



2. Tap 

to disable the auto-wiper.
3. Tap 

to send the wiper yoke to the other end of the chamber.
4. Wait until the wiper motor stops (you can see the progress of the wiper in the Home screen) and then switch off the power to the cabinet using the isolator.



PRIOR to carrying out ALL maintenance tasks on the UV chamber, ensure that the isolator on the front of the control panel is switched to OFF and LOCKED OFF (see Figure 13).
AVANT de réaliser TOUTE tâche de maintenance sur la chambre UV ou sur le panneau de commande, assurez-vous que l'isolateur situé à l'avant du panneau de commande est ÉTEINT et qu'il est DÉVERROUILLÉ (reportez-vous à la Figure 13).



ALWAYS ISOLATE AND DRAIN THE CHAMBER BEFORE CHANGING THE QUARTZ TUBE, AND ENSURE NO-ONE IS STANDING IN LINE WITH THE QUARTZ TUBE WHILE THE FITTING IS REMOVED.
TOUJOURS ISOLEZ ET VIDANGEZ LA CHAMBRE AVANT DE RETIRER LA GAINE DE QUARTZ, ET TENEZ-VOUS SUR CÔTÉ DE LA CHAMBRE TOUT EN DÉVISSANT LE RACCORD.

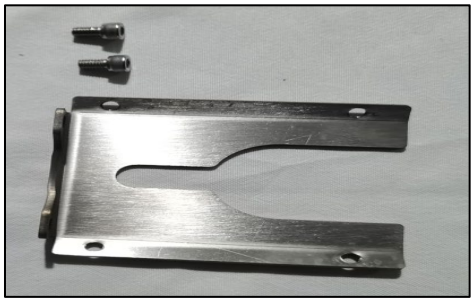
1. Remove all UV lamps by following the procedure in section 5.3.1.
2. Close and lock off the inlet and outlet valves on pipework connected to the unit, if available, to avoid draining down a large section of pipework, and follow your internal procedures to safely depressurise the isolated section of pipework.
3. Carefully remove the drain plug or open the drain valve (if one is fitted) to empty the chamber of liquid.

   Note: Depending on the plant room it may be necessary to have a suitable container available to catch the water released from the drain point. Chamber volumes range from 5 to 100 litres (1.3 to 26.4 U.S. gallons).

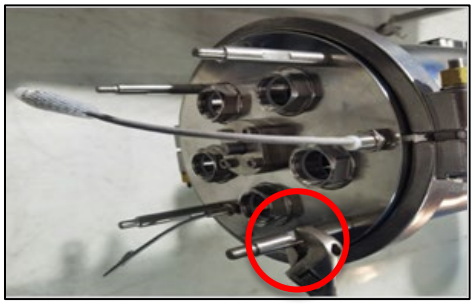


4. Disconnect all the sensor leads from the connectors on the wire harness, and the earth wire from the terminal on the end plate.

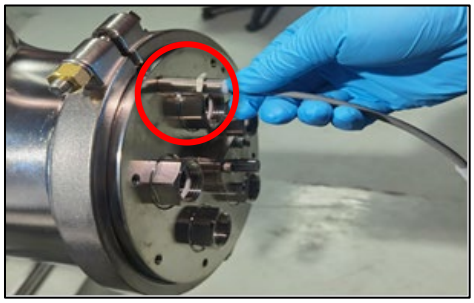
5. Remove the wire harness retaining plate.



6. Remove the auto-wiper connecting rods.



7. Unscrew and remove the temperature sensor from the chamber.



8. **Standing slightly to one side of the chamber**, loosen and remove the fittings at both ends of the chamber using a suitable spanner.

   DO NOT use a pipe grip as this can damage the fitting. Take care not to lose the backing rings inside the fitting.

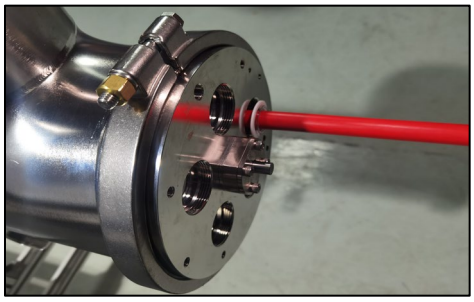


9. The quartz tube is supported at both ends but not in the middle. To assist in removing the tube, a length of plastic piping can be used.

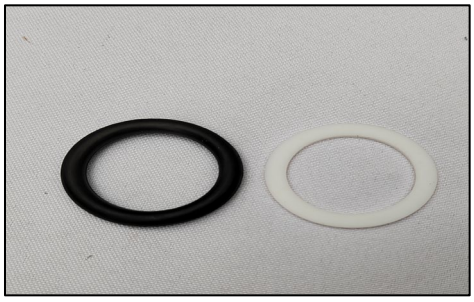
   Prepare a length of plastic pipe with an OD similar to, but less than, the ID of the quartz tube (25 mm / 1 inch OD) and approximately 0.5 metres (1.6 feet) longer than the chamber length.

   Clean the outside of the plastic pipe by wiping with IPA.

   Insert the plastic pipe into the tube.

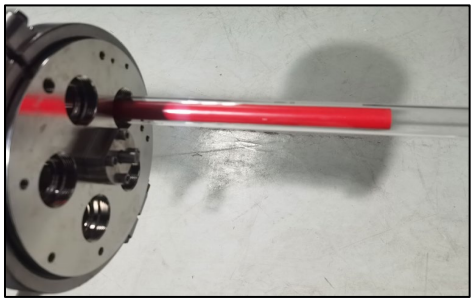


10. Check the quartz tube is free to move. If it is not then 'free' the tube by gently moving it manually. Remove the white PTFE backing ring and the black seal from each end of the tube.



11. Hold the end of the quartz tube with one hand while applying a slight downward pressure on the plastic pipe with the other hand to support the tube, and gently withdraw the quartz tube.

    Take great care that the tube doesn't fall and hit the chamber.



12. Place the tube somewhere clean and dry if it is to be cleaned and reused (refer to section 5.4.2), or dispose of it in a suitable manner.

5.4.2. Cleaning the quartz tube

The quartz tube forms the hydraulic barrier between the UV lamp and the liquid under treatment. It is essential that the tube is kept clean, otherwise there will be a reduction of the UV light which will reduce the effectiveness of treatment.

1. If the quartz tube is to be refitted then it should be cleaned prior to being reinstalled.

   Note: Only handle with cotton or non-marking disposable gloves.

2. If the quartz tube is lightly soiled then wipe the outside with IPA. Use a soft paper tissue for the initial cleaning, then use a lint-free cloth or acid-free tissue for the final clean. If available, carry out a final rinse and clean with clean water.

3. If the quartz tube is highly soiled then clean the outside using an abrasive plastic pad, followed by an IPA clean using a lint-free cloth or acid-free tissue. If available, carry out a final rinse and clean with clean water.

4. If there are iron or carbonate deposits on the quartz tube from hard water, then use 5% citric acid to remove these deposits. Rinse the tube with water, clean with IPA and then, if available, carry out a final rinse and clean with clean water.

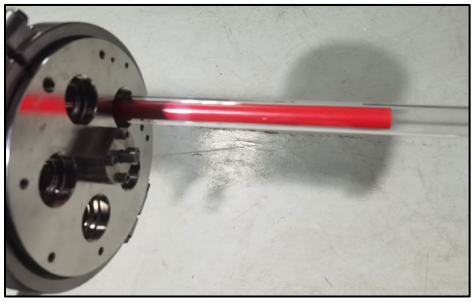
5. If the tube cannot be cleaned then it must be replaced.

Make sure that the inside of the quartz tube is dry before it is installed in the chamber.

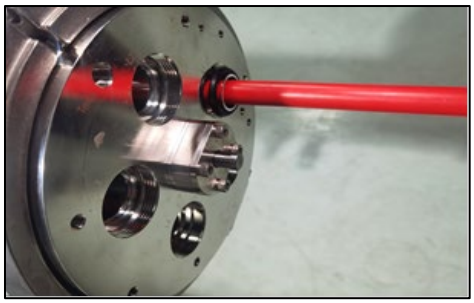
5.4.3. Installing the quartz tube

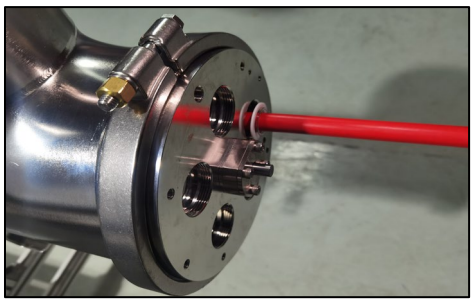
1. Clean the outside of the plastic tube you used for removing the old quartz tube by wiping with IPA. Insert the tube into the chamber so that it projects from both ends.

2. Carefully slide the new tube over the plastic tube and through the chamber.



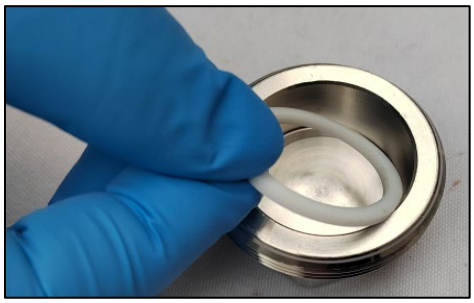
3. Fit a new seal and backing ring over the quartz tube at each end of the chamber.





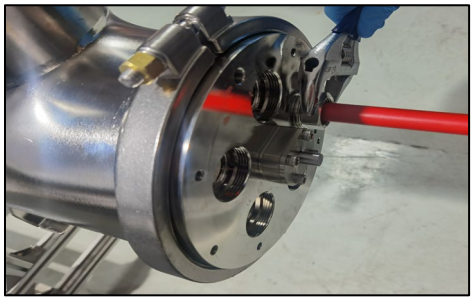
4. Fit a new backing ring (for the quartz tube) into the two fittings (one closed and one open).

   Note: This backing ring is slightly larger than the ID of the fitting so should be fitted as shown then pushed into place. Since it is a 'positive' fit, it is self-retaining.



5. Fit and tighten the fitting using a suitable wrench. DO NOT use a pipe grip as this can damage the fitting.

   Remove the plastic pipe.



6. When all the quartz tubes are replaced as required, refit the drain plug or close the drain valve if one is fitted.



7. **Ensure no-one is standing in line with the quartz fitting.** Open the inlet and outlet valves on the pipework supplying the unit if they have been closed and re-pressurise the hydraulic system. Check the O-ring seal for leaks.

8. If a leak occurs, repeat the tube removal procedure and recheck the O-ring seals and quartz tube sealing surfaces for damage. Replace the seals or tube if necessary.

9. Switch on the power to the cabinet, and if an auto-wiper is fitted, tap Auto Wiper OFF to enable the wiper.

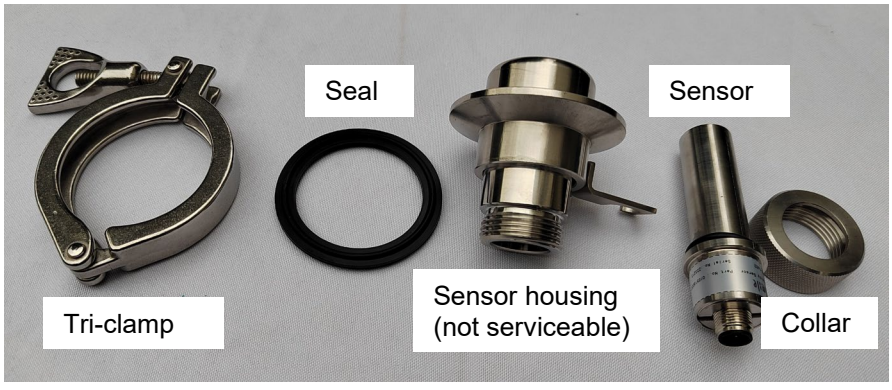
10. Go to the Quartz Tube Maint screen and tap the **Reset** button for the replaced tube(s), or tap **All Tubes Runtime Reset** to set all tube operating hours to zero.

5.5. UV Sensor

5.5.1. Removing the UV sensor and housing

The UV sensor assembly comprises the parts shown in Figure 15 below.

Figure 15: UV Sensor Assembly Parts





PRIOR to carrying out ALL maintenance tasks on the UV chamber, ensure that the isolator on the front of the control panel is switched to OFF and LOCKED OFF (see Figure 13).
AVANT de réaliser TOUTE tâche de maintenance sur la chambre UV ou sur le panneau de commande, assurez-vous que l'isolateur situé à l'avant du panneau de commande est ÉTEINT et qu'il est DÉVERROUILLÉ (reportez-vous à la Figure 13).

ALWAYS ISOLATE AND DRAIN THE CHAMBER BEFORE REMOVING THE UV SENSOR HOUSING.
TOUJOURS ISOLEZ ET VIDANGEZ LA CHAMBRE AVANT DE RETIRER LE BOÎTIER DU CAPTEUR UV.

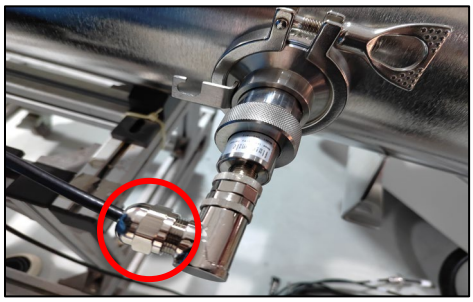
1. Close and lock off the inlet and outlet valves on pipework connected to the unit, if available, to avoid draining down a large section of pipework, and follow your internal procedures to safely depressurise the isolated section of pipework.

2. Carefully remove the drain plug or open the drain valve (if one is fitted) to empty the chamber of liquid.

   Note: Depending on the plant room it may be necessary to have a suitable container available to catch the water released from the drain point. Chamber volumes range from 5 to 100 litres.



3. Disconnect the cable connector from the sensor.



4. Unscrew the knurled collar.



5. Withdraw the sensor from the chamber.



6. Loosen and remove the tri-clamp, and remove the sensor housing from the chamber.



5.5.2. Cleaning the UV sensor and housing window

The sensor quartz and the housing that protrudes into the chamber should be cleaned periodically with IPA. Ideally this will be carried out whenever the quartz tube is being cleaned but the procedures above for removing and fitting the UV sensor have been written so they can also be carried out as stand-alone operations.

1. After removal, the UV sensor and housing should be set in a clean, dry location.



2. The sensor quartz and housing window should be cleaned with IPA using a lint-free cloth or acid-free tissue.



3. If there are deposits which cannot be removed, it may be necessary to replace the entire sensor and/or housing.

The UV sensor housing is not serviceable, so the entire unit must be replaced if the window cannot be cleaned.

5.5.3. Installing the UV sensor and housing

1. Refit the drain plug or close the drain valve if one is fitted.

2. Place the UV sensor housing onto the connection on the chamber with the seal in place and refit the tri-clamp to secure it in place.



3. Refit the sensor into the UV sensor housing and secure it by hand-tightening the knurled collar.



4. Connect the electrical connector to the sensor.



5.6. Auto-Wiper (option)

The auto-wiper system needs to be serviced once a year to replace the service parts and check for wear. The parts detailed should be changed taking note of signs of excessive wear, pitting or abrasions.

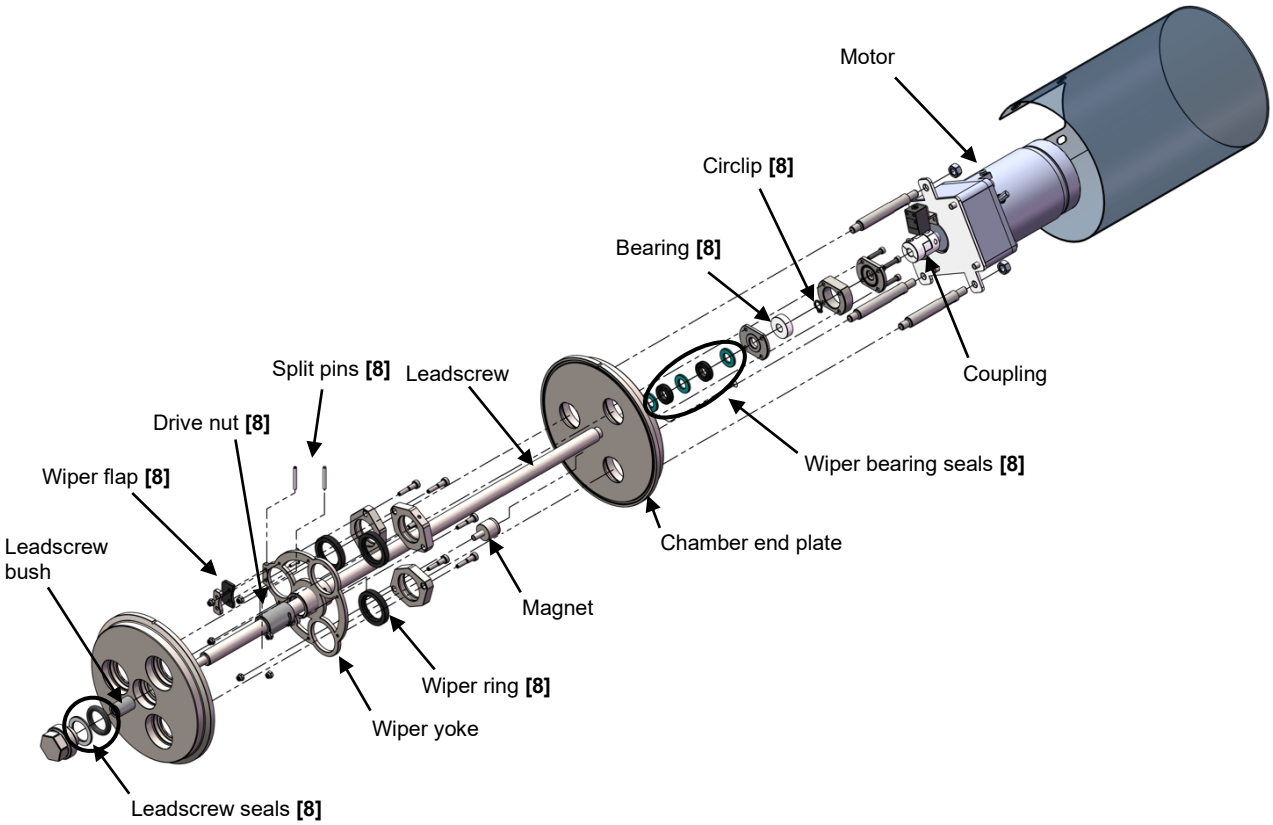


PRIOR to carrying out ALL maintenance tasks on the UV chamber, ensure that the isolator on the front of the control panel is switched to OFF and LOCKED OFF (see Figure 13).
AVANT de réaliser TOUTE tâche de maintenance sur la chambre UV ou sur le panneau de commande, assurez-vous que l'isolateur situé à l'avant du panneau de commande est ÉTEINT et qu'il est DÉVERROUILLÉ (reportez-vous à la Figure 13).



ALWAYS ISOLATE AND DRAIN THE CHAMBER BEFORE PROCEEDING WITH MAINTENANCE ON THE WIPER.
TOUJOURS ISOLEZ ET VIDANGEZ LA CHAMBRE AVANT DE RÉALISER L'ENTRETIEN DU NETTOYEUR.

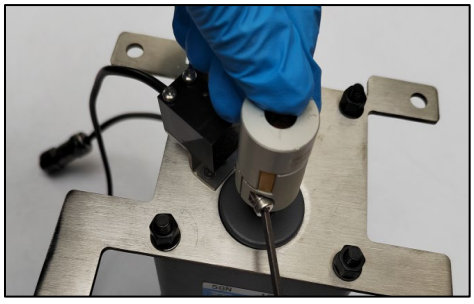
Figure 16: Auto-Wiper Assembly



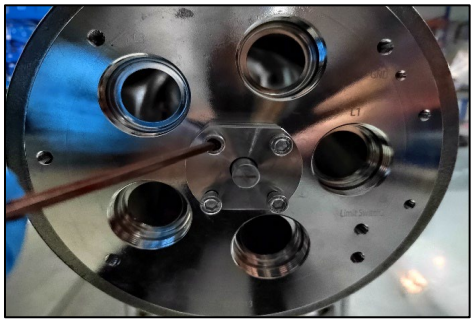
Note: Numbers in square brackets refer to the table of spare parts in section 5.11.

5.6.1. Removing the auto-wiper

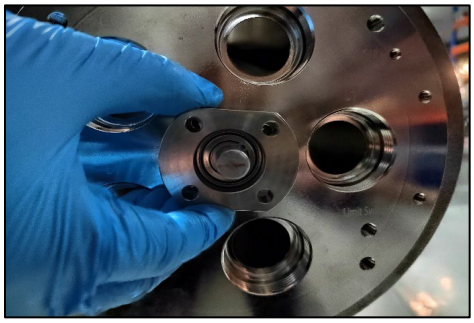
Refer to Figure 14.

1.	Remove the UV lamps (section 5.3.1) and quartz tubes (section 5.4.1).	
2.	Remove the coupling on the leadscrew at the motor end of the chamber.	

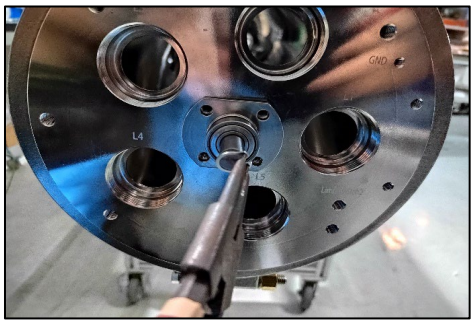
3. Use a hexagonal driver to remove the four screws on the bearing cover.



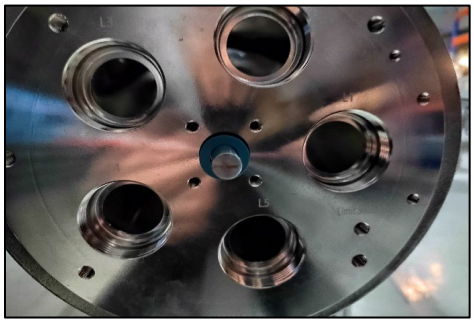
4. Remove the bearing cover and housing.



5. Remove the circlip and bearing. **Note:** The circlip can only be used once and must be replaced when the wiper is serviced.

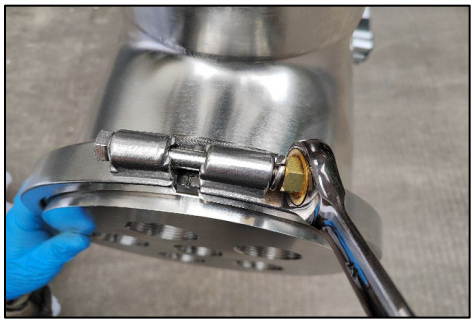


6. Remove the seal covers and O-rings from the end plate.

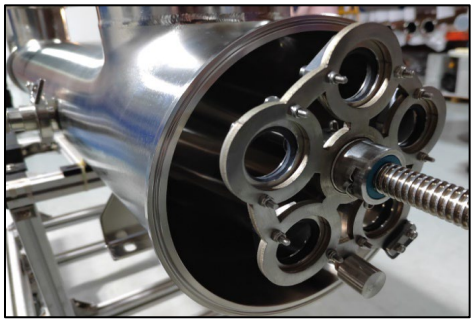


7. Loosen the tri-clamp (or loosen and remove the bolts on larger chambers) and remove the end plate from the motor end of the chamber.

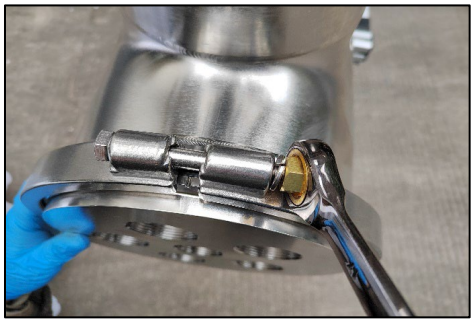
   Make sure all the gaskets and seals are removed with the end plate.



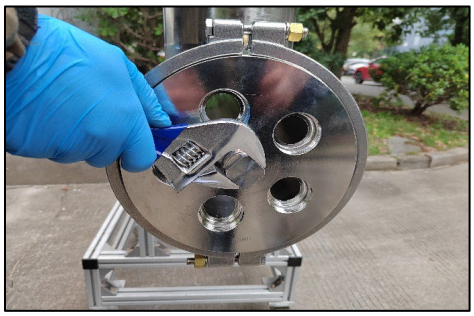
8. Pull the leadscrew and wiper yoke out of the chamber. Unscrew the wiper yoke from the leadscrew.



9. Loosen the tri-clamp (or loosen and remove the bolts on larger chambers) and remove the end plate from the non-motor end of the chamber.

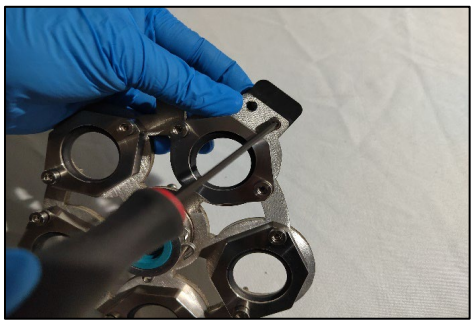


10. Remove the plain leadscrew cap from the end plate and dismantle the bush and seals.

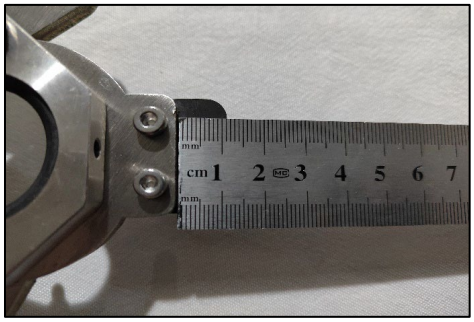


5.6.2. Servicing the auto-wiper

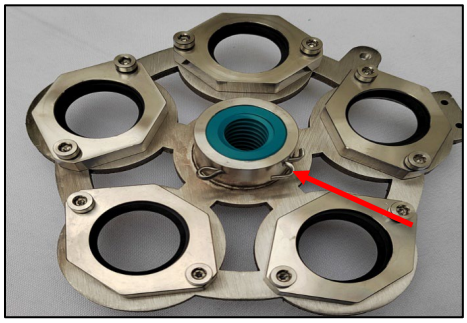
1. Unscrew the retaining plate for the wiper flap.



2. Fit a new wiper flap, making sure that the flap protrudes 10-11 mm beyond the edge of the retaining plate.



3. Use pliers to straighten the split pins on each side of the yoke and remove them.



4. Remove the old drive nut.

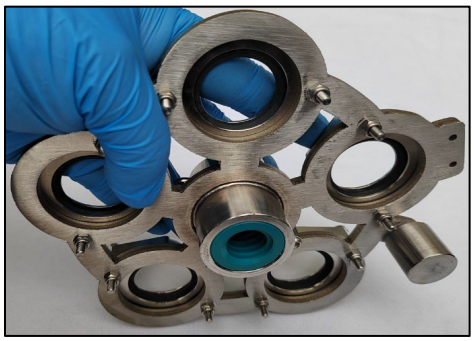


5. Fit the new drive nut:
   - ➢ Align the groove on the drive nut with the holes in the yoke when inserting.
   - ➢ Insert a split pin and bend the ends round using pliers to secure the drive nut in position.



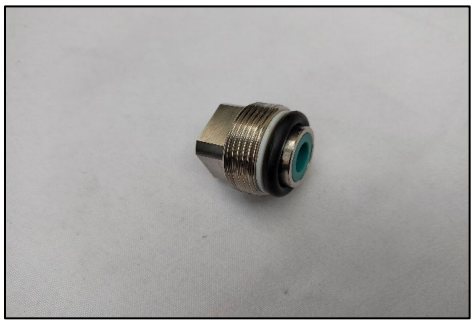


6. Push a suitable tool such as a small screwdriver through the hole on the side of the wiper ring holder to remove the old ring from the yoke. Push a new ring into each holder.



7. Fit a new bush, gasket and O-ring into the leadscrew cap.





5.6.3. Assembling the auto-wiper

1. Replace the seal on the end plate for the non-motor end of the chamber.

   Some chambers may use O-rings to seal the end plate.

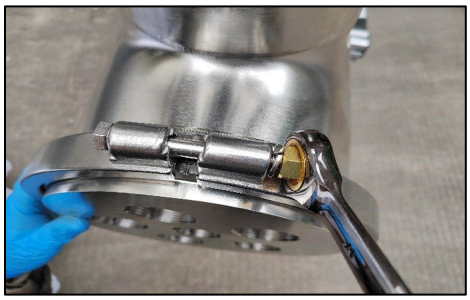


2. Fit the end plate on to the non-motor end of the chamber. Ensure that the grooves on the chamber and end plate are aligned.



3. Install the tri-clamp (or fit the bolts on larger chambers) and tighten to secure the end plate, making sure that the grooves remain aligned.





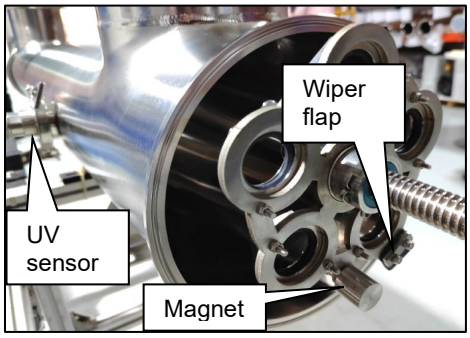
4. Install the leadscrew cap assembly with the O-ring, gasket and bush.



5. Screw the wiper yoke on to the leadscrew and insert the leadscrew into the chamber so that the end of the leadscrew sits in the bush at the far end of the chamber.

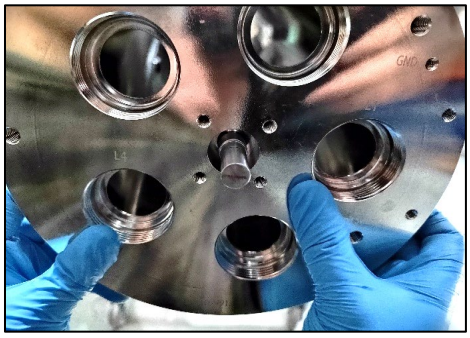
   Note: Position the wiper yoke near to the motor end of the leadscrew to make it easier to insert the leadscrew into the bush.

   Check that the yoke is oriented so that the magnet is aligned with the position of limit switch 2 on the end plate, and the wiper flap is aligned with the UV sensor. If necessary, remove the UV sensor and housing and move the wiper yoke until the flap appears through the open port.



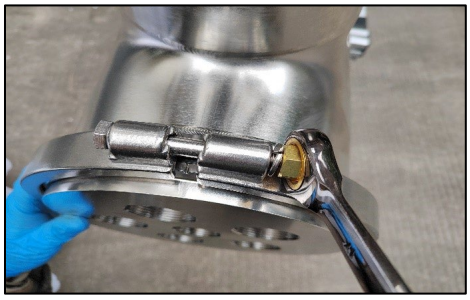
6. Install the other end plate on to the chamber so that the leadscrew passes through the central hole.

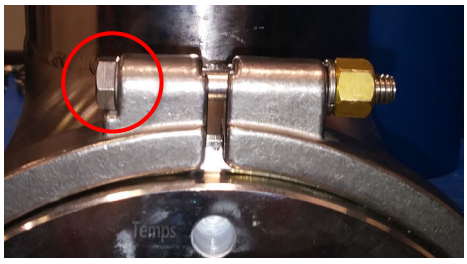
   Ensure that the grooves on the chamber and end plate are aligned.



7. Install the tri-clamp to secure the end plate, making sure that the grooves remain aligned and that the bolt head is next to the recess of the tri-clamp (see image).

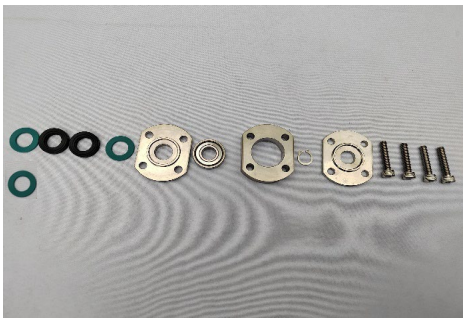
   Note: Larger chambers use bolts to hold the end plate. You will need to hold the end plate in position while fitting the tri-clamp or bolts.





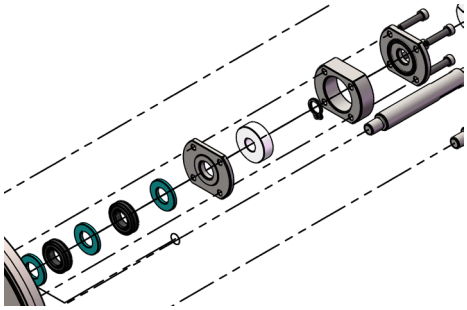
8. Assemble the components for the wiper bearing assembly, using new gaskets and O-rings from the wiper service kit.

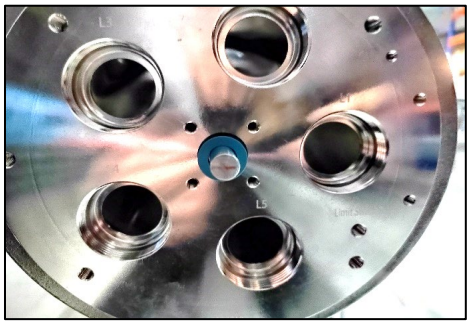
   Lubricate the gaskets and O-rings liberally with suitable grease for your application.



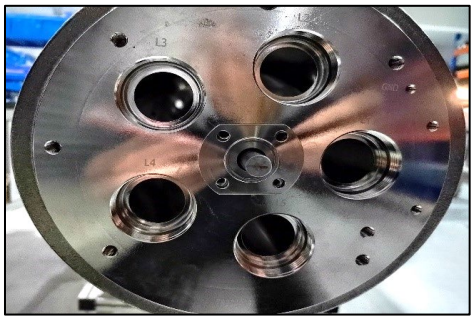


9. Install the gaskets and O-rings into the end plate as shown.

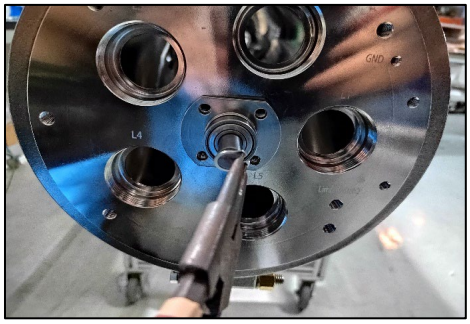




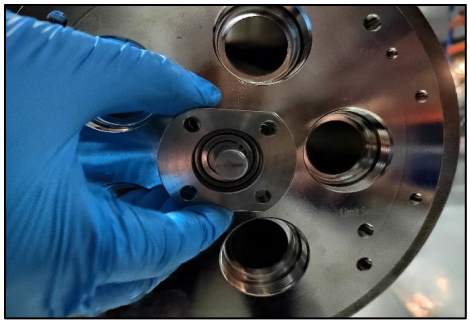
10. Fit the seal cover over the final gasket.



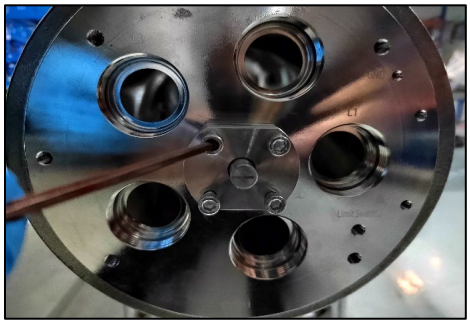
11. Fit the bearing and the circlip.



12. Fit the bearing housing.



13. Fit the bearing cover and secure with the four screws.



14. Fit the quartz tubes into the UV chamber (see section 5.4.3). Wet the outside of each tube with clean water to help it pass through the wiper rings.
15. **Ensure no-one is standing in line with the quartz fittings.** Open the inlet and outlet valves on the pipework supplying the unit and re-pressurise the hydraulic system. Check for leaks.
16. If a leak occurs, remove the leaking quartz tube and recheck the O-ring seals and tube sealing surfaces for damage. Replace the seals or tube if necessary and reassemble.
17. When the chamber is sealed, test the operation of the wiper manually (see section 4.3).
18. Go to the Wiper Maint screen and tap the **Reset** button to set the wiper cycles to zero.
19. Install the UV lamp (see section 5.3).

5.6.4. Replacing the limit switches

Each limit switch and its connector are replaced as a single assembly.

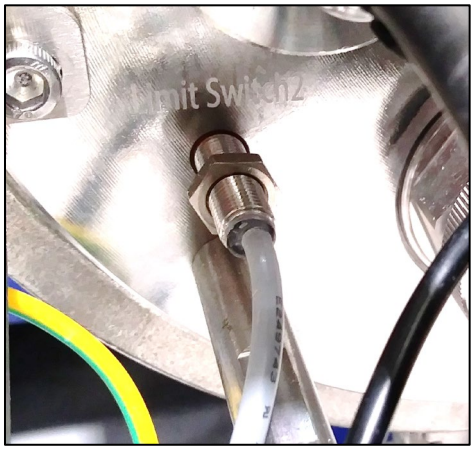
1. Disconnect the limit switch connectors from the cables in the wiring harness.
2. To replace limit switch 1, which is attached to the motor bracket, follow the instructions in section 5.3.1 to remove the wiper motor and bracket.
3. Use a small cross-headed screwdriver to remove the limit switch from its bracket and replace with the new component.

   Note: This limit switch measures the number of rotations of the leadscrew to track the distance that the wiper yoke has travelled from its home position near the motor.



4. To replace limit switch 2, which is inserted into the chamber end plate, use a small spanner to unscrew the switch and remove it from the chamber. Fit the new component but do not over-tighten.

   Note: This limit switch records when the wiper yoke has returned to the home position near the motor.

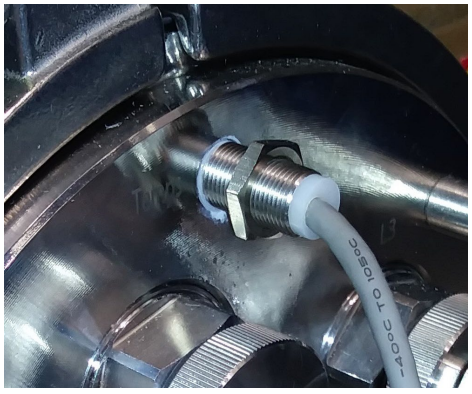


5. Connect the new limit switch cables to the correct connectors in the wiring harness.

5.7. Temperature Sensor

In normal usage, the chamber temperature sensor should not require any maintenance. However, if a sensor becomes faulty then it can be replaced as described below. The sensor and its connector are replaced as a single assembly.

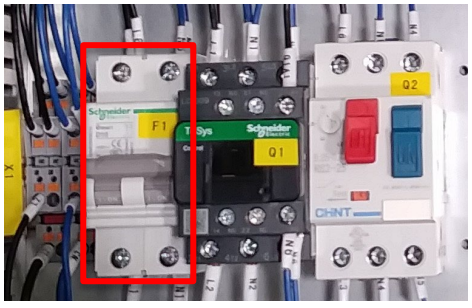
1. Remove the protective cover at the end of the chamber (see section 5.3.1).
2. Disconnect the temperature sensor connector from the cable in the wiring harness.
3. Use a small spanner to unscrew the temperature sensor and remove it from the chamber.



4. Reverse the steps above to fit the new temperature sensor.

5.8. Circuit Breakers

1. F1 (4 A) protects the ballasts from overloads and short circuits.



5.9. Ballast

	**SWITCH OFF AND ISOLATE THE ELECTRICAL POWER SUPPLY BEFORE REPLACING THE BALLAST.** **ÉTEINDEZ ET ISOLEZ L'ALIMENTATION ÉLECTRIQUE AVANT DE REMPLACER LE BALLAST.**

1.	Turn off the power to the control cabinet using the door isolator.	
2.	Pull the terminal blocks out from each end of the faulty ballast. Unscrew the two mounting screws from the backplate and remove the ballast from the control panel.	
2.	Mount the new ballast into the control panel using the mounting screws.	
3.	Insert the terminal blocks into the connectors on the ballast. The ballast input connections are on the right end of the unit, and the output connections are on the left end.	
4.	Switch the power on using the door isolator.	

5.10. Cabinet Air Filters

1. Use a screwdriver to carefully prise off the plastic grills on each side of the control cabinet.



The grill over the air inlet has a slot where you can push with a flat-headed screwdriver to release the catch.



2. Remove the air filter pad and either clean it with a vacuum cleaner, or replace it.

3. Fit the air filter pad in place and carefully clip the grill onto its frame.

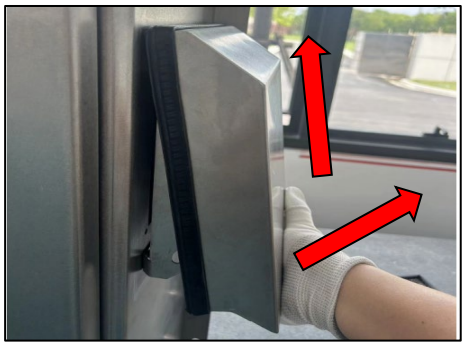


If the ingress protection of your system is rated to IP56 / NEMA 4X, a protective cover has to be removed before you can gain access to the air filters:

1. Insert a hand under the cover and lift it straight upwards.



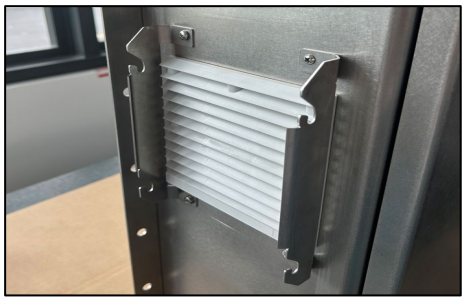
2. Lift the lower edge of the cover outwards to unhook it, and then remove the cover from the frame.



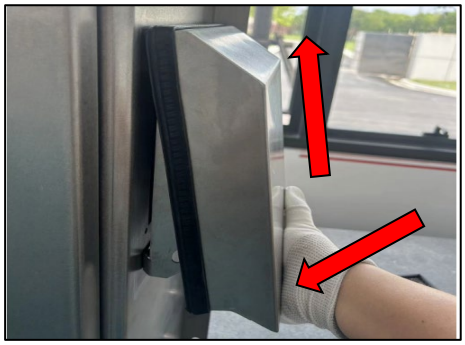


3. Use a screwdriver to carefully prise off the plastic grills, remove the air filter pad and either clean it with a vacuum cleaner, or replace it.

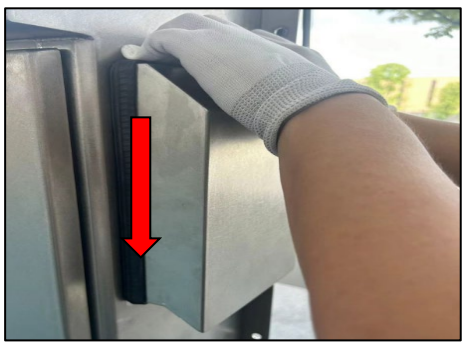
   Fit the air filter pad in place and carefully clip the grill onto its frame.



4. Hang the protective cover on to the hooks.



5. Press down firmly on the cover until you feel that it stops moving.



5.11. Service Parts

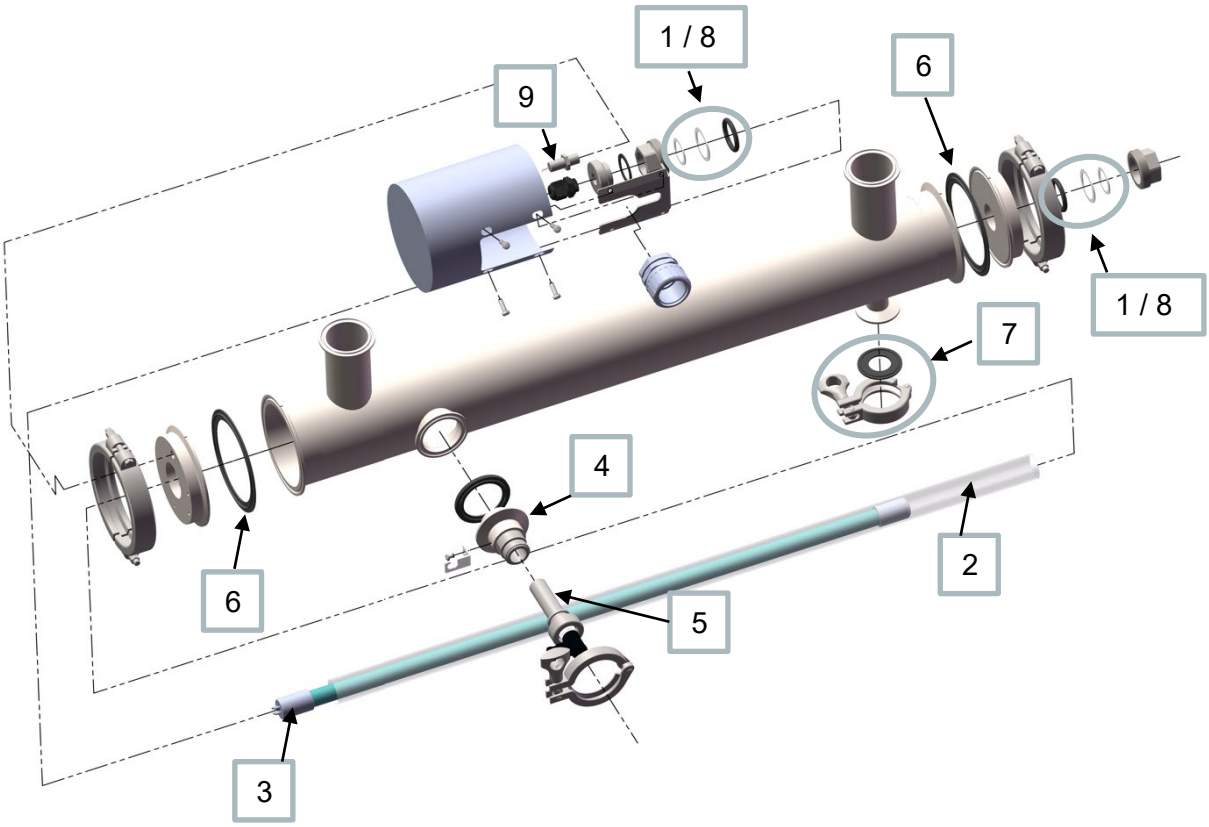
ONLY USE SPARES RECOMMENDED BY THE SYSTEM MANUFACTURER TO AVOID COMPROMISING SAFETY AND PERFORMANCE, AND/OR INVALIDATING THE WARRANTY.

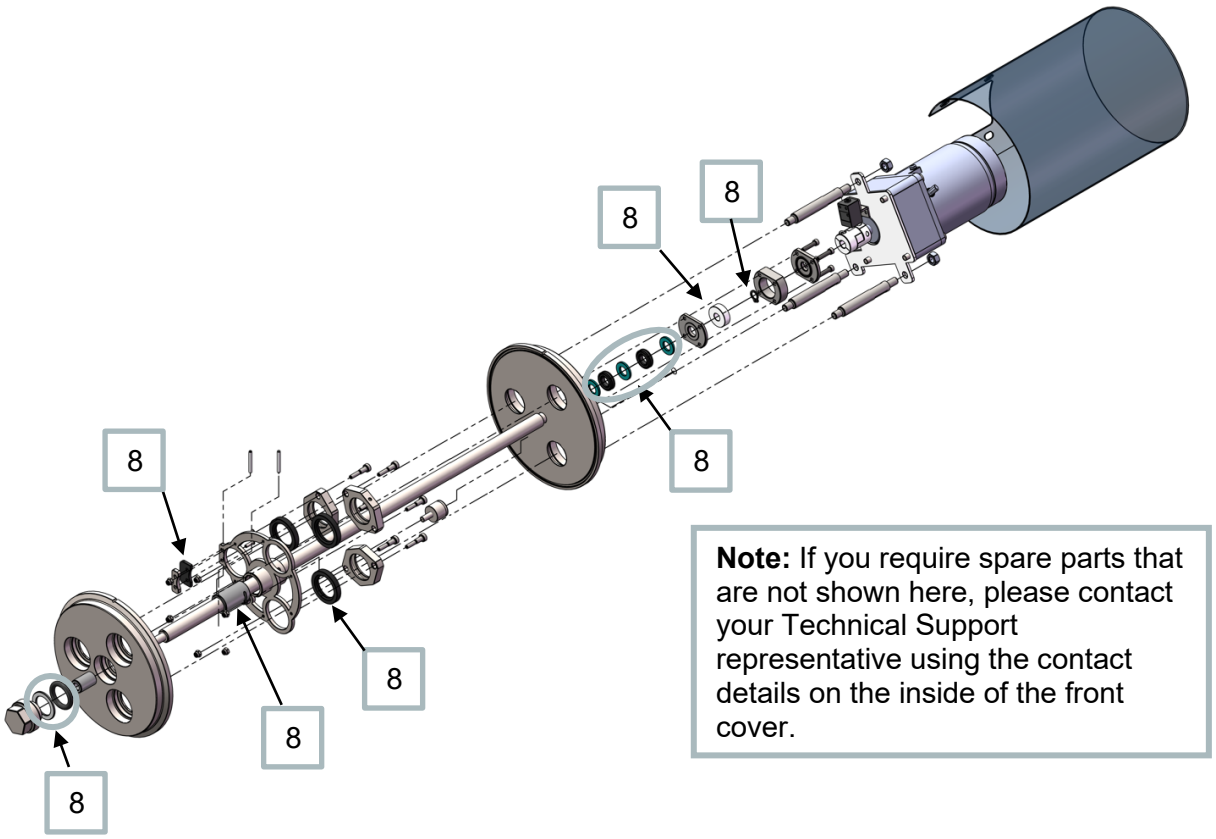
UTILISEZ SEULEMENT LES PIÈCES DE RECHANGE RECOMMANDÉES PAR LE FABRICANT DU SYSTÈME POUR ÉVITER DE COMPROMETTRE LA SÉCURITÉ ET LES RÉSULTATS ET/OU D'INVALIDER LA GARANTIE.

Part	Models	Part Number
[1] Single Quartz Tube Seal Service Kit	All	180072
[2] Quartz Tube	4	322158-0594
	8, 15	322158-1096
	32 – 380	322158-1635
[3] UV Lamp	4	131000-0008
	8	131000-0009
	15	131000-0006
	32 – 380	131000-0007
[4] UV Sensor Housing (not serviceable, replace entire unit)	4 – 125	160682-0150
	215 – 380	160682-0200
[5] UV Sensor	All	170044
[6] Chamber Seal Service Kit	4 – 32	180075
	40 – 125	180076
	215 – 225	180077
	330 – 380	180078
[7] Tri-clamp Drain Spares Kit	All	120376-0010
[8] Wiper and Seal Service Kit (includes seals for all quartz tubes)	40	180073-0040
	85	180073-0085
	125	180073-0125
	215	180073-0215
	225	180073-0225
	330	180073-0330
	350	180073-0350
	380	180073-0380
[9] Temperature Sensor	All	680057
Wiper motor	40	420035-0220 (220 V) 420035-0110 (110 V)
	85 – 380	420035-1220 (220 V) 420035-1110 (110 V)
Limit Switch 1	40 – 380	680061
Limit Switch 2	40 – 380	680062
Wiper Coupling Assembly	40	160724-0001
	85 – 125	160724-0002
	215 – 380	160724-0003

Part	Models	Part Number
Cabinet Air Filter Pad (IP54 / NEMA 12) (2 required per cabinet)	4 – 40	750033-0001
	85 – 380	750033-0002
Cabinet Air Filter Pad (IP56 / NEMA 4X) (2 required per cabinet)	4 – 40	750033-0011
	85 – 380	750033-0012
Ballast	4, 8	650082-0140
	15 – 380	650082-0330

Notes:
Recommended stock holding: 1 lamp, 1 tube and 1 of each service kit required for your chamber (replace as used). Reference numbers in square brackets refer to exploded drawings in Figure 14, Figure 16 and below.





Note: If you require spare parts that are not shown here, please contact your Technical Support representative using the contact details on the inside of the front cover.

Note: Numbers refer to the table of spare parts in section 5.11.

5.12. Clean-In-Place

A Clean-In-Place (CIP) process can be performed without diverting the flow away from the UV system, providing that the temperature does not exceed the rated CIP temperature of the chamber (refer to the data sheet and system labelling for further details).

1.	Check that a wipe cycle is not in progress (section 4.3).
2.	Switch off the lamps either remotely or using the control panel (section 4.2).
3.	Turn off the system using the door isolator switch.
4.	Run the Clean-In-Place process in the plant.
5.	Once CIP is completed, allow the UV system to cool down to below 40ºC (104ºF).
6.	Turn the door isolator switch to ON.
7.	Clear any temperature trips in the system (section 4.4).
8.	Switch on the lamps (section 4.2).

For guidelines and recommendations for performing CIP on your UV system, refer to the CIP guidelines available from your supplier.

6. Alarms and Fault-Finding

6.1. Fault Flowcharts

Figure 17 gives fault-finding options for several faults that may occur when switching on the system after installation. Figure 18 gives the steps that should be followed if a warning or trip alarm is displayed when the system is switched on. Figure 19 gives a procedure to isolate and correct the cause of a low or failed UV intensity warning.

	**ENSURE ELECTRICAL POWER IS ISOLATED BEFORE REMOVAL OR REPLACEMENT OF ANY COMPONENTS. REFER TO THE SAFETY SECTION.** **ASSUREZ-VOUS QUE LE COURANT ÉLECTRIQUE EST ISOLÉ AVANT ENLÈVEMENT OU REMPLACEMENT DES TOUS ÉLÉMENTS. CONSULTEZ LA SECTION SÉCURITÉ.**
	**ELECTRICAL MEASUREMENTS MUST BE CARRIED OUT BY QUALIFIED PERSONNEL.** **MESURES ÉLECTRIQUES DOIVENT ÊTRE EFFECTUÉES PAR DU PERSONNEL QUALIFIÉ.**

If in doubt, contact the supplier of your system for further assistance.

Figure 17: System Fault Flowchart

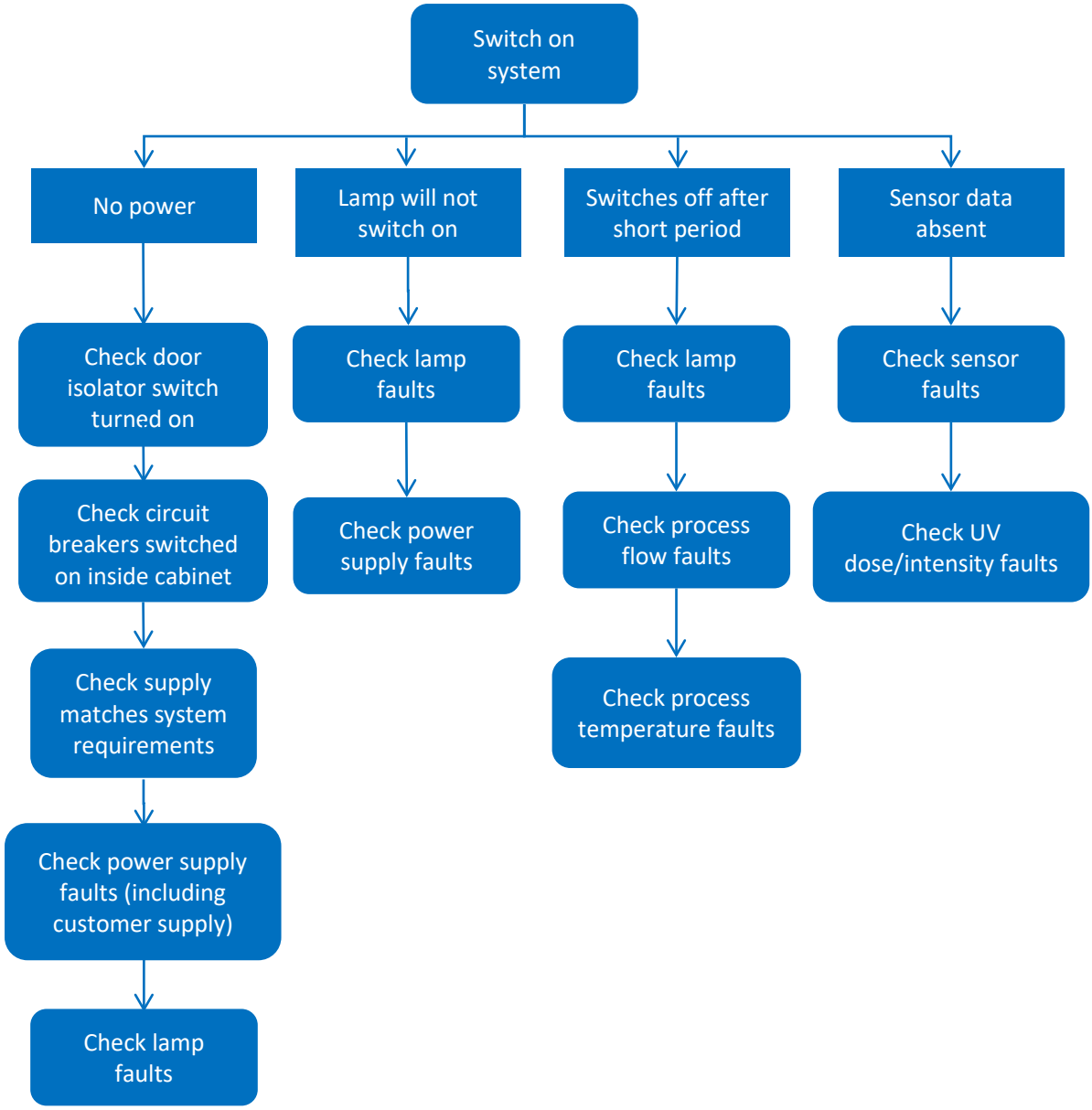


Figure 18: System Warnings and Trips Flowchart

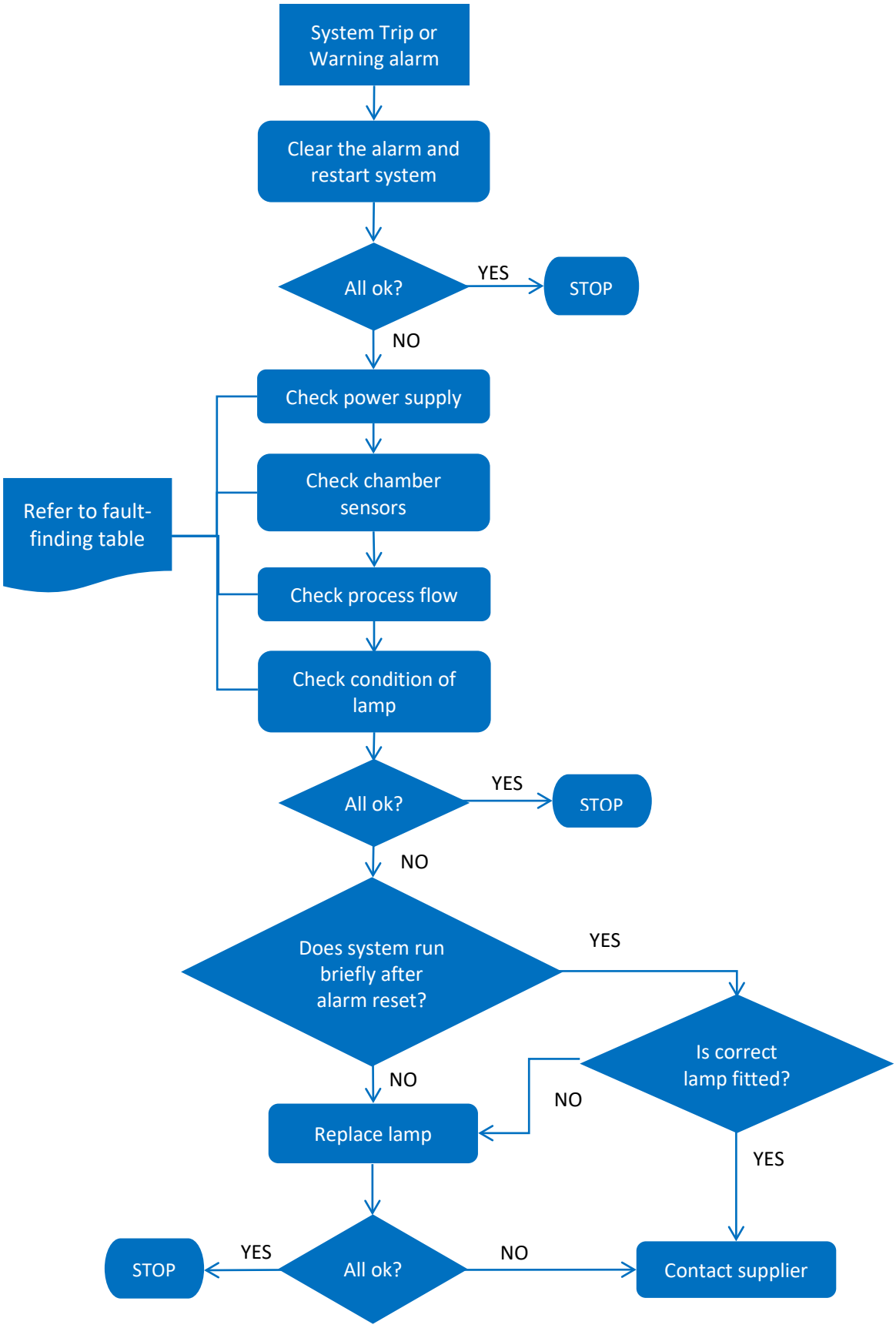
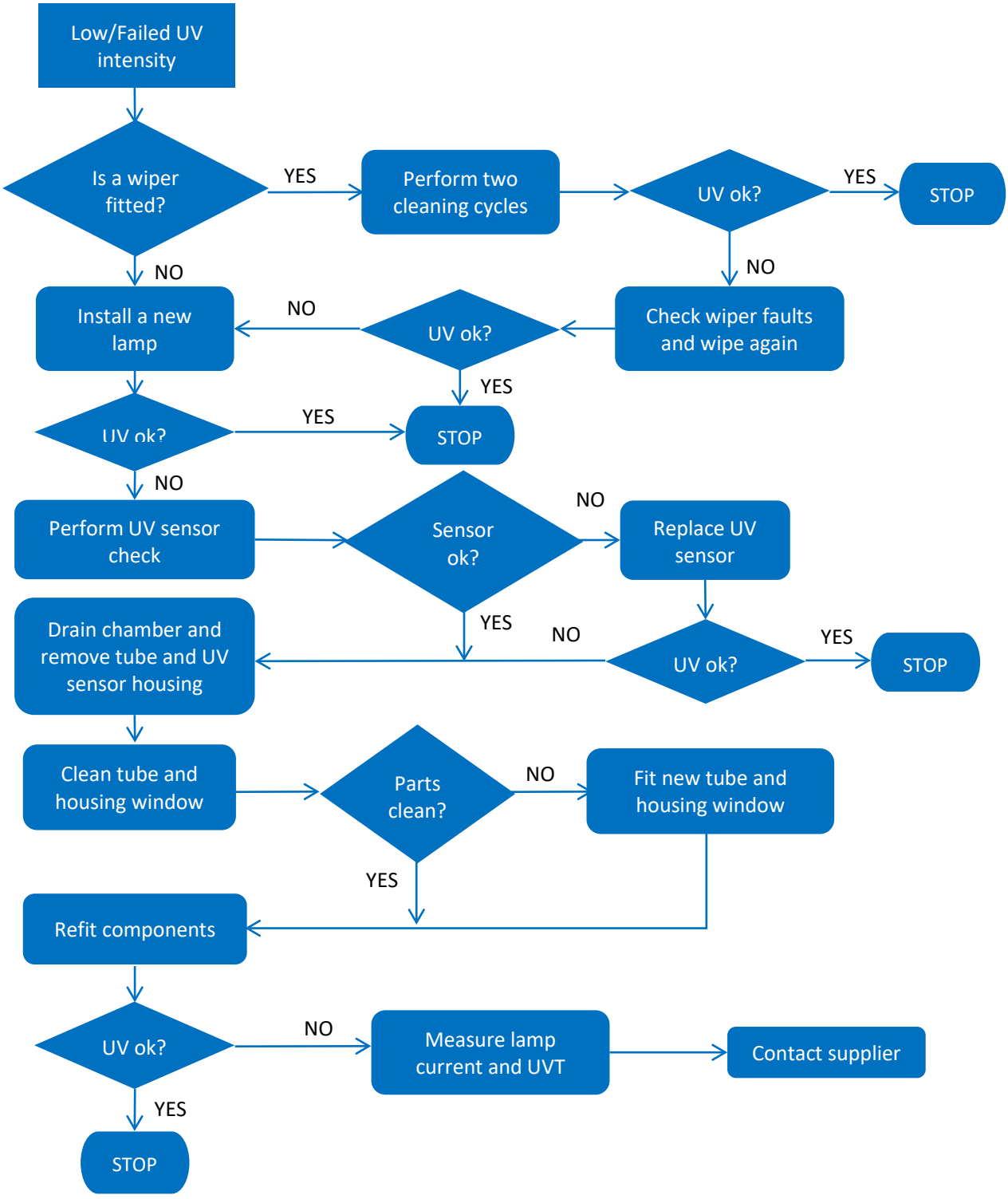


Figure 19: Low/Failed UV Intensity Fault Flowchart



6.2. System Alarms

The tables below list common alarms and faults and their possible causes, with advice on how to confirm the cause and suggested actions to rectify the problem.

Table 4: Fault Finding Table

FAULT	CAUSE	ACTION
[Text Display not lit.]	Loss of 24 V supply.	1. Turn on electrical supply. 2. Reset circuit breaker if necessary. 3. Replace power supply/wiring connection. 4. Turn on isolator.
Lamp cooling	Lamp has just been switched off and cannot be switched on immediately.	Wait for 1 minute before switching lamps on.
Lamp failure	1. Lamp defective. 2. Ballast defective. 3. Wiring/connector loose. 4. Chamber overheated, system tripped. 5. Water leaking into quartz tube.	
Lamp life end	Lamp running time has reached limit set by supplier.	Replace UV lamps.
Tube life end	Quartz tube running time has reached limit set by supplier.	Replace quartz tubes.
Low dose alarm	1. Contamination of quartz tube. 2. UVT of process liquid has decreased. 3. UV intensity sensor inaccurate. 4. Contamination of UV sensor housing window. 5. Lamp has reached limit of running hours. 6. Overheating of control cabinet, ballasts switched to low power.	1. Check process conditions. 2. Remove and clean quartz tubes. 3. Check UV sensor reading using a reference sensor. 4. Clean UV sensor and housing window. 5. Replace UV lamp. 6. Remove and clean filters in cabinet.
Dose output signal can not be measured	1. Dose output span not set. 2. Dose output connection loose or missing.	1. Set up dose output in software. 2. Check output from cabinet to site control system is correctly connected.
Intensity sensor error	1. UV sensor connector loose. 2. UV sensor defective.	1. Check UV sensor connected correctly. 2. Replace UV sensor.
High temperature warning	1. Temperature sensor connector loose. 2. Temperature sensor defective. 3. Chamber overheating due to low flow of process liquid. 4. Local overheating of chamber due to air pockets.	1. Check process conditions. 2. Check temperature sensor connected correctly. 3. Bleed air out of chamber. 4. Replace temperature sensor.
Local Start failed	System cannot be controlled in Local mode.	1. Check if alarm needs to be acknowledged on control panel. 2. Check that system set to Local mode. 3. Check if component has failed in cabinet.

FAULT	CAUSE	ACTION
Remote failed	System not under remote control.	1. Check that cabinet is switched on. 2. Check if lamp is in cooling stage. 3. Check if alarm needs to be acknowledged on control panel. 4. Check remote control input to cabinet is connected correctly. 5. Check that system set to Remote mode. 6. Check if component has failed in cabinet.
Wiper cannot return to home position/Wiper cannot leave home position	1. Wiper coupling disconnected. 2. Wiper circuit breaker open. 3. Wiper parameters incorrect. 4. Wiper blocked by foreign body in chamber. 5. Wiper limit switch defective.	1. Check motor coupling connected under protective cover. 2. Reset wiper circuit breaker in cabinet. 3. Contact supplier for assistance with changing wiper parameters. 4. Remove wiper mechanism and check for blockages. 5. Replace limit switch.
Wiper fault	1. As for previous fault. 2. Defective wiper component (motor, bearing, limit switch).	1. Perform checks for previous fault. 2. Check if wiper motor operates; replace if necessary. 3. Check that limit switches operate using magnets; replace if necessary. 4. Service the wiper mechanism.
Wiper life end	Number of wipe cycles completed has reached limit set by supplier.	Replace serviceable parts on wiper mechanism.
UVT meter error	1. UVT meter connector loose. 2. UVT meter defective.	1. Check input to cabinet from UVT meter is correctly connected. 2. Replace UVT meter.
Flow meter error	1. Flow meter connector loose. 2. Flow meter defective.	1. Check input to cabinet from flow meter is correctly connected. 2. Replace flow meter.
Temp sensor error	1. Temperature sensor connector loose. 2. Temperature sensor defective.	1. Check connector on temperature sensor cable is connected firmly. 2. Replace temperature sensor.
Ballast error	1. Ballast connector loose. 2. Ballast defective.	1. Check connectors on ballast(s) are connected firmly. 2. Replace ballast(s) as indicated by red lights on the side of the ballast.*

* In normal operation, a green light is displayed on the side of the ballast. In the event of a fault with the ballast or the connected lamp, a red light is displayed.

7. Lamp / Quartz Breakage During Operation

In the unlikely event that a quartz tube fails during operation, and if the lamp breaks as a result, then a lamp failure or low UV alarm will be triggered.

If you do not have an automatic divert valve[1] in your process, the lamp and quartz materials could be released into the process stream. In this case, please take the following steps as soon as a failure is identified:

1. Shut down the process.
2. Quarantine any product that has been made with water treated by the UV since the failure occurred.
3. Isolate and drain the UV chamber.[2]
4. Open the chamber and inspect the failed lamp and quartz tube.
5. Ascertain the extent of the damage and material loss. Photographic records are always useful for future review.

Take the following steps to place the system back into service:

1. Drain the process pipework, tanks, chambers etc.[2]
2. Flush out all areas of the process downstream from the UV chamber at high velocity.[2]
3. Examine the pipework, especially any areas where debris could be trapped (valves, fittings, low spots, filters, etc.) to ensure that all quartz pieces have been removed.
4. If applicable, run a CIP process on the affected pipework. Follow any other local regulations for cleaning the pipework in your process.
5. Once you have satisfied yourself that the system is clean, perform a rinse and test the liquid to check that it complies with your water quality requirements.[3]
6. Investigate the cause of the failure and take measures to prevent a reoccurrence.
7. Check that service replacement schedules for quartz tubes and lamps have been followed. Old quartz can become brittle and could be the root cause of a failure.
8. Contact your supplier for further advice if necessary.
9. Order the spare parts required to repair the UV chamber from your supplier. Remember to replace all electrical parts that have been exposed to the process liquid to prevent future issues.
10. Once the parts are available, contact your supplier to arrange for a service engineer to visit and recommission the system.

Notes:

1. With high risk processes where the product cannot be quarantined or is intended for immediate consumption, consider installing with an automatic dump valve downstream that is activated by the low UV or water leak alarm signals. There needs to be sufficient pipe length between the UV chamber and dump valve to ensure that the valve diverts the flow before contaminated water reaches the valve. This length will depend on the process flow velocity and the time taken for the valve to respond and operate.
2. Water drained or flushed from the system may contain traces of quartz and/or mercury. Waste should be handled according to your local regulations.
3. Particular attention should be paid to quartz particles and mercury. A medium-pressure lamp contains ~150 mg of mercury and an amalgam lamp contains ~25 mg (0.001 oz).

8. Decommissioning

1.	Ensure the system is thoroughly clean. We recommend that a CIP cycle is run where possible.
2.	Perform an assessment of the safest method of supporting and lowering the items while they are detached from the supporting fixtures. Use suitable and properly maintained lifting equipment operated by trained personnel.
3.	Dispose of the components according to local regulations. Pay particular attention to the UV lamp which contains a small quantity of mercury. On request, your supplier representative can supply details of suitable disposal companies.

8.1. Temporarily Suspending Operation

If you intend to suspend operation of the UV treatment system *and switch off the flow of liquid through the chamber* for more than a few days, it is important to follow the steps below to reduce the chance of corrosion occurring inside the chamber and avoid faults when the system is restarted.



It is not recommended to leave the UV chamber standing full of stationary water, as this can cause corrosion and fouling to begin more rapidly than when water is flowing. The risk depends on many factors which are specific to each application. Please assess the risk as part of your system's maintenance procedures and contact your system supplier if you have any questions.

Before the system is shut down:

1.	Isolate and completely drain the UV chamber.
2.	Flush the chamber out thoroughly with clean water. This is especially important if you are treating contaminated water (such as in wastewater systems) or water containing additives (such as in swimming pool systems).
3.	Allow the chamber to dry.

Before refilling the chamber and restarting the system:

1.	Inspect the inside of the chamber to check for any fouling of surfaces by limescale or other deposits. Pay particular attention to the quartz tubes and the UV sensor housing.
2.	If there are any signs of fouling on the surfaces, we recommend that the chamber is given a complete service to clean or, if necessary, replace components BEFORE the system is switched on. Failure to do so may cause the system to malfunction when it is restarted.

Appendix A: System Record Table

Use this table to record lamp changes, modifications, updates, etc.

Date	Action

Appendix B: Declarations of Conformity

FM29365

EU DECLARATION OF CONFORMITY

We:

Hanovia Ltd (t/a Nuvonic), 780 Buckingham Avenue, SLOUGH, Berkshire SL1 4LA, United Kingdom

(EU Address: Nuvonic GmbH, Hungenbach 1D, D-51515 Kürten, Germany)

declare under our sole responsibility that the products listed below:

ECO Ultraviolet Fluid Treatment Systems
(see table below for model numbers)

to which this declaration relates are in conformity with the provisions of the EU Electromagnetic Compatibility (EMC) Directive 2014/30/EU and comply with the following standards:

EN IEC 61000-6-2:2019	Electromagnetic compatibility (EMC) Part 6-2: Generic standards – Immunity standard for industrial environments
EN IEC 61000-6-4:2019	Electromagnetic compatibility (EMC) Part 6-4: Generic standards – Emission standard for industrial environments

The products are also in conformity with the provisions of the EU Low Voltage Directive 2014/35/EU and comply with the following standard:

EN 61010-1:2010 + A1:2019	Safety requirements for electrical equipment for measurement, control and laboratory use

The products are also in conformity with the provisions of the EU Pressure Equipment Directive 2014/68/EU and comply with the following standards:

EN 13445-1:2014 + A1:2014	Unfired pressure vessels - Part 1: General
EN 13445-2:2014 + A3:2018	Unfired pressure vessels - Part 2: Materials
EN 13445-3:2014 + A8:2019	Unfired pressure vessels - Part 3: Design
EN 13445-4:2014	Unfired pressure vessels - Part 4: Fabrication
EN 13445-5:2014 + A1:2018	Unfired pressure vessels - Part 5: Inspection and testing

Issued: 1st October 2023, Slough, UK

Signed for and on behalf of Hanovia Ltd (t/a Nuvonic)

Andy Sidwell – Technical Director

FM29365

UK DECLARATION OF CONFORMITY

This declaration is issued under the sole responsibility of the manufacturer:

Hanovia Limited (t/a Nuvonic)
780 Buckingham Avenue, SLOUGH, Berkshire, SL1 4LA, UK

and hereby declares that:

ECO Ultraviolet Fluid Treatment Systems
(see table below for model numbers)

have been demonstrated to fulfil the objectives referred to in the listed sections of the following UK Statutory Instruments:

SI 2016 No. 1101*	The Electrical Equipment (Safety) Regulations 2016
SI 2016 No. 1091*	The Electromagnetic Compatibility Regulations 2016
SI 2016 No. 1105*	The Pressure Equipment (Safety) Regulations 2016

*As amended by EU Exit Regulations

and hereby declares that the equipment named above is in conformity with the relevant United Kingdom designated standards:

EN 61010-1:2010 + A1:2019	Safety requirements for electrical equipment for measurement, control, and laboratory use - Part 1: General requirements
EN IEC 61000-6-2:2019	Electromagnetic compatibility (EMC) Part 6-2: Generic standards – Immunity standard for industrial environments
EN IEC 61000-6-4:2019	Electromagnetic compatibility (EMC) Part 6-4: Generic standards – Emission standard for industrial environments
EN 13445-1:2014 + A1:2014	Unfired pressure vessels - Part 1: General
EN 13445-2:2014 + A3:2018	Unfired pressure vessels - Part 2: Materials
EN 13445-3:2014 + A8:2019	Unfired pressure vessels - Part 3: Design
EN 13445-4:2014	Unfired pressure vessels - Part 4: Fabrication
EN 13445-5:2014 + A1:2018	Unfired pressure vessels - Part 5: Inspection and testing

The UKCA marking is applied accordingly.

Signed:

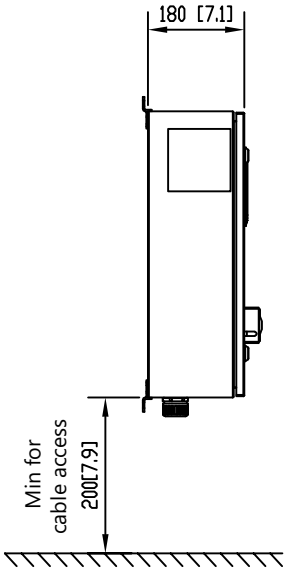
Name: Andy Sidwell
Position: Technical Director
Location: Slough, UK
Date: 1st October 2023

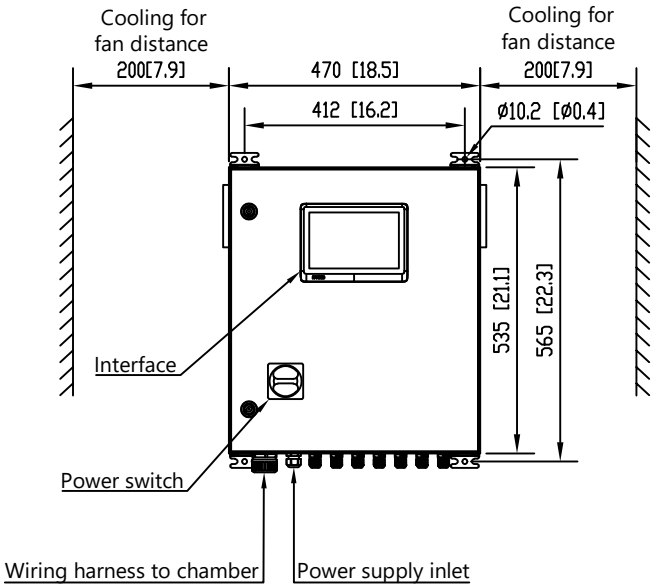
Comparison of Model Numbering

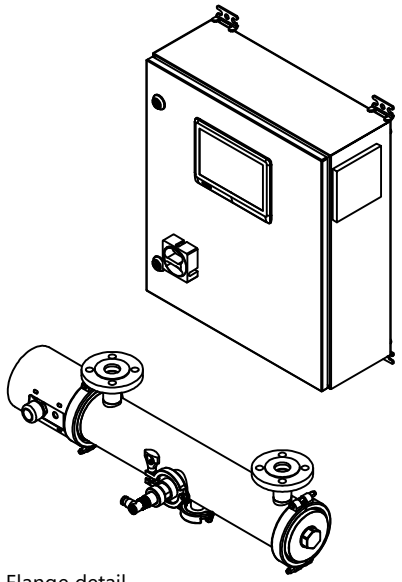
ECO Model Number	Original Model Number from Compliance Testing
4	01-B-N-100-S
8	01-B-N-100-M
15	01-C-N-100-M
32	01-C-N-100-L
40	01-C-N-170-L
85	02-C-N-170-L
125	03-C-N-170-L
215	04-C-N-220-L
225	05-C-N-220-L
330	05-C-N-275-L
350	06-C-N-275-L
380	07-C-N-275-L

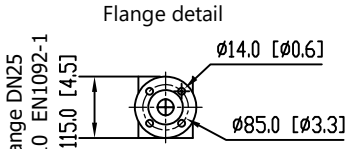
Appendix C: Drawings

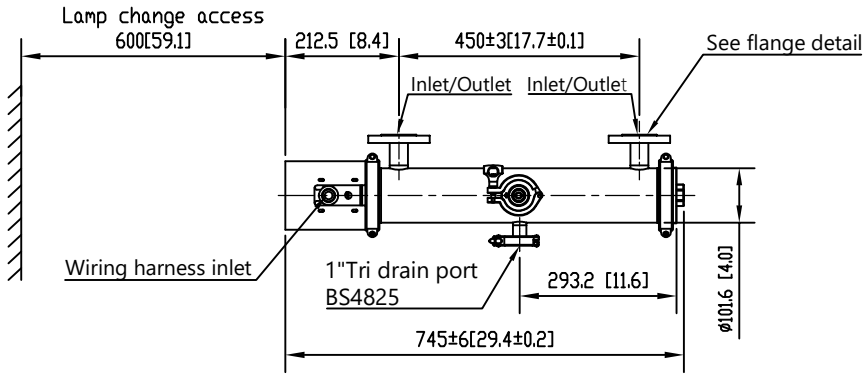
MAX OPERATING PRESSURE: 10 BAR (145 PSI)
CHAMBER WEIGHT: 10.5 KG (23.2 LBS)
CHAMBER WEIGHT IN OPERATION: 14.4 KG (31.8 LBS)

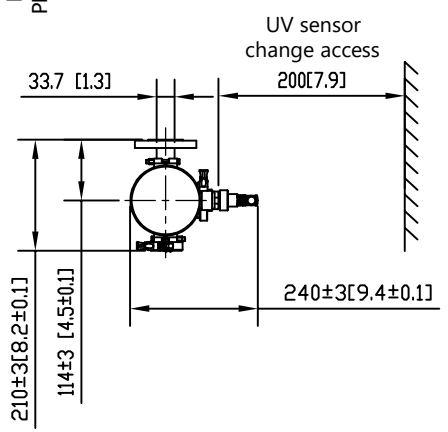








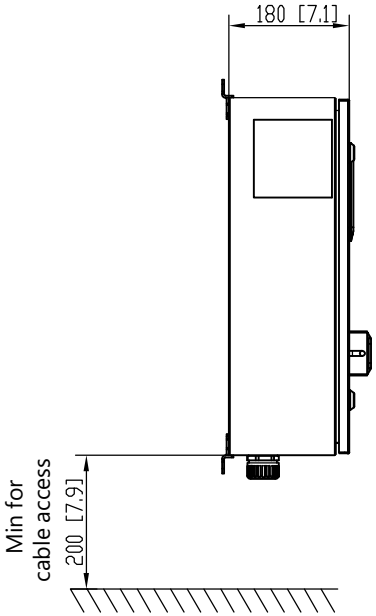


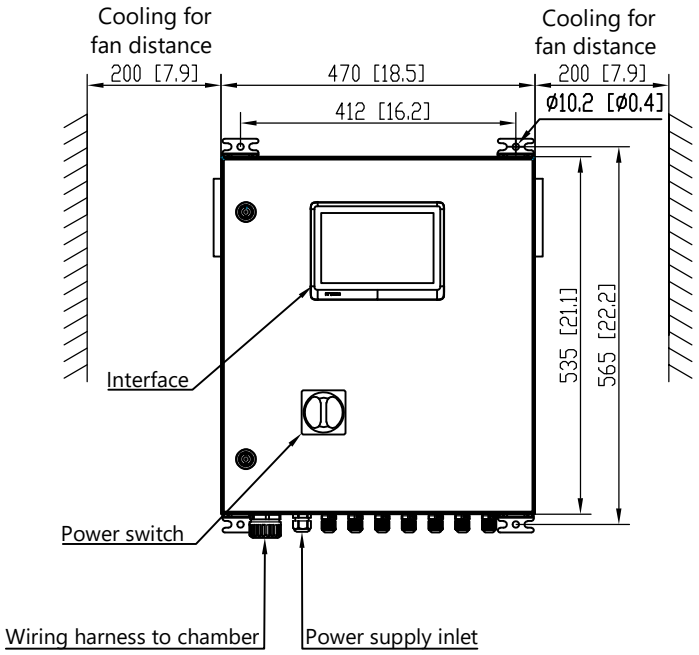


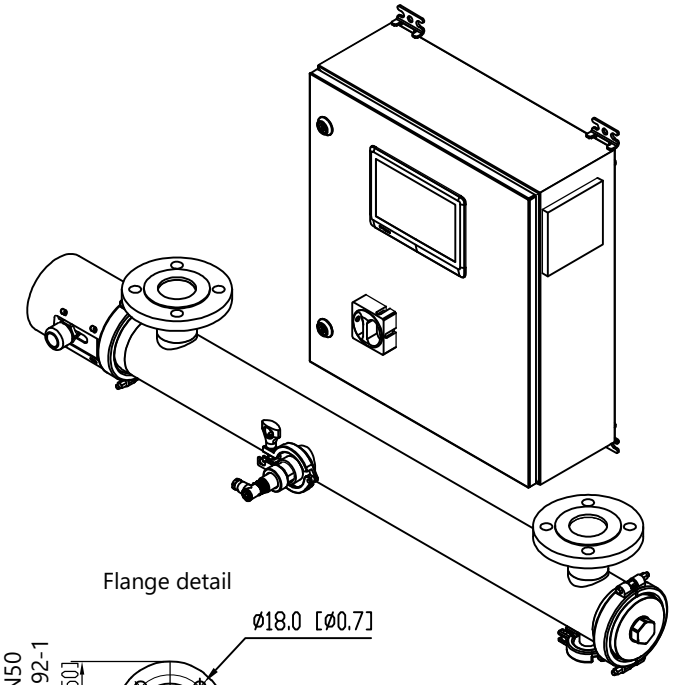
TITLE: ECO 4 (PN10 NW)	DRN	Walker	DATE	2023/7/20
	CKD	Walker	APPRD	Raymond
DRAWING No:- 820337-1A12E-01				

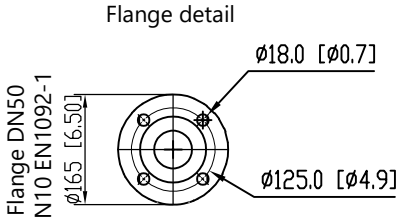
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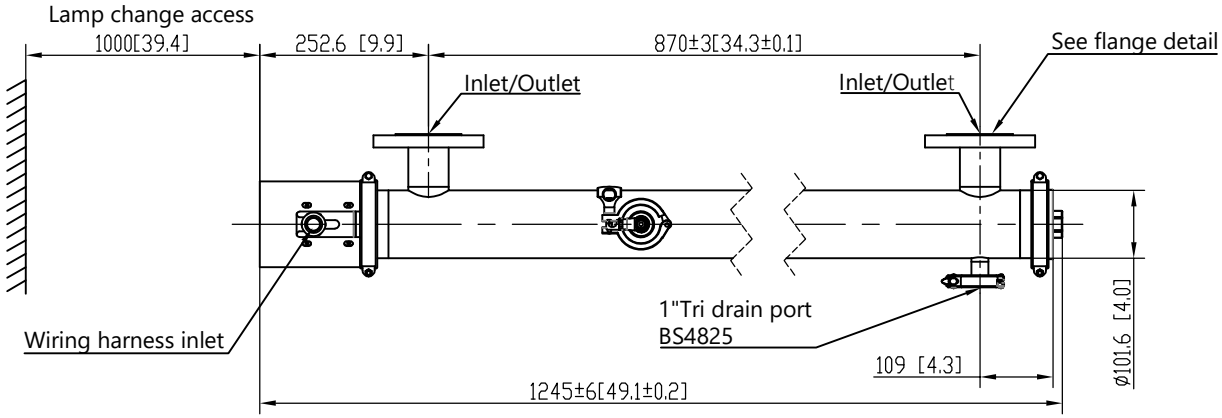
MAX OPERATING PRESSURE: 10 BAR (145 PSI)
CHAMBER WEIGHT: 18.5 KG (40.7 LBS)
CHAMBER WEIGHT IN OPERATION: 24 KG (53.9 LBS)

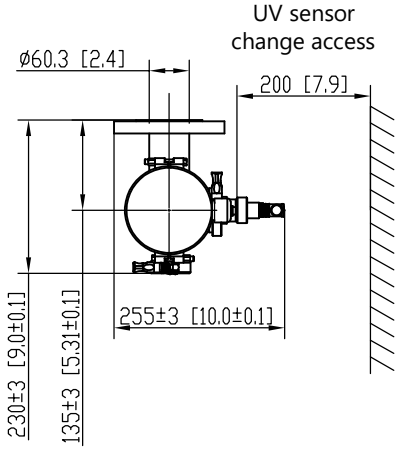










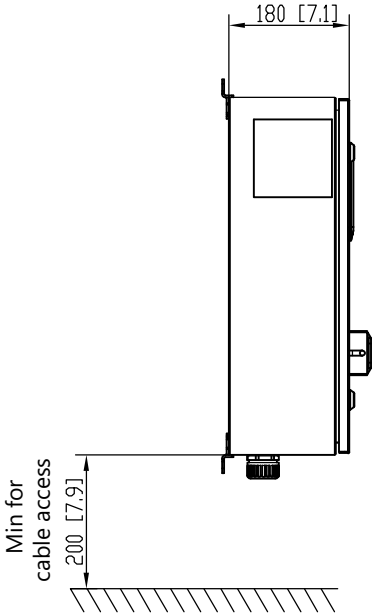


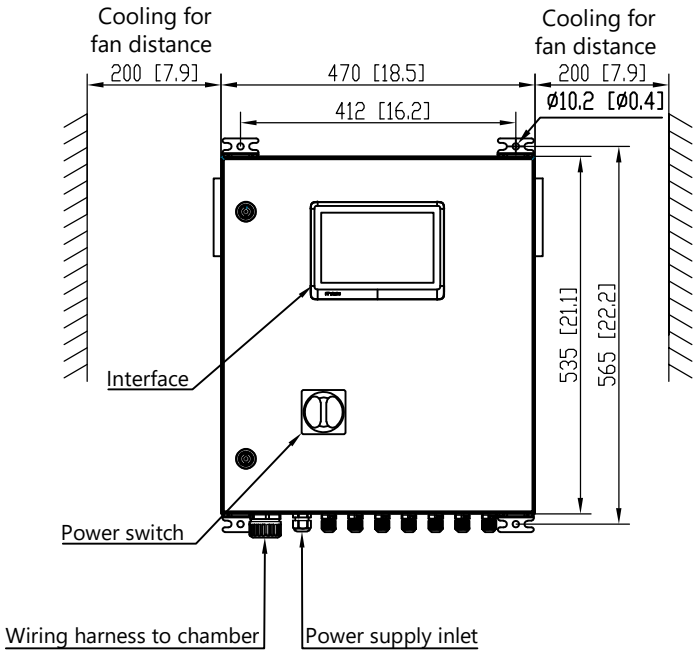
TITLE: ECO 8 (ANSI NW)	DRN	Walker	DATE	2023/7/20
	CKD	Walker	APPRD	Raymond

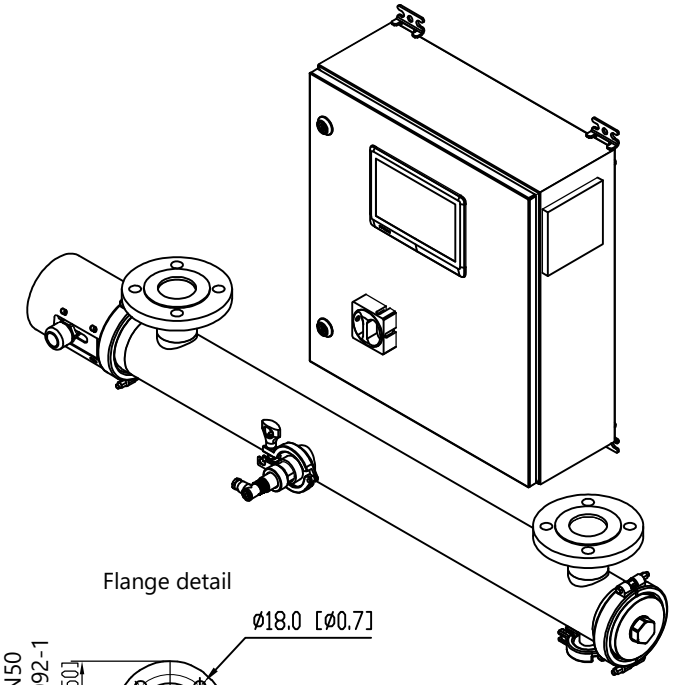
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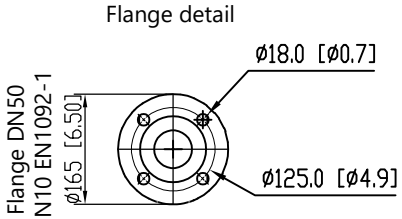
DRAWING No:- 820338-1C12E-01

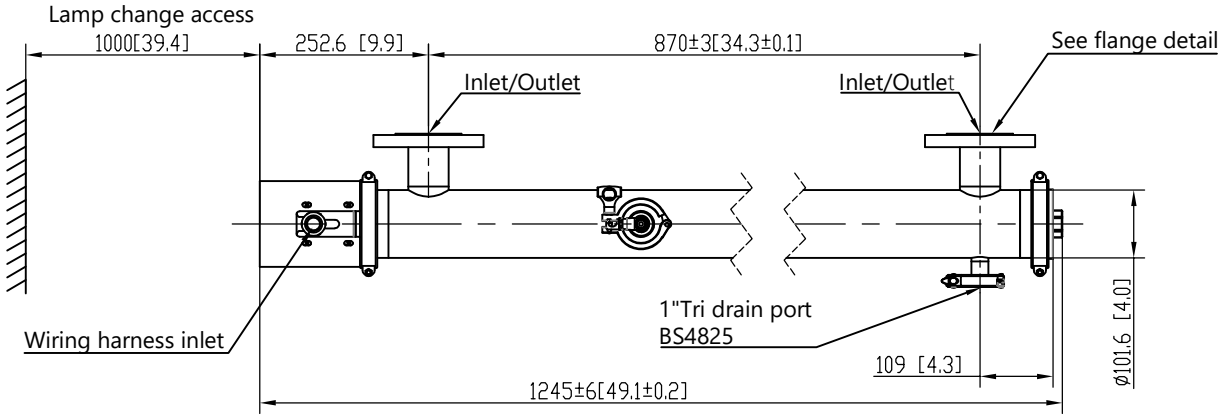
MAX OPERATING PRESSURE: 10 BAR (145 PSI)
CHAMBER WEIGHT: 18.5 KG (40.7 LBS)
CHAMBER WEIGHT IN OPERATION: 24 KG (53.9 LBS)

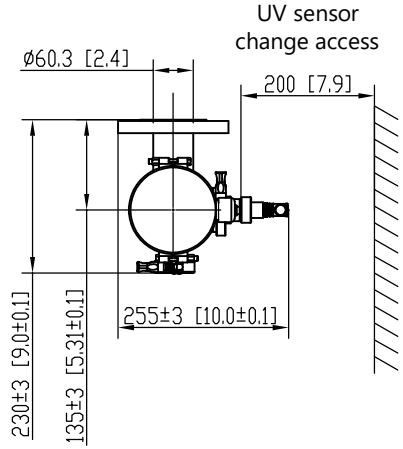








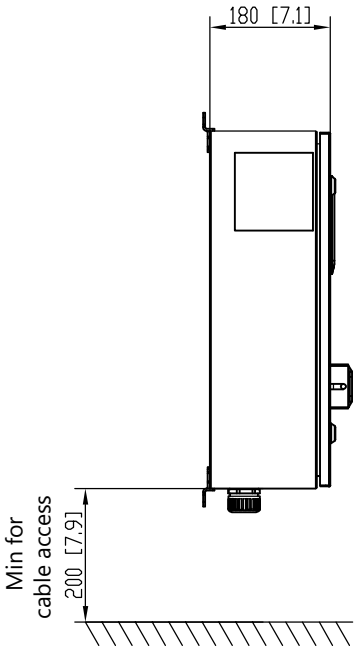


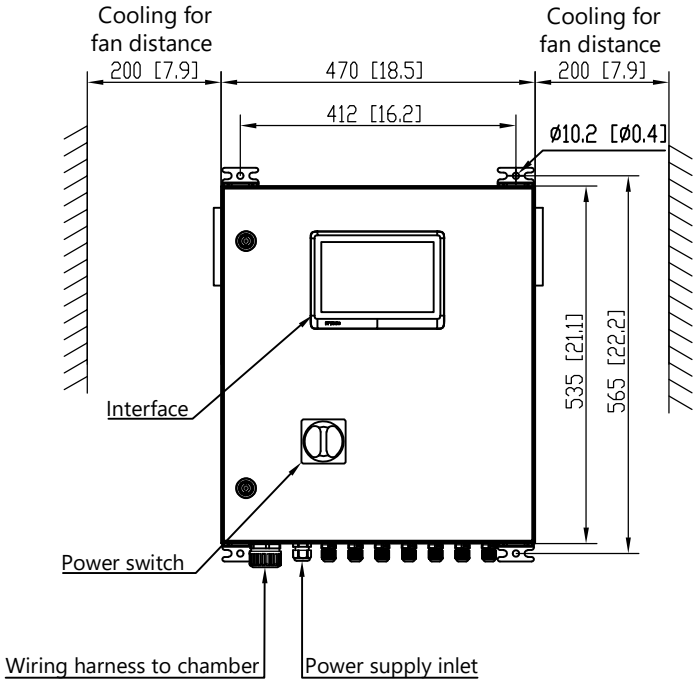


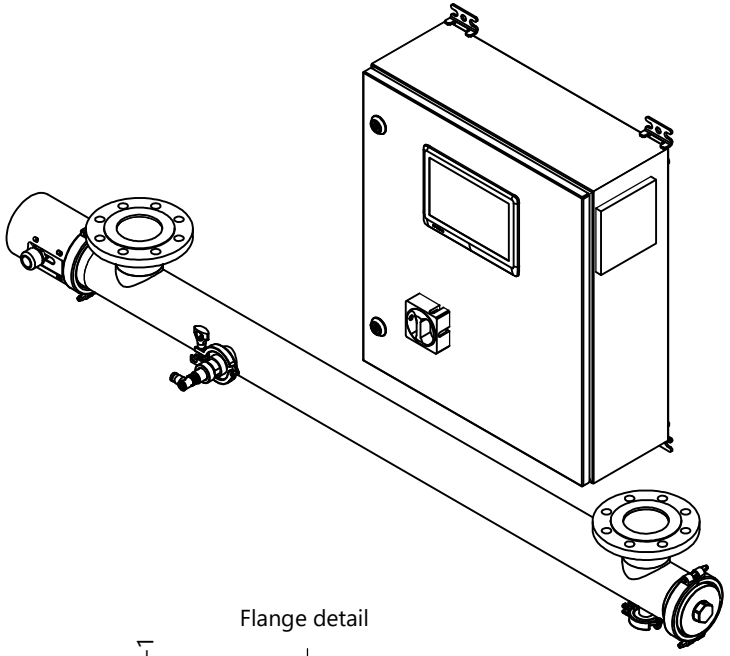
TITLE: ECO 15 (PN10 NW)	DRN	Walker	DATE	2023/7/20
	CKD	Walker	APPRD	Raymond
DRAWING No:- 820339-1C12E-01				

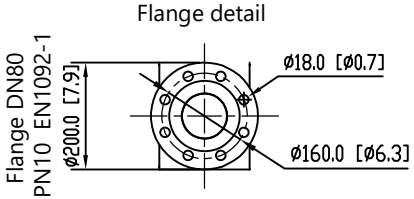
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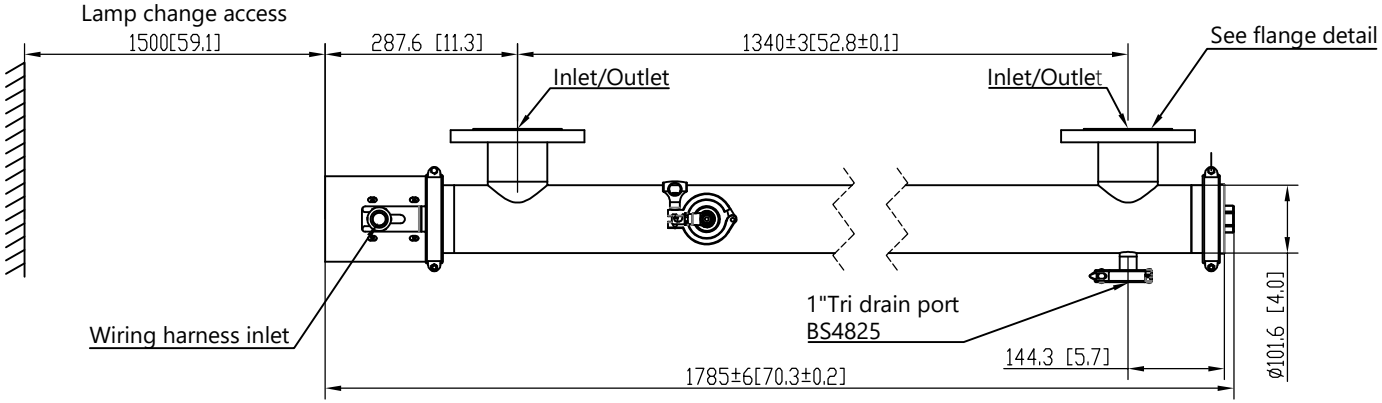
MAX OPERATING PRESSURE: 10 BAR (145 PSI)
CHAMBER WEIGHT: 24.3 KG (53.6 LBS)
CHAMBER WEIGHT IN OPERATION: 37 KG (81.6 LBS)

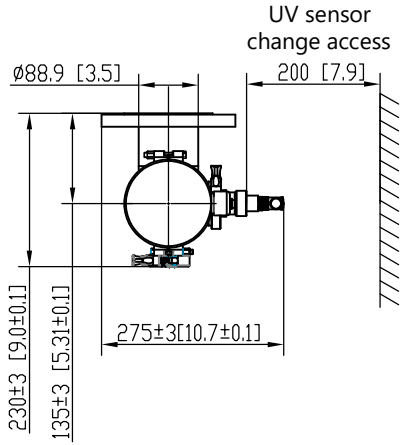








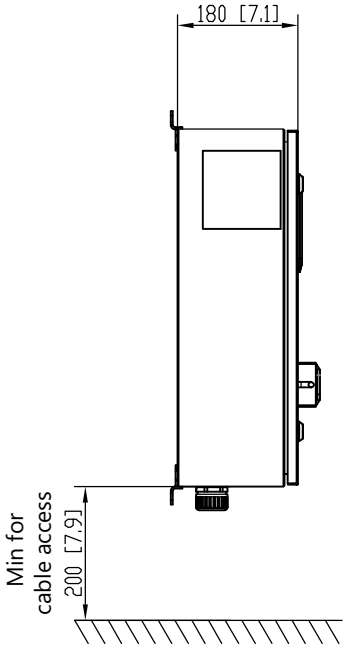


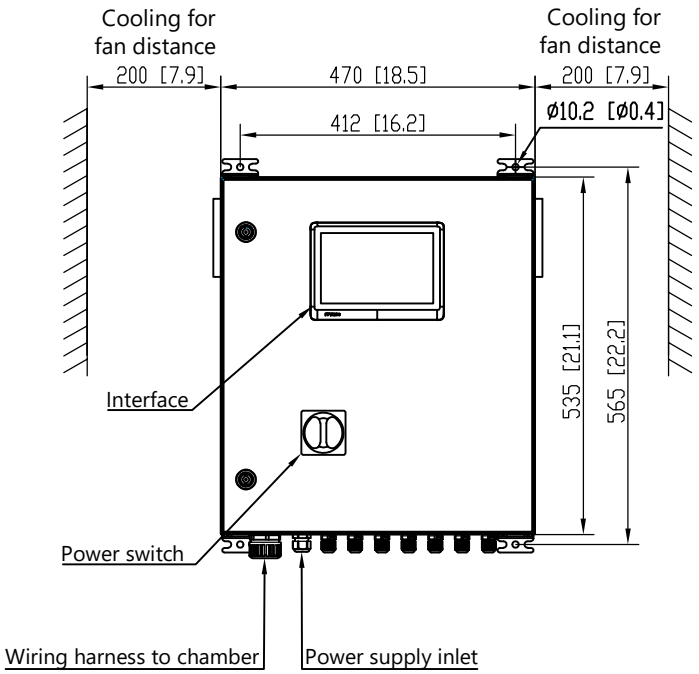


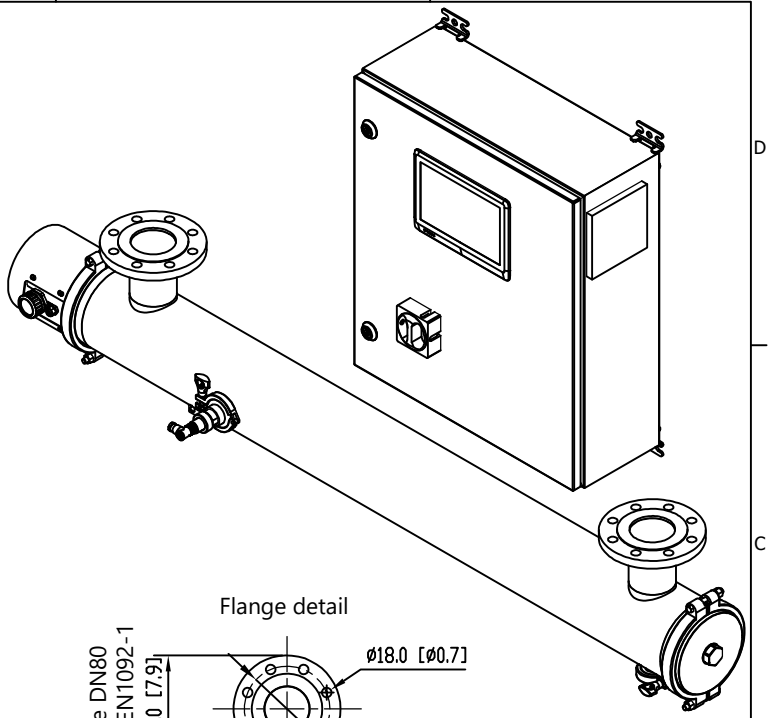
TITLE: ECO 32 (PN10 NW)	DRN	Walker	DATE	2023/7/20
	CKD	Walker	APPRD	Raymond
DRAWING No:- 820340-1E12E-01				

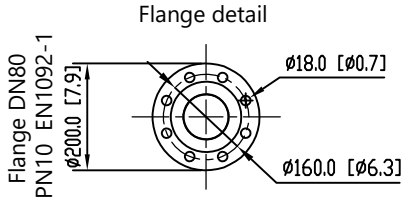
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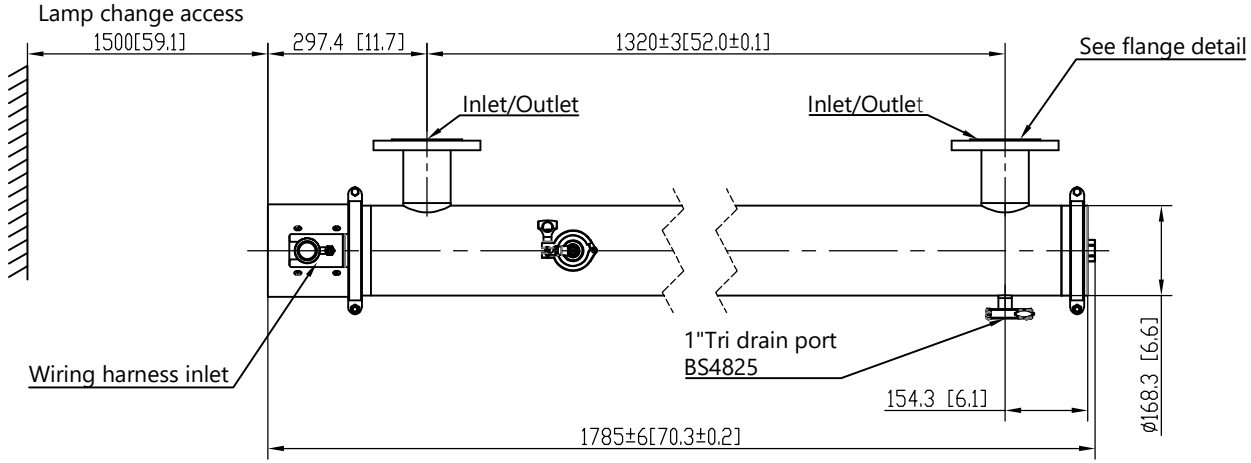
MAX OPERATING PRESSURE: 10 BAR (145 PSI)
CHAMBER WEIGHT: 42.7 KG (94.1 LBS)
CHAMBER WEIGHT IN OPERATION: 74.7 KG (164.6 LBS)

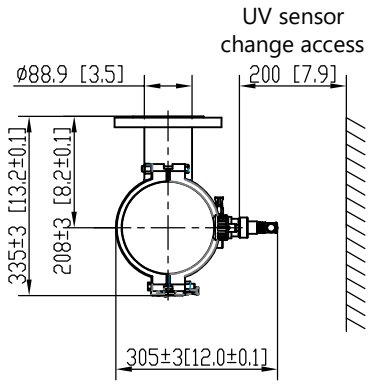










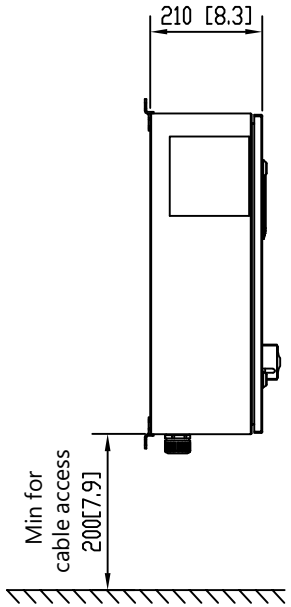


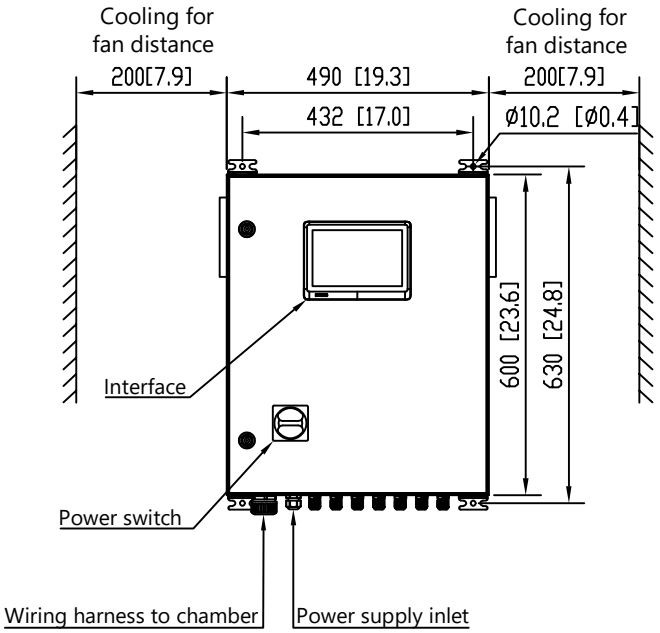
NUVONIC	TITLE: ECO 40 (PN10 NW)	DRN	Walker	DATE	2023/7/20
		CKD	Walker	APPRD	Raymond

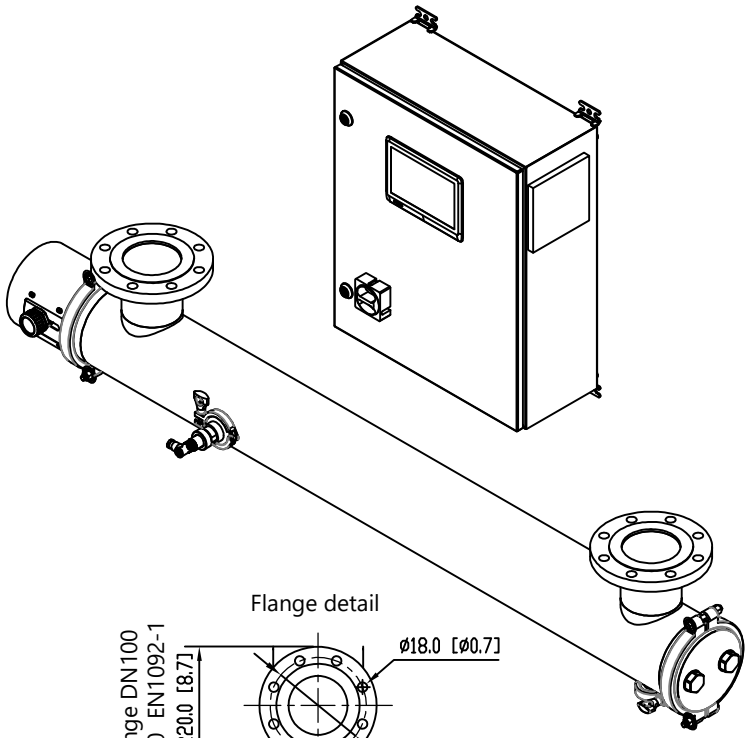
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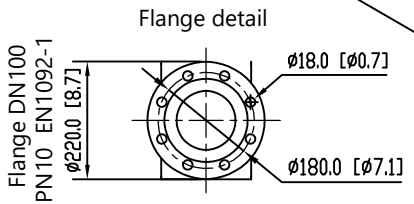
DRAWING No:- 820341-1E12E-01

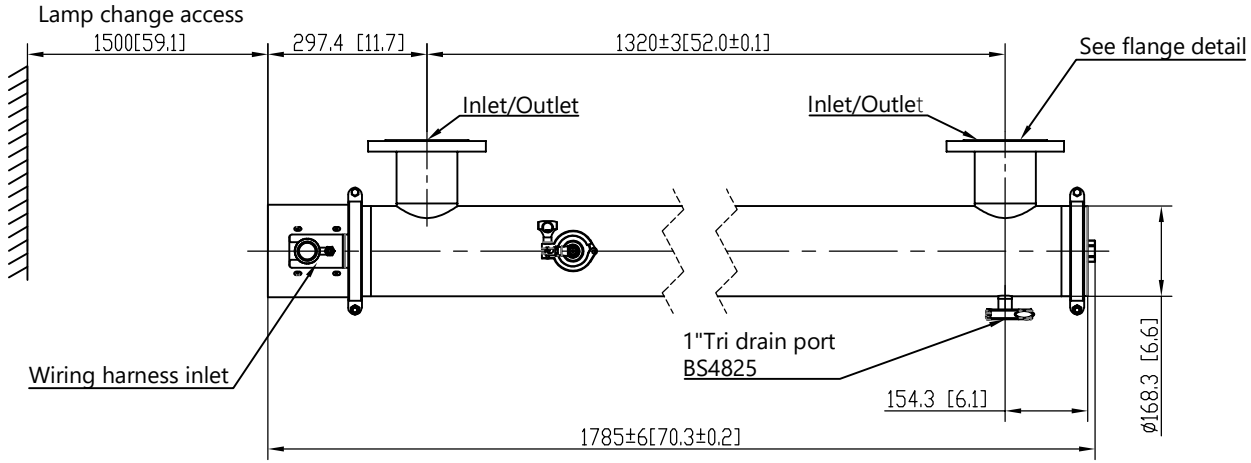
MAX OPERATING PRESSURE: 10 BAR (145 PSI)
CHAMBER WEIGHT: 42.7 KG (94.1 LBS)
CHAMBER WEIGHT IN OPERATION: 74.7 KG (164.6 LBS)

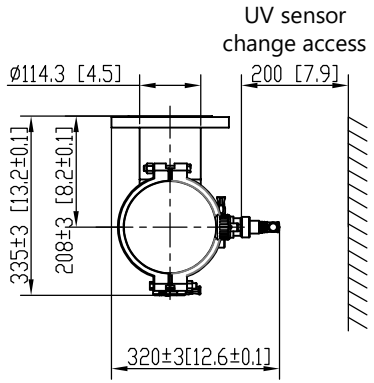








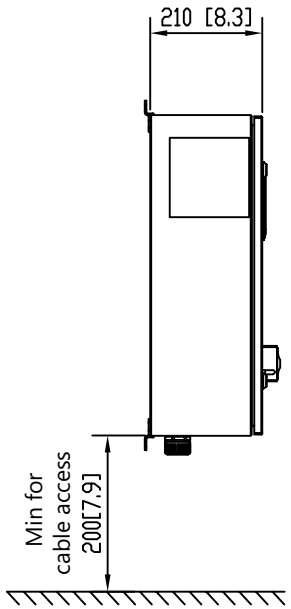


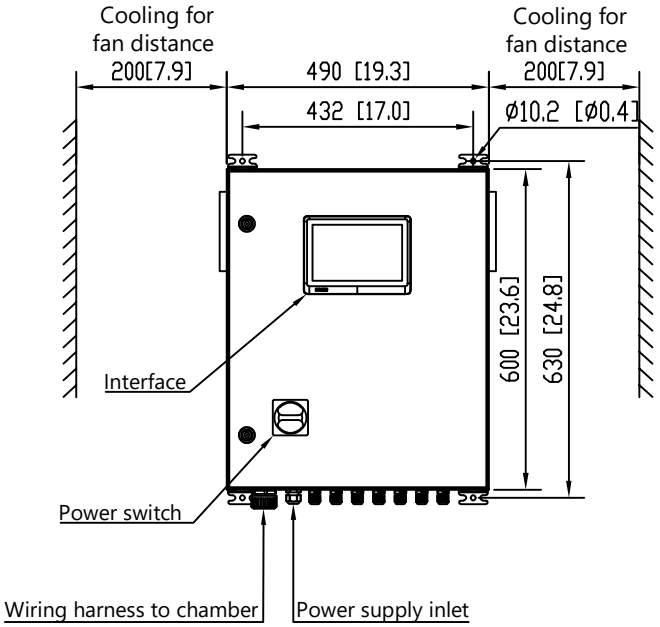


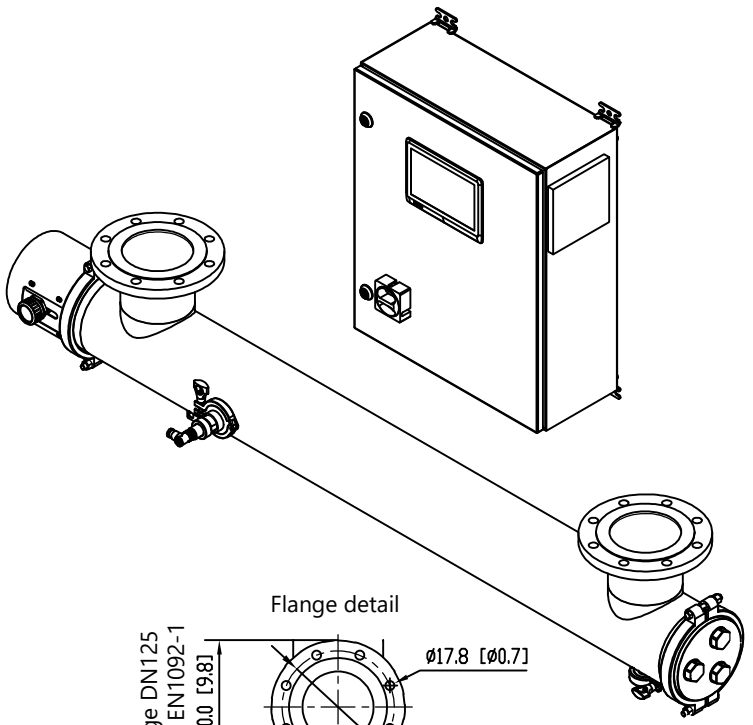
TITLE: ECO 85 (PN10 NW)	DRN	Walker	DATE	2023/7/20
	CKD	Walker	APPRD	Raymond
DRAWING No:- 820342-1G12E-01				

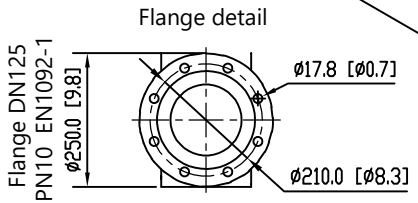
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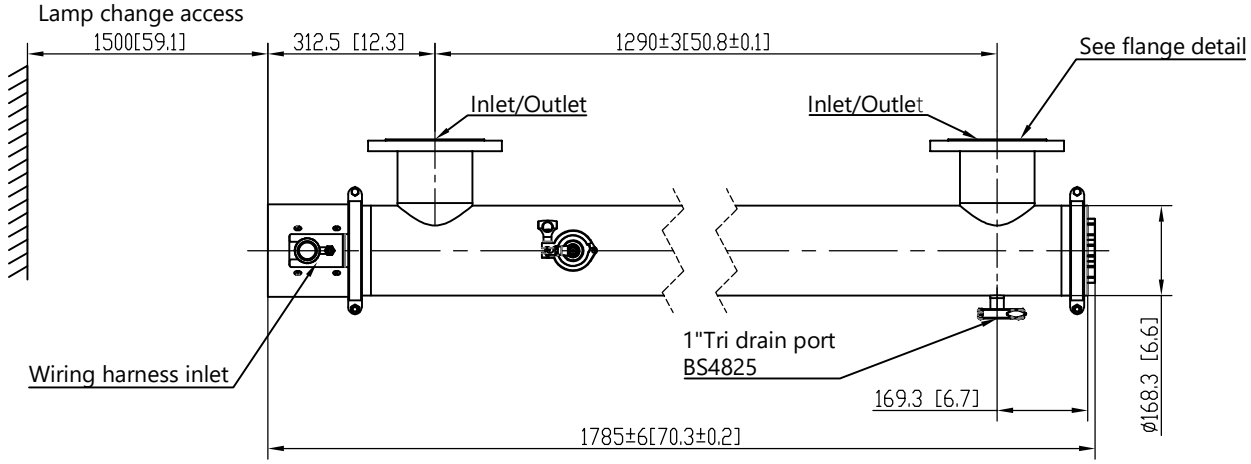
MAX OPERATING PRESSURE: 10 BAR (145 PSI)
CHAMBER WEIGHT: 43.5 KG (95.8 LBS)
CHAMBER WEIGHT IN OPERATION: 76.5 KG (168.6 LBS)

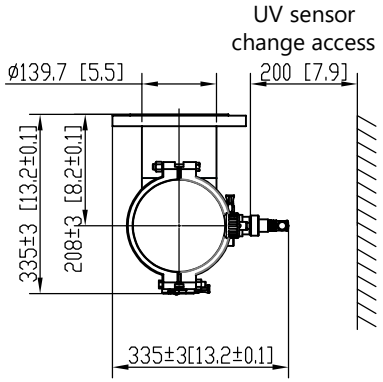






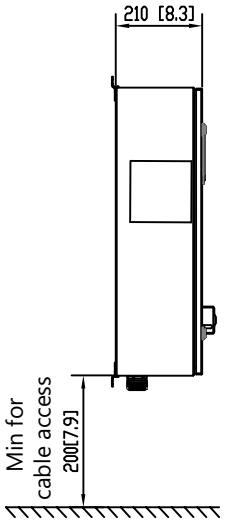


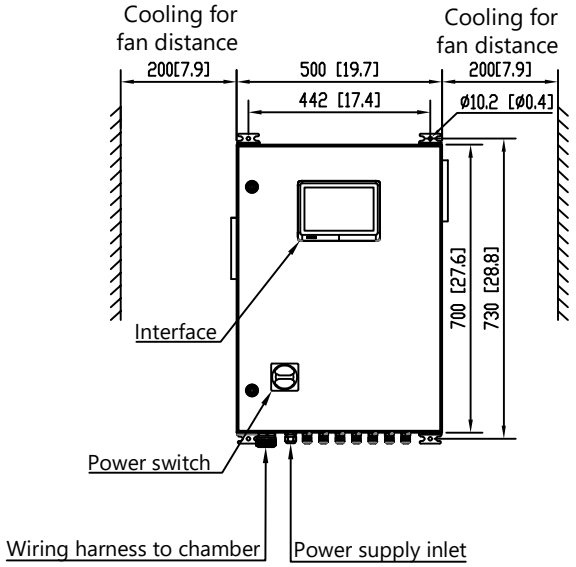


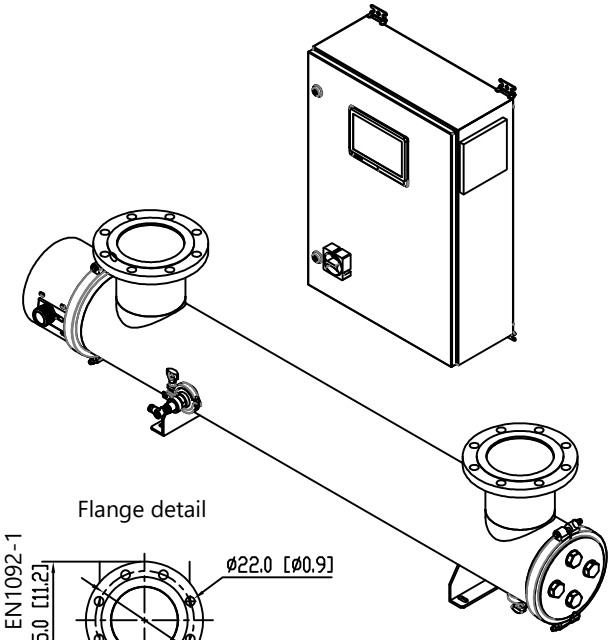


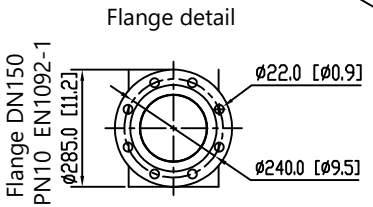
TITLE: ECO 125 (PN10 NW)		DRN	Walker	DATE	2023/7/20
		CKD	Walker	APPRD	Raymond
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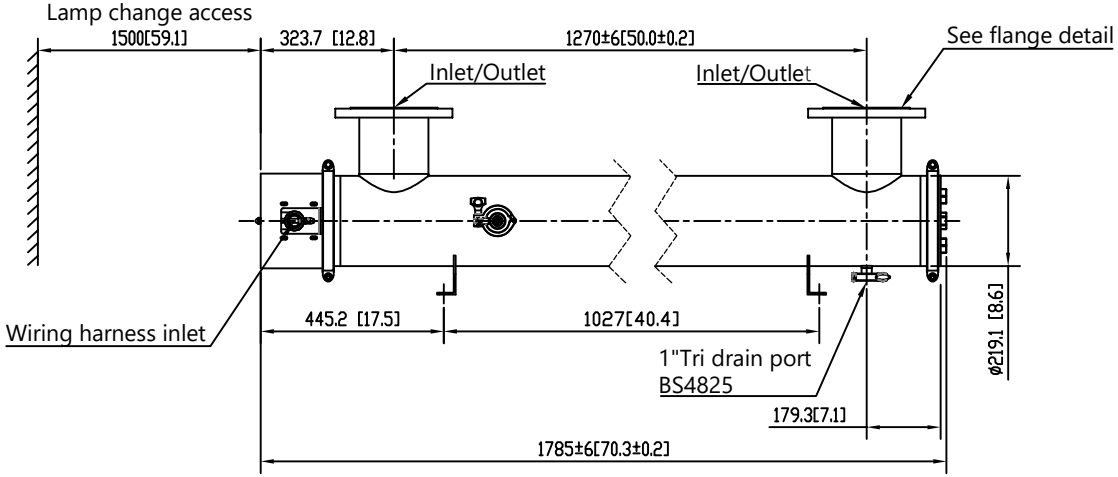
MAX OPERATING PRESSURE: 10 BAR (145 PSI)
CHAMBER WEIGHT: 67.2 KG (148.2 LBS)
CHAMBER WEIGHT IN OPERATION: 123.5 KG (272.3 LBS)

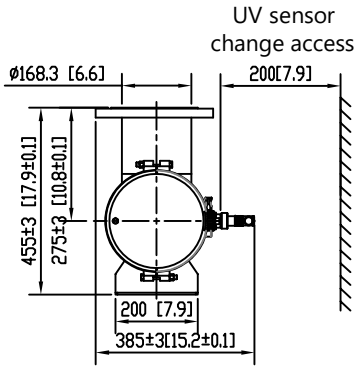


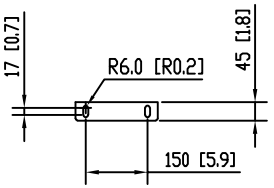










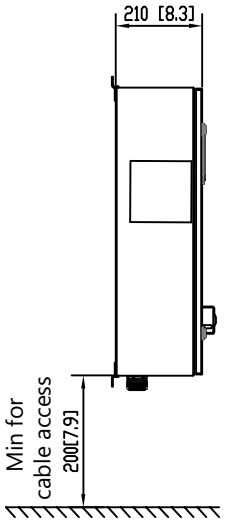


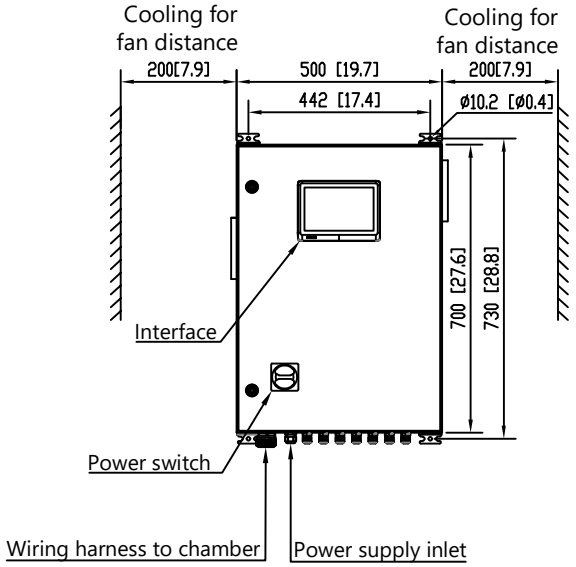
TITLE: ECO 215 (PN10 NW)	DRN	Walker	DATE	2023/7/20
	CKD	Walker	APPRD	Raymond

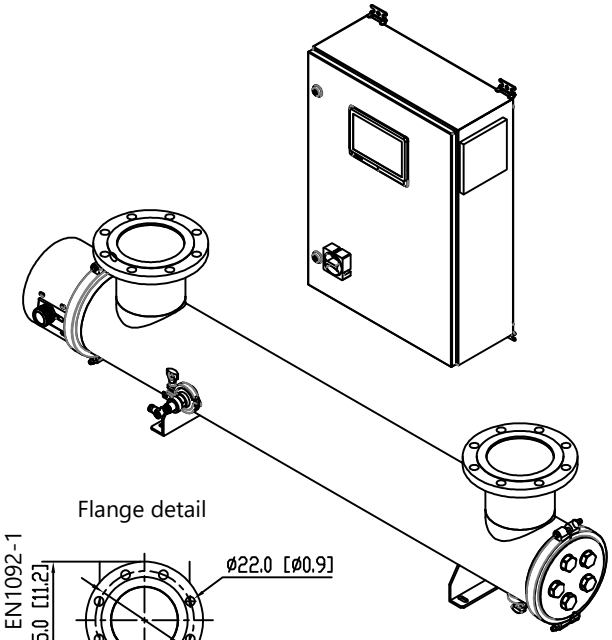
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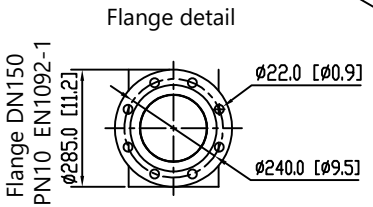
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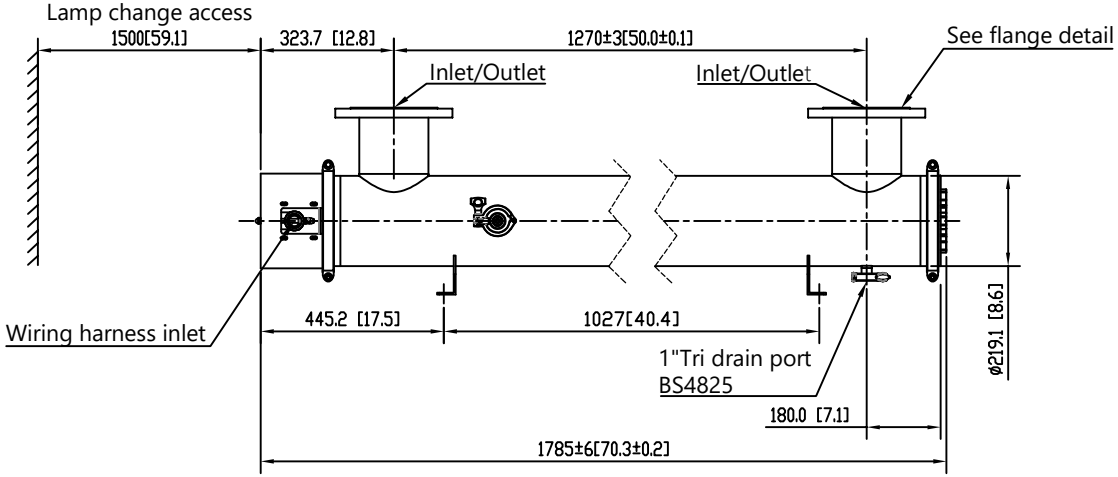
MAX OPERATING PRESSURE: 10 BAR (145 PSI)
CHAMBER WEIGHT: 67.2 KG (148.2 LBS)
CHAMBER WEIGHT IN OPERATION: 123.5 KG (272.3 LBS)

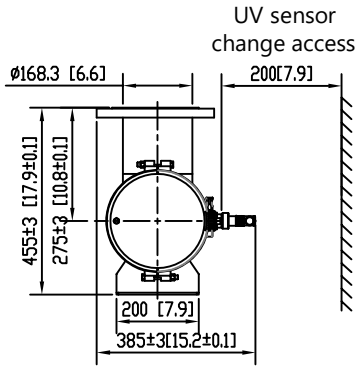


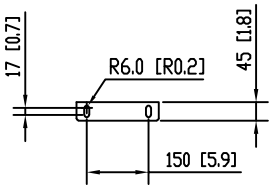










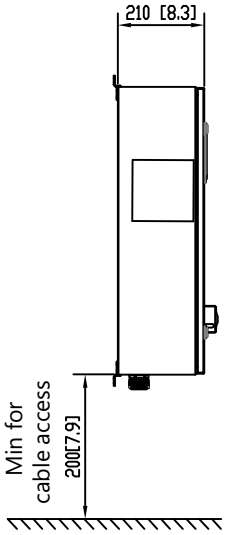


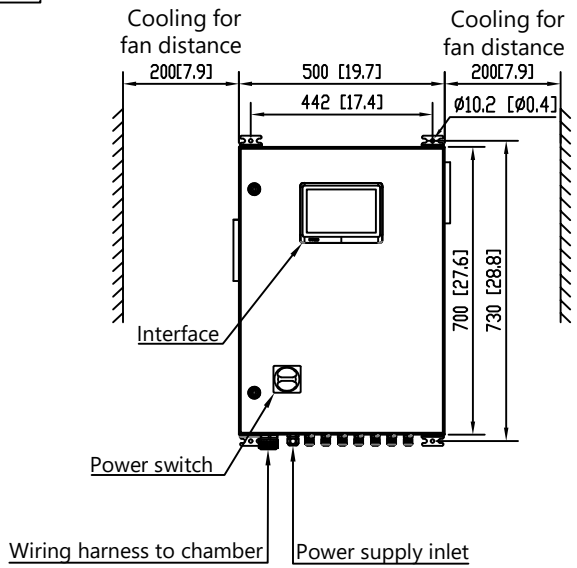
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	CKD	Walker	APPRD	Raymond

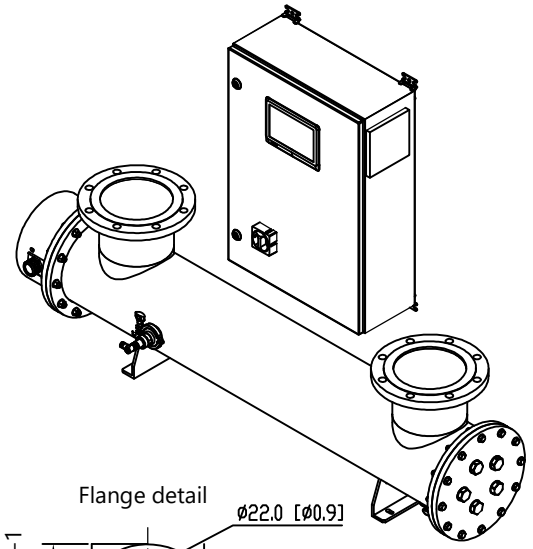
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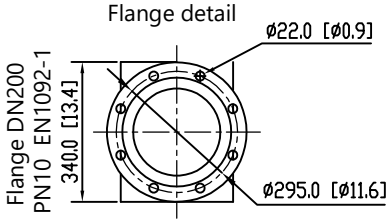
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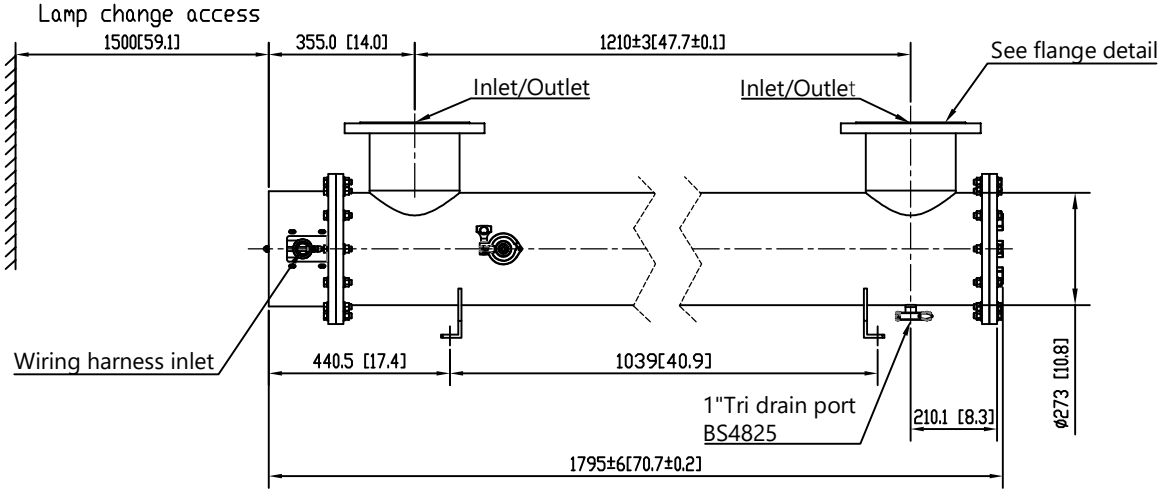
MAX OPERATING PRESSURE: 10 BAR (145 PSI)
CHAMBER WEIGHT: 127.2 KG (280.4 LBS)
CHAMBER WEIGHT IN OPERATION: 215.6 KG (475.3 LBS)

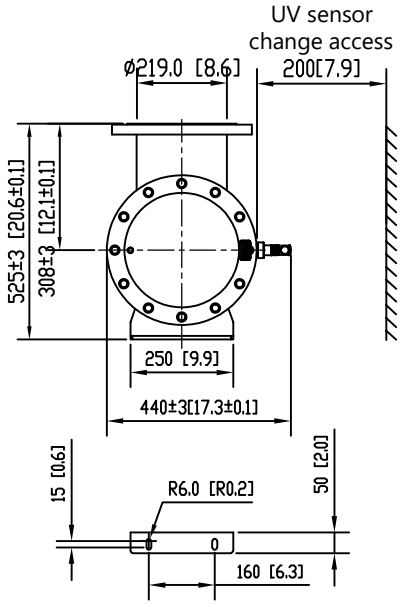










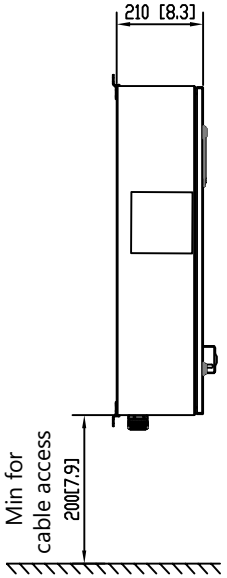


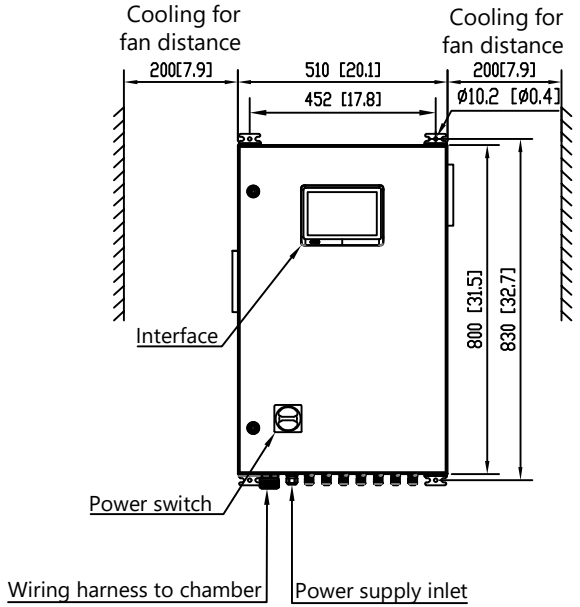
TITLE: ECO 330 (PN10 NW)		DRN	Walker	DATE	2023/7/20
		CKD	Walker	APPRD	Raymond

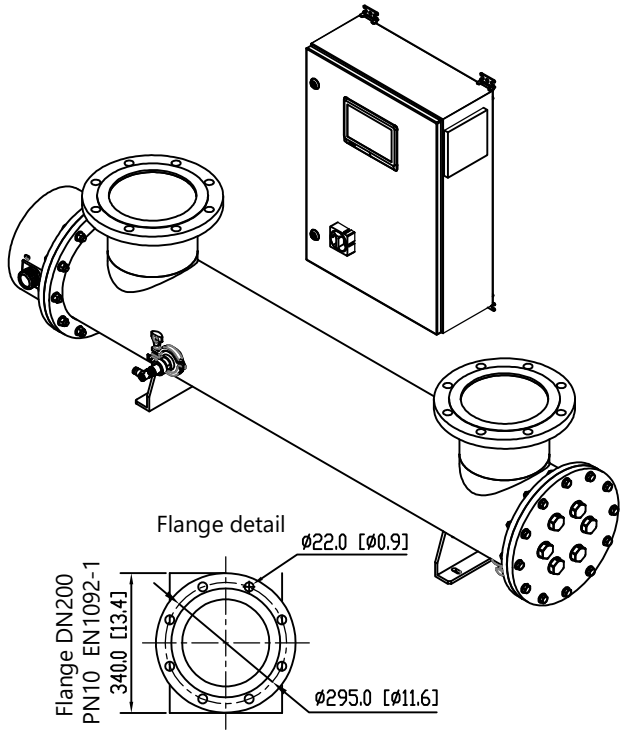
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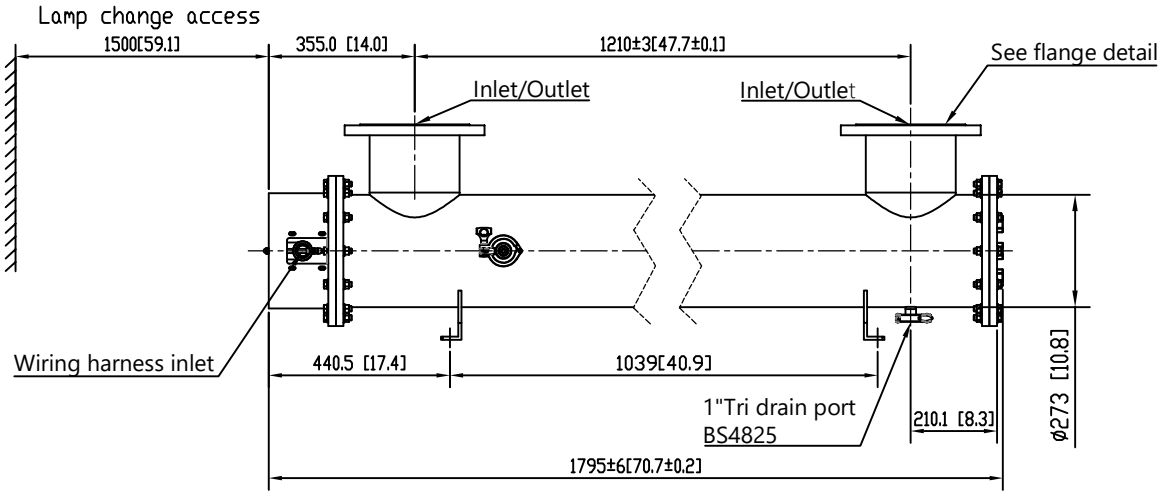
DRAWING No:- 820346-1M12E-01

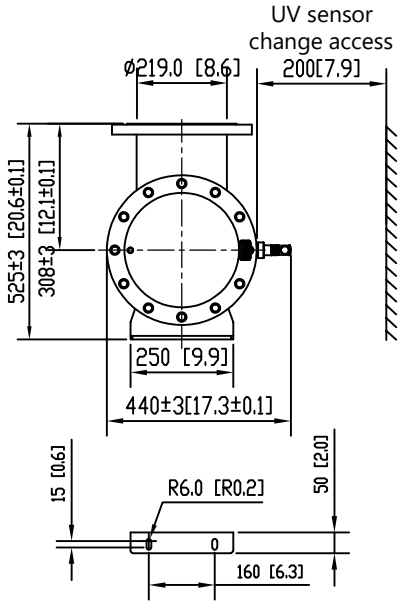
MAX OPERATING PRESSURE: 10 BAR (145 PSI)
CHAMBER WEIGHT: 127.2 KG (280.4 LBS)
CHAMBER WEIGHT IN OPERATION: 215.6 KG (475.3 LBS)







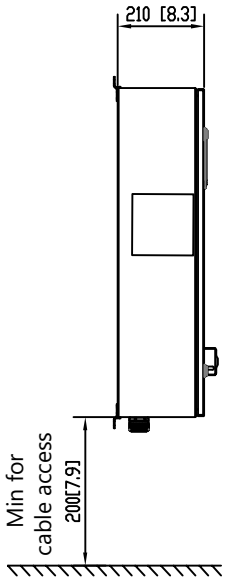


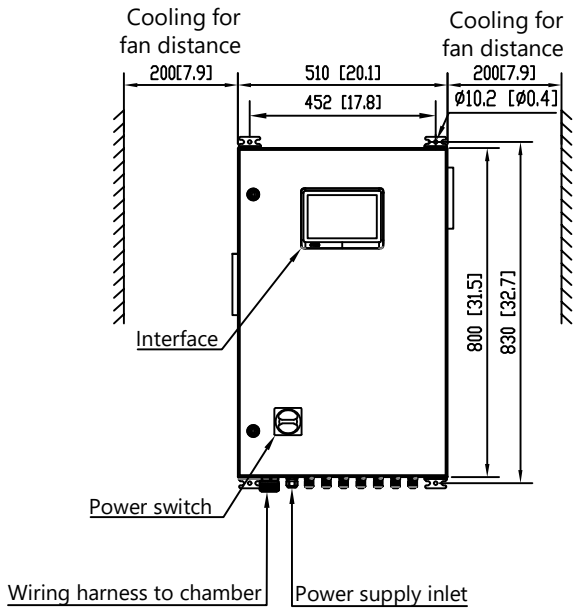


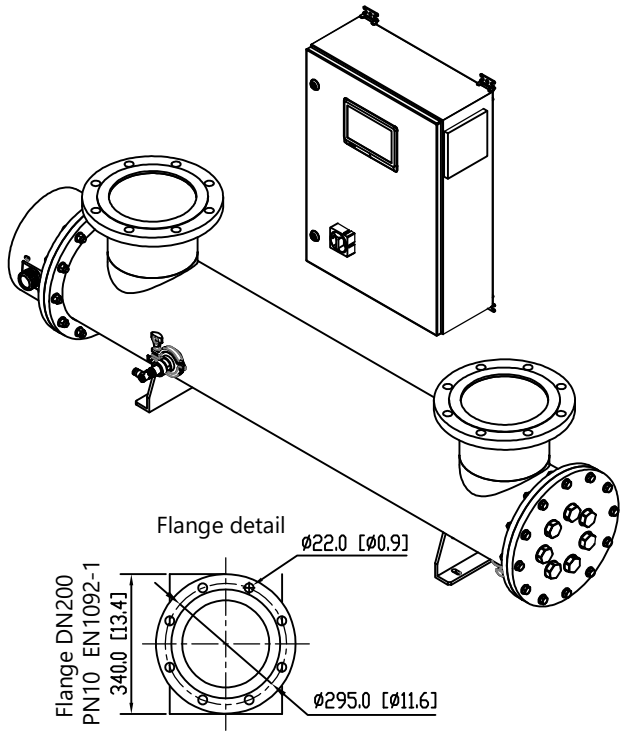
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	CKD	Walker	APPRD	Raymond
DRAWING No:- 820347-1M12E-01				

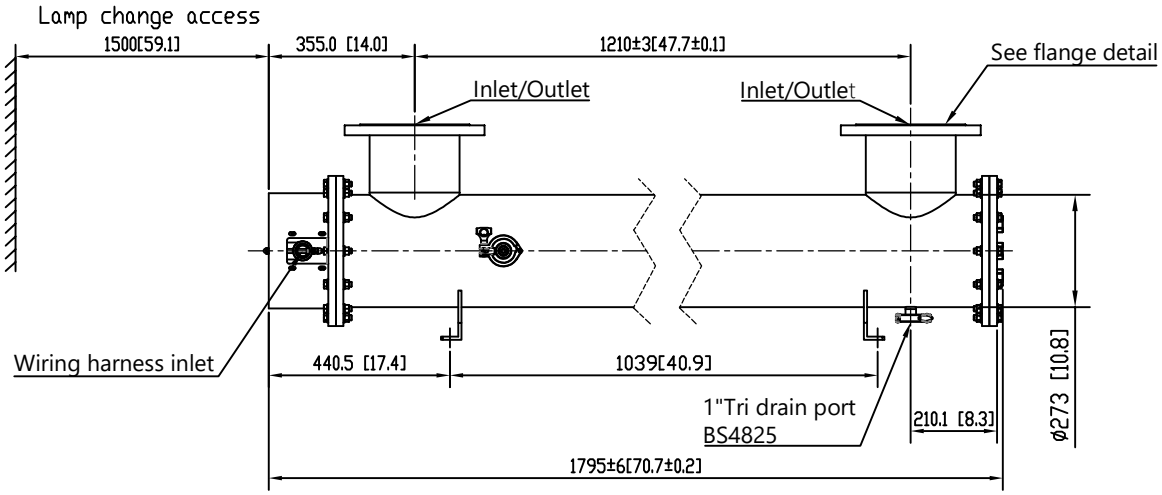
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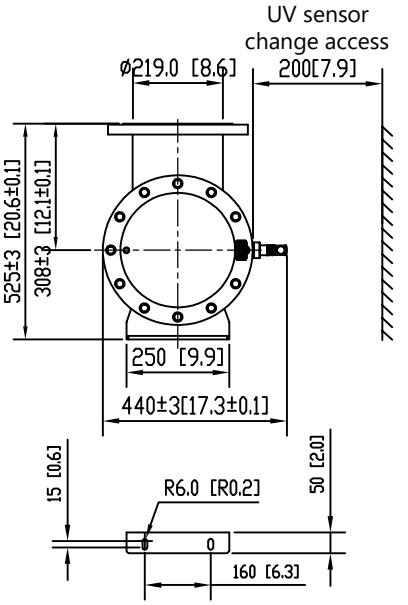
MAX OPERATING PRESSURE: 10 BAR (145 PSI)
CHAMBER WEIGHT: 127.2 KG (280.4 LBS)
CHAMBER WEIGHT IN OPERATION: 215.6 KG (475.3 LBS)







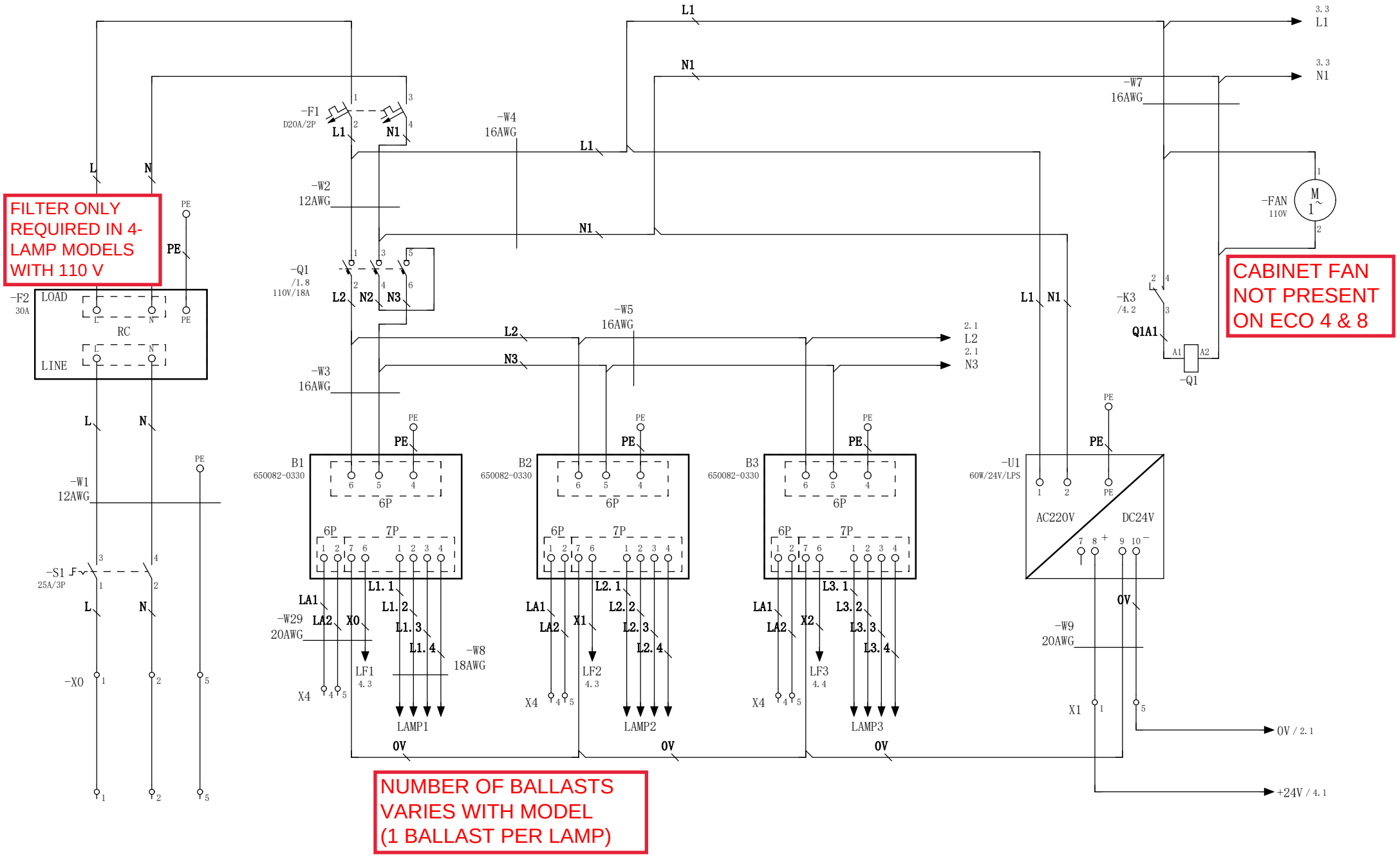




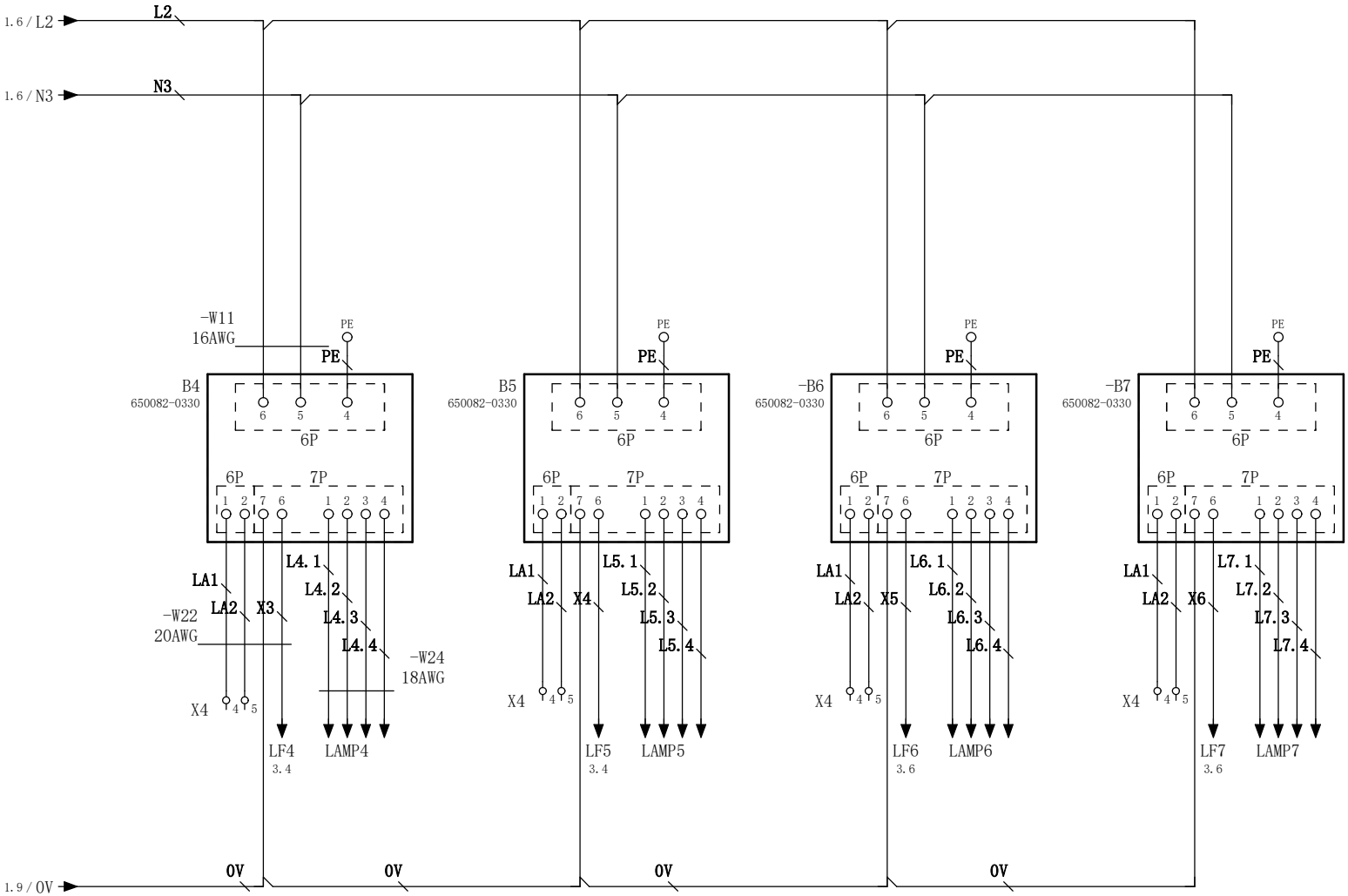
TITLE: ECO 380 (PN10 NW)	DRN	Walker	DATE	2023/7/20
	CKD	Walker	APPRD	Raymond
DRAWING No:- 820348-1M12E-01				

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UNWIPED SYSTEMS

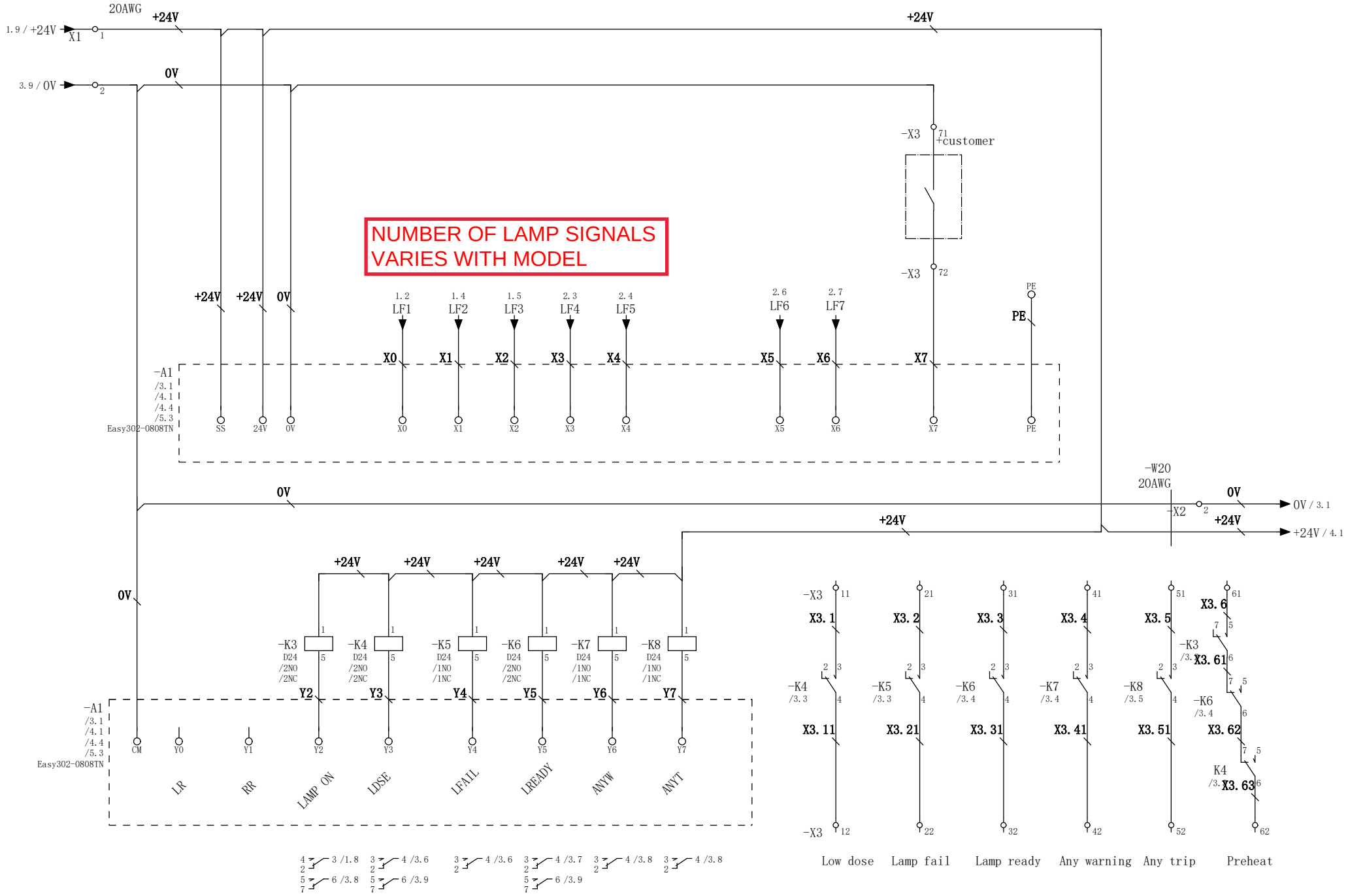


			Date	2024-9-4			E-XX-04-U1 Page1/5	
			Design	Christian				
			Prove	Raymond				Page 1
			Version	V3				Total 5



NUMBER OF BALLASTS
VARIES WITH MODEL
(1 BALLAST PER LAMP)

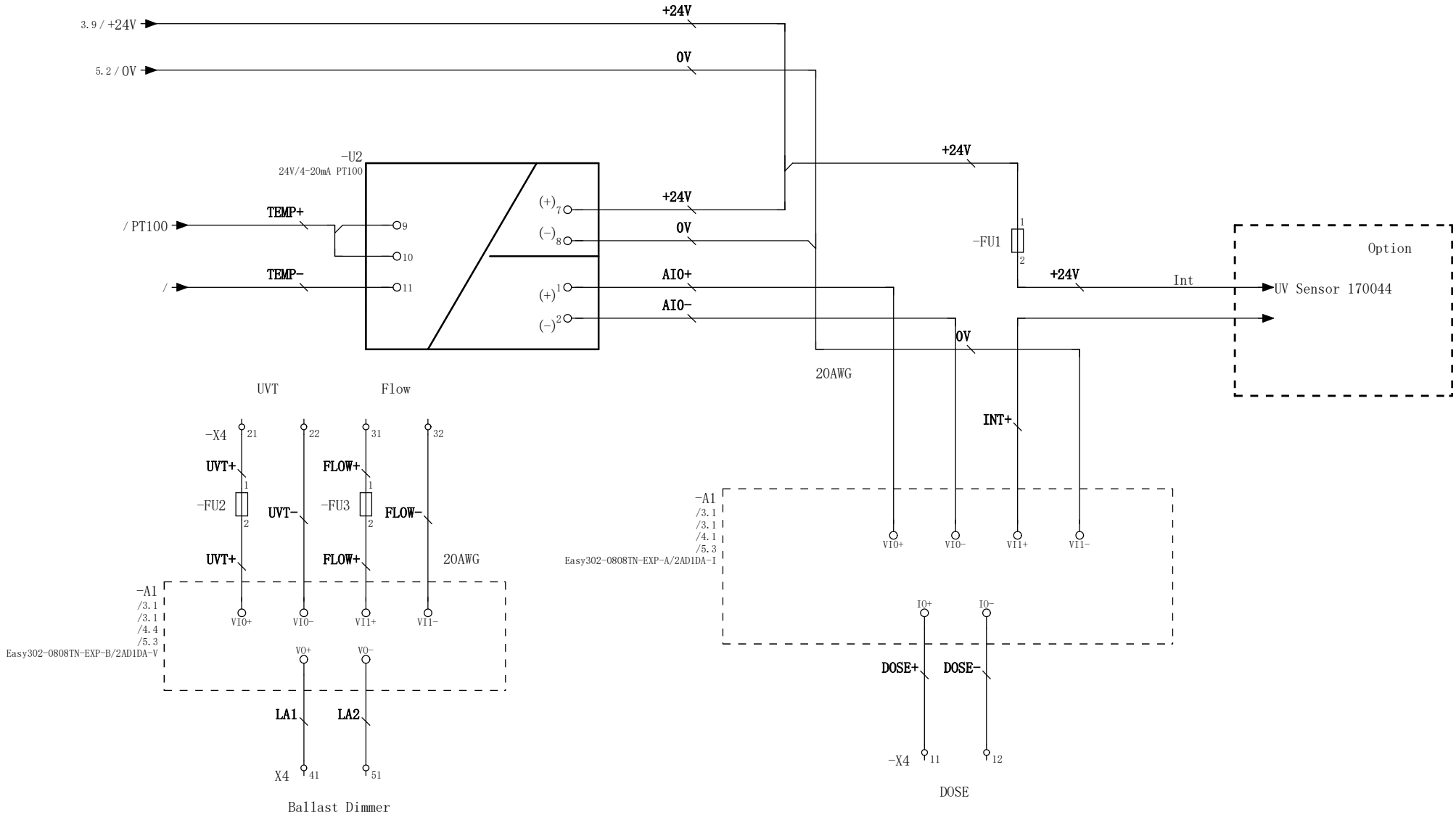
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			Design	Christian				
			Prove	Raymond				Page 2
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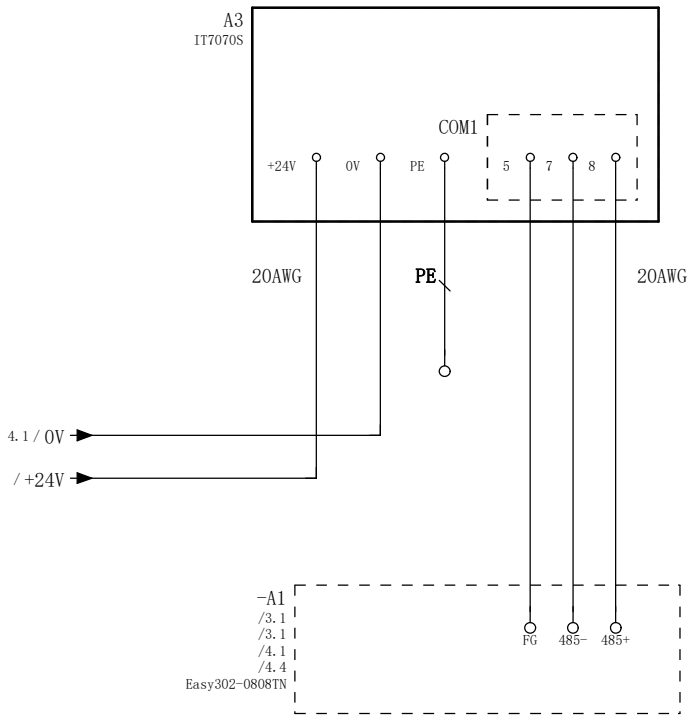
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Design	Christian
Prove	Raymond
Version	V3

E-XN-07-C1 Page3/5

Page	3
Total	5

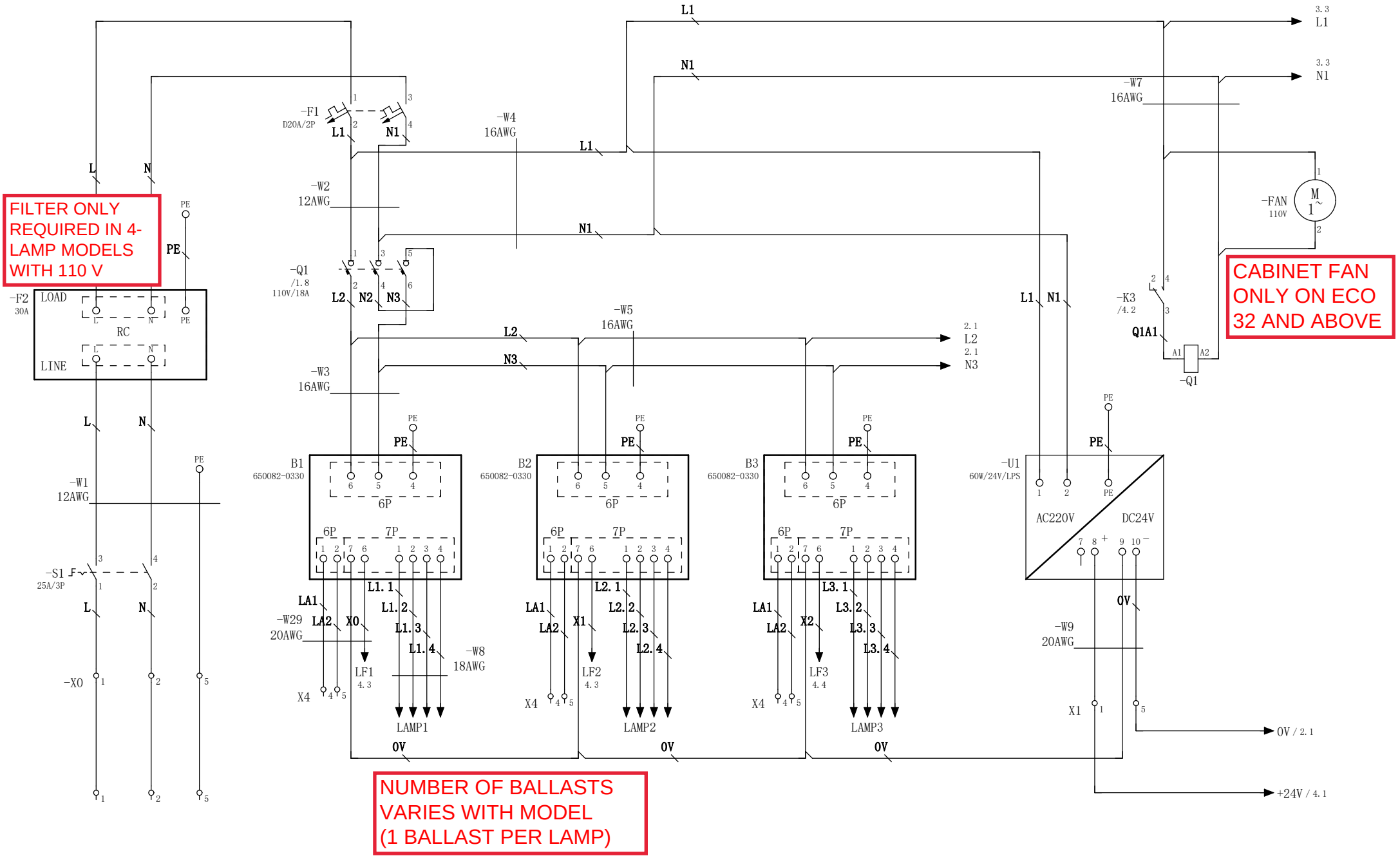


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			Design	Christian					
			Prove	Raymond				Page	4
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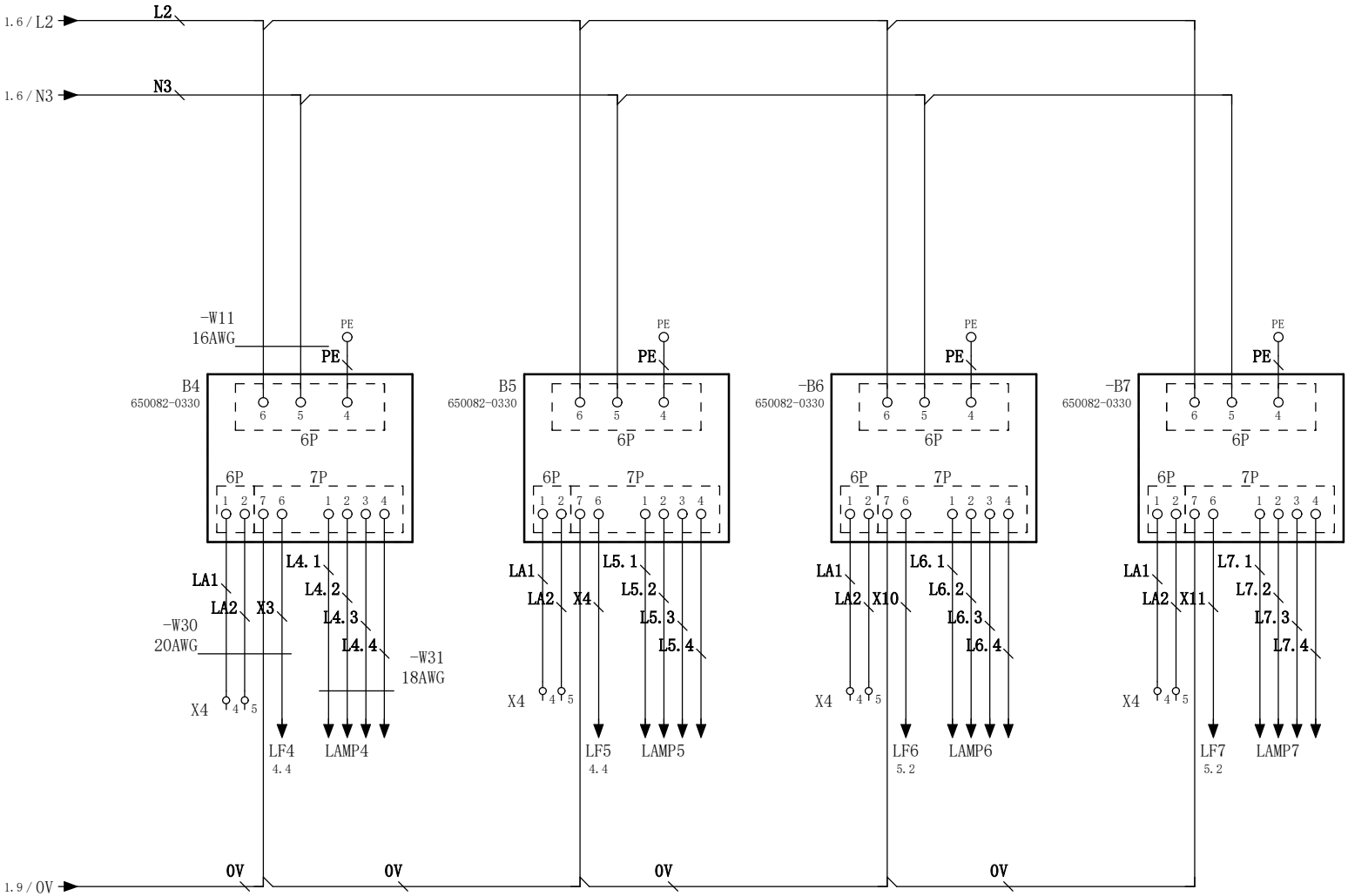


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			Design	Christian				
			Prove	Raymond				Page 5
			Version	V3				Total 5

WIPED SYSTEMS



			Date	2024-9-4			E-XX-04-U1	Page1/7	
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			Version	V3				Total	7

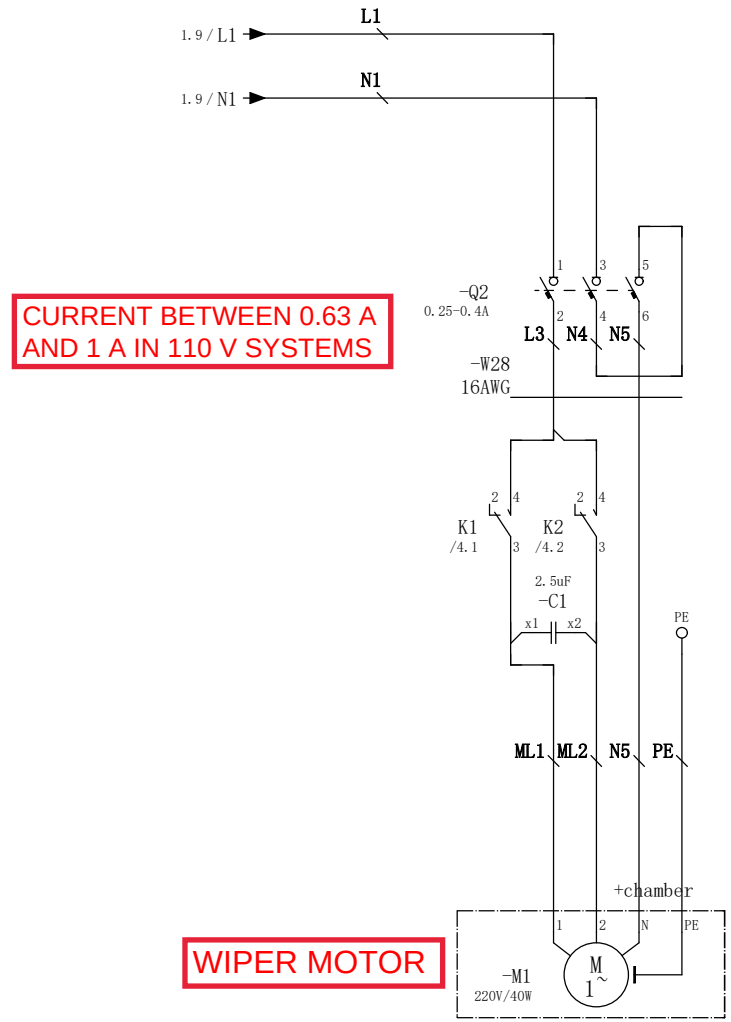


NUMBER OF BALLASTS
VARIES WITH MODEL
(1 BALLAST PER LAMP)

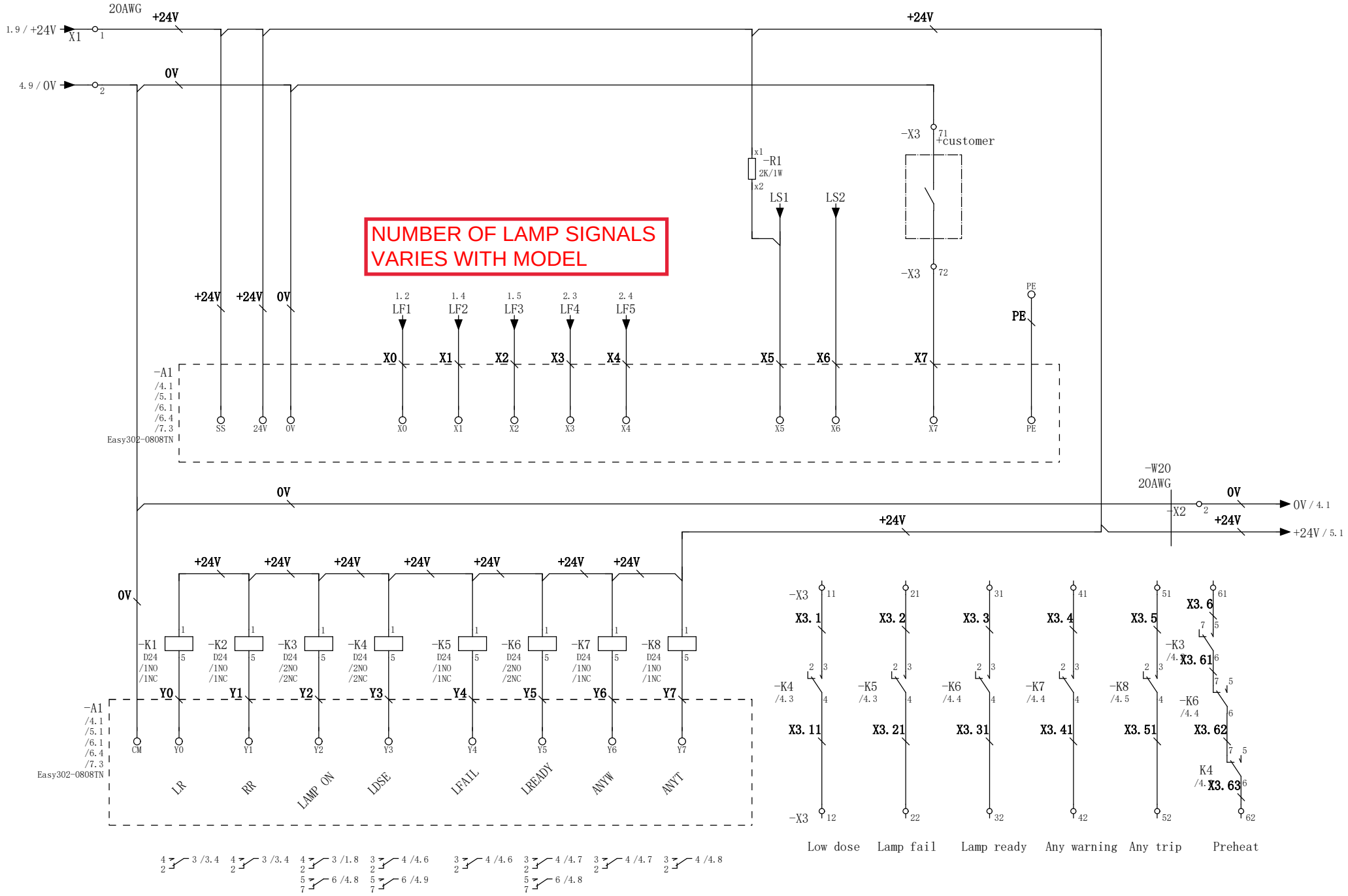
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E-XW-07-C1 Page2/7

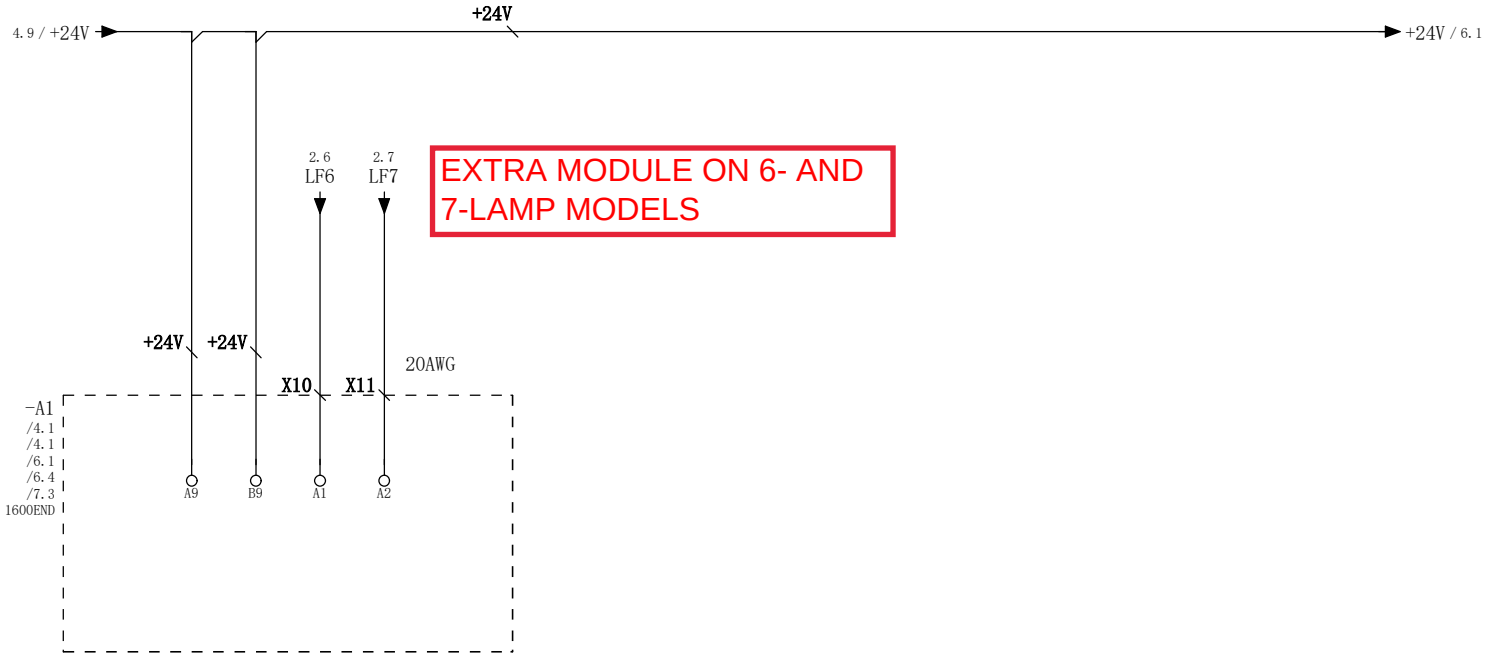
Page 2
Total 7



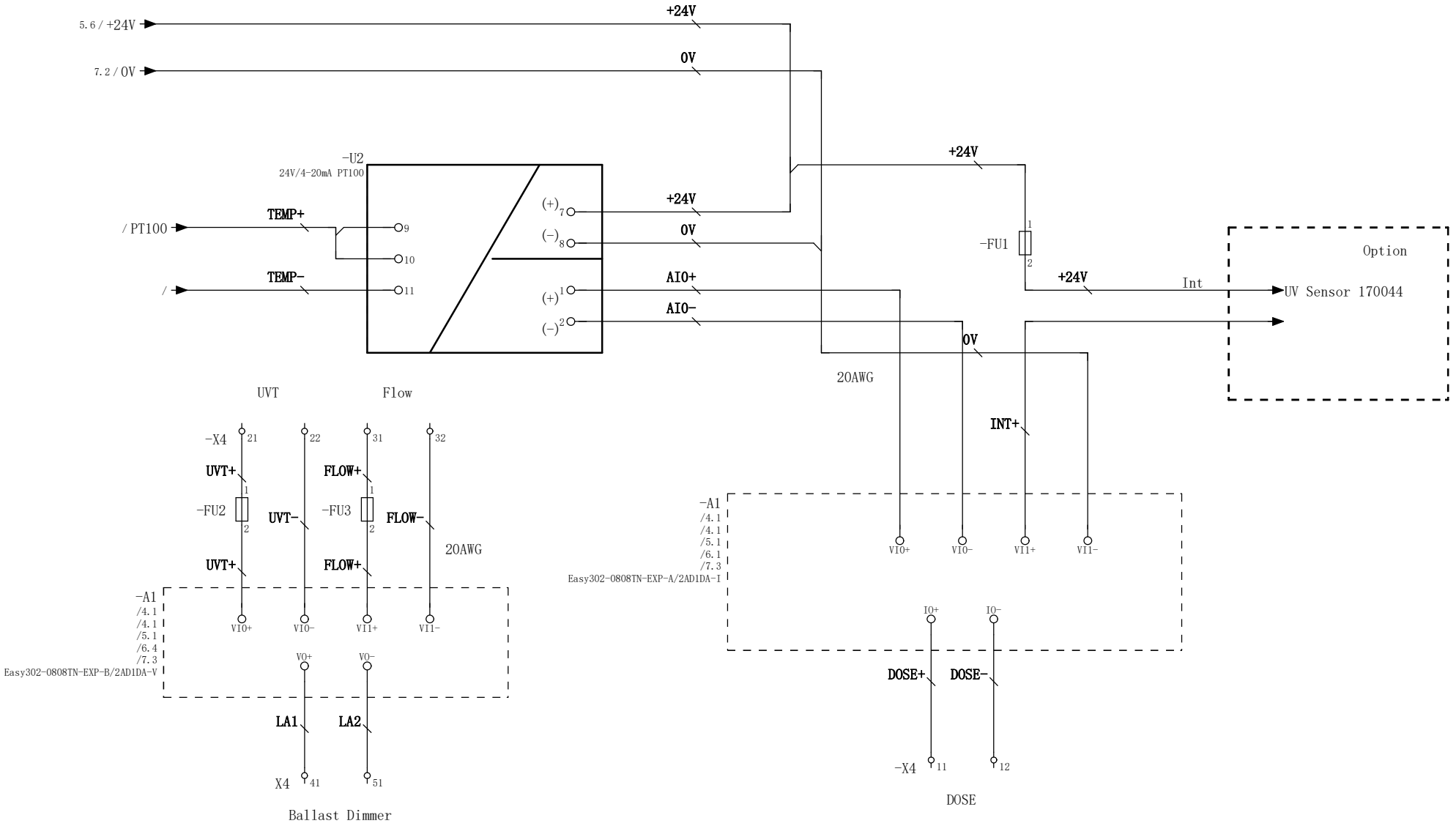
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			Version	V3				Total	7



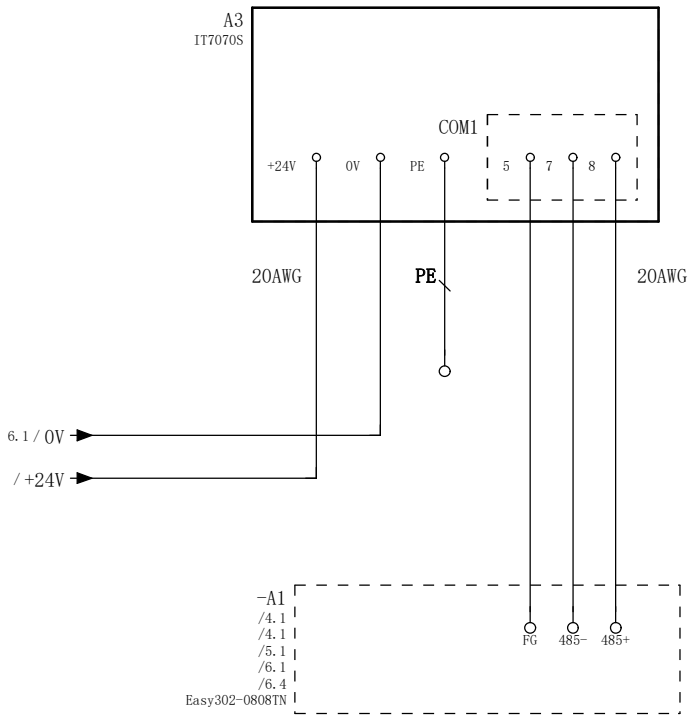
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			Version	V3				Total	7



			Date	2024-9-4			E-XW-07-C1 Page5/7	
			Design	Christian				
			Prove	Raymond				Page 5
			Version	V3				Total 7



			Date	2024-9-4			E-XW-07-C1 Page6/7	
			Design	Christian				
			Prove	Raymond				Page 6
			Version	V3				Total 7



			Date	2024-9-4			E-XW-07-C1 Page7/7	
			Design	Christian				
			Prove	Raymond				Page 7
			Version	V3				Total 7